

Housing Like Irish Weather?

Volatility Regimes and Asymmetric Transmission
in the Irish Housing Market



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Abstract

This paper examines whether the transmission of demand conditions to housing-market outcomes in Ireland is regime-dependent. Regimes are identified from demand-block dynamics — modified domestic demand, rent-to-income, unemployment, and CPI — rather than from house prices, separating identification from the outcomes being explained. A hierarchical Bayesian Markov-switching VAR is estimated under a novel Student- t shrinkage prior (the $\mathfrak{J}\mathfrak{A}$ prior), which nests the Minnesota prior as a limiting case and estimates tail thickness jointly with the VAR coefficients. Regime uncertainty is propagated across blocks by embedding a differentiable Hamilton filter inside the No-U-Turn Sampler, analytically marginalising the discrete state sequence at each leapfrog step. This eliminates the generated-regressor bias that arises when posterior-mean smoothed probabilities are treated as fixed inputs in later stages. A block-specific mixing weight α_b is introduced to measure the contribution of each block’s own dynamics to regime classification beyond the demand signal. The estimates indicate that credit and asset-market transmission are state-dependent in both magnitude and persistence. The asset block exhibits a single 28-quarter high-volatility episode spanning 2008Q2–2015Q1, five times longer than the corresponding demand episode, consistent with a financial accelerator sustaining stress well beyond the initiating shock. Credit transmission is asymmetric across regimes with the shadow-rate response roughly twice as large under S_1 , but stress episodes are short-lived. The supply block is largely regime-invariant, consistent with binding non-price constraints on construction. Residential property prices exhibit only modest regime separation. Linear models that average across regimes therefore understate policy elasticities in precisely the periods where accurate calibration matters most.

Keywords: Irish housing market; Markov-switching VAR; Bayesian shrinkage; regime asymmetry; MCMC; impulse response analysis.

JEL Codes: C11, C32, E44, R31.

1. Introduction

The Irish housing market has experienced significant structural transformation over the past twenty-five years. During the early 2000s, the Celtic Tiger era saw a credit-fuelled construction expansion which was associated with a marked increase in residential property prices, with real prices approximately doubling between 2000 and 2007. This period was characterised by lax lending practices leading to credit expansion, strong income growth, and inward migration from these pull factors, which generated sustained demand-side pressures. These dynamics reversed sharply however, during the Global Financial Crisis (2008 to ca.2010), which constituted a large negative macroeconomic downturn, resulting in a demand shock and a pronounced output contraction; unemployment rose to around 15 per cent, credit supply tightened sharply, and residential property prices declined by approximately 55 per cent from peak to trough by 2012. The subsequent sovereign debt crisis amplified these adjustment dynamics and led to Ireland's entry into an EU-IMF stabilisation programme in 2010. The recovery that followed was uneven, reflecting both deleveraging and supply-side rigidities. The institutional framework for mortgage lending was subsequently reconfigured with the introduction of macroprudential policy in 2015, aimed at containing systemic risk and moderating credit-driven demand. The COVID-19 pandemic represents a further structural break, disrupting construction activity and inducing shifts in housing demand. By 2024, nominal residential property prices had returned to their pre-crisis peak, causing affordability constraints to intensify markedly.

The sequence of booms, crises, and policy changes makes it unlikely that transmission mechanisms linking demand, credit, supply, and asset markets remained constant. A single-regime empirical framework is therefore likely to mischaracterise important features of the housing cycle.

The central empirical task of this paper is to identify these regime shifts and characterise how they alter the propagation of shocks through the housing system. The standard Markov-switching VAR framework faces two problems in this context. First, existing work often conflates regime identification with the dependent variable of interest. Regimes are extracted from house-price or asset-return fluctuations and then used to characterise

the dynamics of those same dependent variables (Hamilton and Raj, 2002; Guidolin and Timmermann, 2007), making it difficult to separate the source of the regime from the behaviour it is meant to explain.

Second, multi-stage hierarchical frameworks routinely pass posterior-mean smoothed regime probabilities from an earlier stage as fixed inputs to later stages. These probabilities are estimated quantities, not observed data, so conditioning on them suppresses posterior uncertainty in downstream parameters, leading to the generated-regressor problem identified by Pagan (1984). Consequently, uncertainty is understated particularly during stress periods, where policy interpretation is most sensitive to it.

The paper addresses these two problems through a block-recursive identification strategy (Zha, 1999), in which regimes are anchored to demand-block dynamics — where household formation and mortgage eligibility are determined — and downstream blocks are conditioned on the resulting latent state signal. Embedding a differentiable Hamilton filter inside the No-U-Turn Sampler then marginalises the discrete regime sequence analytically at every leapfrog step, so posterior uncertainty flows through the full estimation cascade rather than being truncated at the stage boundary.

Empirical results indicate that credit and asset-market transmission are state-dependent in magnitude and duration, whereas the supply block is largely regime-invariant and the univariate price block exhibits only modest regime separation. Linear models that average across regimes therefore understate policy elasticities during high-volatility periods, which has direct implications for the calibration of macroprudential tools.

Section 1.1 summarises the contributions and Section 1.2 situates the paper in the related literature. Sections 2.1 and 2.9 cover data, methodology, and robustness. Sections 3 and 4 report the results and their macroprudential implications. Section 5 concludes.

1.1 Contributions

This paper makes three contributions to the literature. First, regimes are identified from demand-block dynamics (modified domestic demand, rent-to-income, unemployment, and CPI) rather than from house prices, separating identification from outcomes. The identification ordering reflects the empirical regularity that income and labour-market

conditions shape household formation and mortgage eligibility before feeding through to construction activity or asset prices (Kelly et al., 2018; Egan and McQuinn, 2024b). This ordering is a maintained assumption; the robustness section examines its sensitivity.

Second, a hierarchical Student- t shrinkage prior (the $\mathfrak{JA}$ prior) is proposed for Markov-switching VARs. The global regularisation hyperparameters τ_A (shrinkage intensity), lag-decay rate φ , and tail thickness ν_A are estimated jointly with the VAR coefficients. Unlike the Horseshoe (Carvalho et al., 2010) and the adaptive hierarchical prior of Chan (2019), the $\mathfrak{JA}$ prior imposes monotonic lag decay and estimates tail thickness from the data, balancing regularisation in stable periods with flexibility should the data warrant larger deviations. It nests the Minnesota prior of Litterman (1986) as a limiting case ($\nu_A \rightarrow \infty$, $\varphi \rightarrow 1$; Appendix B.4).

Third, a differentiable Hamilton filter is embedded within the No-U-Turn Sampler (Hoffman and Gelman, 2014), re-evaluating the demand-block transition matrix and regime mixing weight α_b at every leapfrog step. This eliminates the generated-regressor bias (Pagan, 1984) that arises when posterior-mean smoothed probabilities are plugged in as fixed regressors in later stages, ensuring that regime uncertainty is fully reflected in all downstream credible intervals.

1.2 Related Literature

This paper sits at the intersection of four strands of literature: Markov-switching VAR modelling, Bayesian prior specification for VARs, the econometrics of multi-stage regime-switching estimation, and the empirical literature on the Irish housing market.

1.2.1 Markov-Switching Models and Regime-Dependent Transmission

The Markov-switching framework of Hamilton (1989) provides the standard approach for modelling discrete parameter shifts across latent states; Later work by Ang and Timmermann (2012) surveyed its application to financial markets and document substantial changes to inferences about return predictability and cross-asset correlations once latent states are admitted. Applications to housing markets have established that regime heterogeneity is economically substantive rather than merely statistical.

From a housing cycle perspective, work by Simo-Kengne et al. (2013) in the South

African housing market modelled with an MS-VAR has demonstrated that the impulse response of house prices to a contractionary monetary shock is substantially larger in bear regimes than in bullish regimes, an asymmetry consistent with a financial accelerator effect that amplifies shocks during downturns. Further evidence from the US corroborates this finding: [Baradaran Motie and Shahraki \(2023\)](#) find from threshold vector autoregressions (TVARs) that output, monetary, and credit shocks exhibit demonstrable effects on house prices when modelled as regime-dependent, with almost all shocks exhibiting larger and more persistent impacts when the housing market is in distress.

Turning to multi-national evidence, [André et al. \(2022\)](#) estimate a Bayesian MS-VAR for the US, UK, and Canada and find that house price levels respond negatively to monetary tightening. Yet the sensitivity is not uniform across sentiment states, motivating a search for regime variation in demand and supply conditions rather than the price variable itself. This pattern repeats even from a macroprudential perspective: [Alessandri and Mumtaz \(2019\)](#) show from a Threshold Vector Autoregression (TVAR) that output impacts are approximately five to six times larger during financial distress than in expansionary periods, identifying regimes based on credit conditions.

The closest antecedent in the Irish housing economic literature is work from [Egan and McQuinn \(2023\)](#), who estimate a Markov-switching VAR by maximum likelihood for the period 1981 to 2019. They document a complex time-varying relationship between housing supply and house price levels, demonstrating that the regime structure reflects substantive changes in market dynamics rather than a statistical artefact.

From a credit perspective, [Egan and McQuinn \(2024a\)](#) find that credit conditions have acted exogenously as a major driver behind the dynamics of post-2012 GFC Irish house prices. This credit and housing relationship is further supported by [Kelly et al. \(2018\)](#) who estimate from loan-level data that a ten per cent increase in credit availability raises property prices by approximately 1.5 per cent. Notably, [Kelly et al. \(2018\)](#) conclude that Ireland's Loan-to-Value (LTV) and Loan-to-Income (LTI) limits materially affected price dynamics and that the Loan-to-Income (LTI), Loan-to-Value (LTV) and Debt Service Ratio acted as guardrails against forward guidance and macroprudential policy, which

is consistent with a regime-like transmission as credit conditions were loosened during the boom years (2003–2007) and tightened sharply during the crisis, signalling to lenders and borrowers that the business cycle had shifted. The 2015–16 macroprudential reforms further altered this relationship as documented by Egan and McQuinn (2024a), which is theoretically justified as Gatt (2023) find that a rule tightening more aggressively during credit booms dominates symmetric alternatives in reducing tail risks to output and inflation, a result relevant to the asymmetric impulse responses in Section 3.

Supply in the Irish housing market has faced a sluggish response resulting in a structural ‘disequilibrium’ as documented by Egan and McQuinn (2024a). This point has also been furthered by Günnewig-Mönert and Lyons (2024) who find that the responsiveness of new housing supply to prices fell sharply during the 2000s and has only *partially recovered* post-GFC 2012.

1.2.2 Bayesian VAR Prior Specification

Regularisation of VAR estimation in small-to-medium samples has long relied on the Minnesota prior of Litterman (1986), which shrinks parameters to reduce cross-variable variance, along with the assumption that the first lag is approximately a random walk and earlier lags are not as informative as recent ones, becoming shrunken toward zero.

A key motivation for this paper’s approach builds on the earlier work of Giannone et al. (2015), who demonstrated in a BVAR context that treating shrinkage hyperparameters as unknown quantities can substantially reduce prior subjectivity and improve out-of-sample forecasting accuracy.

Three broad families of alternative priors have since emerged. The spike-and-slab prior of George and McCulloch (1993), formalised as Stochastic Search Variable Selection (SSVS), assigns each coefficient to a two-component mixture: a tight spike concentrated near zero and a diffuse slab, with a latent binary indicator governing inclusion.

This delivers exact sparse posterior modes and coherent posterior inclusion probabilities, but the discrete indicator space induces a highly multimodal posterior that degrades MCMC mixing in high-dimensional settings, as documented by Malsiner-Walli and Wagner (2019). Additionally, by construction it does not exploit the temporal ordering that

structures VAR lag decay.

In comparison to the $\mathfrak{JA}$ prior, the spike-and-slab prior cannot achieve the same continuity, as the discrete spike-slab indicator must be conditioned on the latent regime state, further inflating the already-multimodal posterior in a switching context (Malsiner-Walli and Wagner, 2019).

The continuous global-local Horseshoe prior (Carvalho et al., 2010; Follett and Yu, 2019) avoids this discrete limitation and competes well with Minnesota when genuine sparsity is present (Cross et al., 2020), yet it shrinks uniformly across all coefficients and therefore cannot encode the lag-decay structure inherent in VAR dynamics, a limitation noted explicitly by Chan (2019).

Moreover, the $\mathfrak{JA}$ prior shares conceptual similarities with the Horseshoe prior of Carvalho et al. (2010) and with regularisation shrinkage techniques widely used within the global-local shrinkage literature, for instance sparse Bayesian regularisation employed by Polson and Scott (2010).

Both priors employ hierarchical shrinkage toward zero with data-adaptive intensity and allow occasional large deviations when supported by the likelihood. The $\mathfrak{JA}$ prior differs in four respects.

First, the Student- t marginal distribution on coefficients places greater probability mass in the tails than the Horseshoe, making it more tolerant of large coefficient values during crisis episodes or structural breaks. This heavy-tailed behaviour is governed by the degrees-of-freedom parameter ν_A , which is learned from the data rather than fixed ex ante.

Second, the lag-decay structure encoded in φ imposes temporal ordering on the shrinkage pattern, with more distant lags shrunk more aggressively than recent lags. This contrasts with the Horseshoe, which applies global-local shrinkage without an explicit lag-decay structure tied to VAR dynamics.

Third, as a direct consequence of the shrinkage structure, the Student- t prior suppresses spurious regimes: when the data provide little evidence for a genuinely distinct regime, posterior mass on regime-specific coefficient deviations is pulled toward zero without

imposing hard restrictions on the transition matrix (see Remark B.1).

The $\mathfrak{JA}$ prior relates to Chan (2019)’s adaptive hierarchical prior, which addresses shrinkage heterogeneity across equations while retaining Gaussian tails. Estimation under an uninformative $\Gamma(2, 0.1)$ prior still allows flexibility while accommodating financial and economic data presumed to be leptokurtic (Juárez and Steel, 2010).

Fourth, it imposes a Student- t distribution on coefficients and applies it within a hierarchical Markov-switching VAR. This enables tail thickness to be estimated jointly with the VAR coefficients via the NUTS sampler. The uninformative $\Gamma(2, 0.1)$ prior primarily serves to accommodate heavy-tailed behaviour during crisis periods without being strictly a pre-specification; rather, it constitutes a reasonable estimate of degrees of freedom for financial and economic data presumed to be leptokurtic (Juárez and Steel, 2010). This property is also relevant given that prior performance in macroeconomic forecasting varies substantially with the time period and variable type (Gruber and Kastner, 2023).

These features make the $\mathfrak{JA}$ prior attractive for regime-switching models, where the prior must balance regularisation in tranquil periods with flexibility in volatile episodes, while respecting the persistence structure of macroeconomic time series.

The $\mathfrak{JA}$ prior nests the Minnesota specification as a limiting case and admits the Tikhonov modal correspondence and Huber-type robustness properties, which are formalised in Propositions B.5–B.6.

1.2.3 Regime Uncertainty in Multi-Stage Estimation

There often exists a structural inferential problem in hierarchical regime-switching modelling frameworks: estimated regime probabilities are routinely passed as fixed inputs to later stages. These smoothed probabilities are estimated objects rather than observed data, so conditioning on them suppresses posterior uncertainty in all downstream parameters, inducing generated-regressor bias, a problem first described by Pagan (1984) and later by Oxley and McAleer (1993), who show that this generated-regressor bias is pervasive and materially large, even when first-stage estimation is consistent. Within the MS-VAR context, the problem is compounded because regime probabilities depend non-linearly on

all model parameters. Still, the standard sandwich variance correction of [White \(1980\)](#), [MacKinnon and White \(1985\)](#), and [Newey and West \(1987\)](#) accounts for linear first-stage estimation error but does not capture this nonlinearity, so the understatement of posterior variance within impulse responses cannot be corrected at the inference stage.

This paper builds on a growing body of literature which uses gradient-based samplers and techniques enabling their compatibility with discrete latent variables: [Correia et al. \(2020\)](#) develop exact marginalisation via sparse differentiable mappings for neural network training; further work from [Lai et al. \(2023\)](#) highlights the importance of developing efficient marginalisation techniques, finding that automatic marginalisation often improves chain mixing in a probabilistic programming framework. Additionally, from a computational perspective, [Margossian et al. \(2020\)](#) report substantial computational gains from marginalising latent Gaussian variables under Hamiltonian Monte Carlo. This paper applies the same principle to Markov-switching econometrics, using a reparameterisation technique that analytically marginalises the discrete regime sequence at every leapfrog step, so that the demand block’s full posterior uncertainty enters each downstream block’s MCMC draws rather than being collapsed to a point estimate. This also removes the need to compute a full joint mixture model; the process is formally described in [Section 2.7](#).

2. Data and Methodology

2.1 Data

Table 1: Variable Definitions, Transformations, and Block Assignments.

Symbol	Description	Trans.	Source
<i>Demand block (D)</i>			
Y_t	Modified domestic demand	$\Delta \log$	CSO
ρ_t	Rent-to-income ratio	$\Delta \log$	ECB/OECD
u_t	Unemployment rate	Δ	FRED/CSO
π_t	CPI inflation	$\Delta \log$	CSO
<i>Supply block (S)</i>			
V_t	Construction volume index	$\Delta \log$	CSO
N_t	Planning permissions (dwellings)	Δ	CSO
w_t	Unit labour cost index	$\Delta \log$	ECB
$p_t^\dagger$	Construction cost index [†]	Δ	CSO
<i>Credit block (C)</i>			
r_t^m	New mortgage interest rate (% p.a.)	L	ECB
λ_t	Loan-to-deposit ratio	$\Delta \log$	CBI
f_t	Household net lending	Δ	CSO
$r_t^\ddagger$	Wu–Xia shadow rate ^b	Δ	Wu–Xia
<i>Asset block (A)</i>			
sp_t	Sovereign spread (IE 10Y – US 10Y)	Δ	OECD/FRED
p_t^e	Share prices (Equity) (inverted) ^c	$-\Delta$	OECD/FRED
C_t	SovCISS Sovereign composite indicator of systemic stress	L	ECB
B_t	Central Bank of Ireland total assets	$\Delta \log$	CSO/CBI
<i>Outcome block (R)</i>			
p_t^h	Nominal residential property price index	$\Delta \log$	CSO

Notes. Sample period 2000 Q2–2024 Q4 ($T = 98$). $\tilde{x}_{j,t} = \frac{x_{j,t} - \bar{x}_j}{s_j}$, where $\bar{x}_j$ and s_j denote the sample mean and standard deviation. Transformation codes: $\Delta \log$ = log first-difference; Δ = first difference; L = level. Source abbreviations: CSO = Central Statistics Office (Ireland); ECB = European Central Bank; OECD = Organisation for Economic Co-operation and Development; CBI = Central Bank of Ireland; FRED = Federal Reserve Bank of St. Louis Economic Data; Eurostat = Statistical Office of the European Union. [†] Own construction via chain-linking of monthly CSO Building and Construction Wholesale Price Index releases, rebased to 2021. [‡] Wu–Xia shadow rate used to capture the stance of monetary policy at the effective lower bound.

From a data perspective, five series were constructed prior to inference.

First, a quarterly construction cost index¹, built from monthly CSO Building and Construction Wholesale Price Index releases, aggregated to quarterly frequency, chain-linked across four base-period breaks spanning 1998–2021, and rebased to 2021 = 100, capturing the materials and output costs associated with residential construction.

Second, a rent-to-nominal-income ratio is constructed as an indicator of the relative cost of renting versus owning. This variable captures the financial incentives for households to enter the rental market and serves as a proxy for demand-side pressures in the housing market. Note, however, that this rental series was converted to an index, with rent costs sourced from the OECD while nominal wages were sourced from the ECB’s database series, both rebased to 2021 = 100 for comparability.

Let R_t denote the OECD rent price index and W_t the rebased nominal wage and salary index. The resulting ratio is dimensionless, captures changes in rental costs relative to nominal income, and enters the model as $\Delta \log RTI_t$, approximating the quarterly growth between rents and income

$$RTI_t = \frac{R_t}{W_t} \times 100 \quad (1)$$

Third, to capture the effective monetary policy stance during the Zero Lower Bound (ZLB) period the main refinancing operations (MRO) rate is combined with the Wu–Xia shadow rate (Wu and Xia, 2017), where the effective policy rate r_t^{eff} is set equal to the minimum of the two, ensuring the model reflects both conventional and unconventional ECB monetary policies over the sample period including unconventional measures such as quantitative easing (QE) and forward guidance.

$$r_t^{\text{eff}} = \min(r_t^{\text{MRO}}, r_t^{\text{shadow}}). \quad (2)$$

Fourth, a sovereign 10-year bond spread Δsp_t is constructed as the first difference of the Irish–US 10-year government bond yield spread to capture changes in risk premia and

¹Note, this index is distinct from the unit labour cost series variable which is separately included in the supply block.

serve as a general gauge of market stress:

$$\Delta \text{sp}_t^\dagger = \Delta \left(y_t^{(10, \text{IE})} - y_t^{(10, \text{US})} \right). \quad (3)$$

Fifth, a share price index from the FRED database is constructed as a proxy for asset market conditions, capturing changes in equity valuations that may influence household wealth and borrowing capacity. For comparability the series is sign-adjusted, with the resulting variable entering the model as $-\Delta p_t^e$.

$$-\Delta p_t^e \quad (4)$$

[†]The upright multi-letter notation sp_t is distinct from the latent regime indicator $s_t \in \{0, 1\}$.

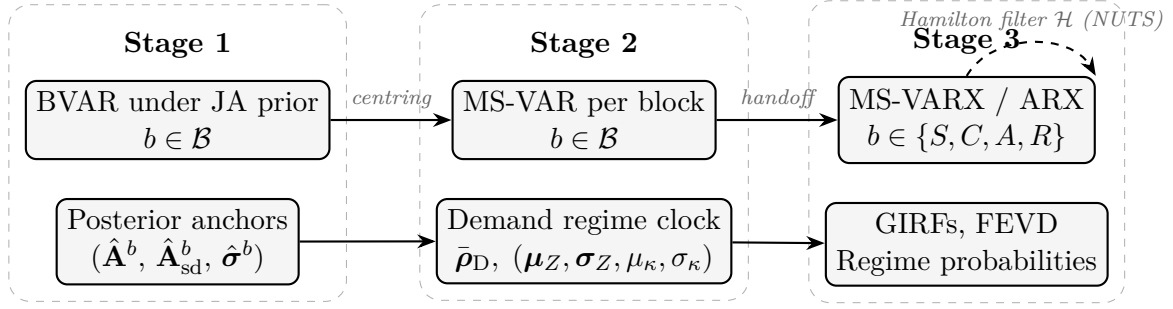


Figure 1: Three-stage hierarchical estimation pipeline. Stage 1 fits block-specific BVARs under the JA prior to obtain posterior anchors. Stage 2 estimates block-specific MS-VARs centred on Stage 1 objects, with the demand block providing the regime clock via a moment-matched handoff. Stage 3 estimates MS-VARX/ARX models for downstream blocks, integrating demand regime uncertainty at each NUTS leapfrog step via the differentiable Hamilton filter.

Notation. Throughout this paper, scalars are set in italic lowercase ($x, \sigma_{r,j}, \nu_A, x_A, x_B$) vectors are denoted in bold lowercase ($\mathbf{y}_t, \mathbf{a}_0^b, \boldsymbol{\varepsilon}_t^b, \boldsymbol{\theta}^b$) while matrices in bold uppercase ($\mathbf{A}^b, \mathbf{R}^b, \mathbf{P}, \boldsymbol{\Omega}^b$). Greek letters follow the same rule, hence, bold lowercase vectors ($\boldsymbol{\pi}, \boldsymbol{\rho}, \boldsymbol{\theta}$), bold uppercase for matrices ($\boldsymbol{\Omega}, \boldsymbol{\Sigma}$).

Cardinalities are labelled $n_{\cdot}$: n_b for block dimension, $n_q = 1 + n_b p$ for regressors per equation, with p retained for lag order in deference to the VAR-literature convention $\text{BVAR}(p)$. Calligraphic letters denote operators and distributions ($\mathcal{H}$ Hamilton filter, $\mathcal{L}$ log-likelihood, $\mathcal{N}$ Normal, $\mathcal{T}$ Student- t); typeface distinguishes them from italic scalars with the same letter, so that $\mathcal{H}$ (Hamilton filter operator), $\mathcal{H}_{\text{HMC}}$ (HMC Hamiltonian, Appendix C.1), H (Hurst exponent), and $\bar{h}$ (regime hardness) occupy disjoint symbolic roles.

Ket notation $|\cdot\rangle$ is reserved for vectors on the regime-probability simplex Δ^{K-1} , with the computational basis $\{|k\rangle\}_{k=0}^{K-1}$ indexed by regime state; the filter state $|\pi_t\rangle$, ergodic distribution $\pi(i)$, and smoothed regime probabilities $\rho_{t|T'}^k$ are distinguished by symbol in order to prevent notation confusion, where $k \in \{0, 1\}$ indexes the regime state, $b \in \mathcal{B}$ indexes the block, and $T' = T - p$ denotes the number of usable observations after lag removal; the subscript $t|T'$ denotes smoothing conditional on the full sample. This notation is adopted from Dirac's formalism and is used here as it represents states independently of any particular basis and renders simplex operations including filtering updates, marginalisation, and inner products over regime states more compact and less ambiguous than standard column-vector notation, for instance, its use in differentiable

probabilistic programming contexts is documented by Wang and Scott (2024). Concretely, for a vector $\mathbf{v} = (v_0, \dots, v_{K-1})'$ on the simplex, the ket $|v\rangle$ collects its components, and the inner product $\langle k|v\rangle = v_k$ extracts the k -th element corresponding to regime state k .

2.2 Conceptual Framework: Regimes as Deviations from a Latent Baseline

A notable aspect of the regimes identified by this model is that they are not predetermined economic categories — an idiosyncrasy perhaps, yet grounded in theoretical understanding. The initial BVARs estimated for every block during Section 2.3 serve two roles: they provide the regularisation hyperparameters, and, crucially, they establish a posterior baseline for dynamics.

This Stage 1 posterior serves as a centring object for the regime-dependent models during Stages 2 and 3. Regimes are therefore defined as deviations from this regularised baseline state, which is a priori unknown to be tranquil, crisis, or mixed. The three-stage framework identifies regimes as periods where dynamics shift consistently toward higher or lower volatilities and altered transmission mechanisms relative to the Stage 1 baseline.

At each date t , the system occupies a latent state whose label (low- or high-volatility) emerges from the posterior rather than being imposed ex ante. The Hamilton filter is initialised at the ergodic distribution $\boldsymbol{\pi}^*$ implied by the estimated transition matrix — a standard MCMC initialisation. The baseline is estimated as the Stage 1 unconditional BVAR posterior, shaped jointly by the observed data and the learnt shrinkage hyperparameters (τ_A, φ, ν_A) of the $\mathfrak{JA}$ prior.

The labels *tranquil* and *crisis* describe the direction of deviation from this baseline, not exogenously defined boom-or-bust states; the regime hardness ($\bar{h} = T'^{-1} \sum_t \max_k \rho_t^{k,b}|_{T'}$) measures how sharply the system separates into distinct states.

For $K = 2$ regimes, mean regime hardness is defined as

$$\bar{h} = \frac{2}{T} \sum_{t=1}^T |P(S_t = 1 \mid \mathbf{Y}) - 0.5| \quad (5)$$

High mean regime hardness implies that structural breaks are sharp rather than gradual. For instance, the assets block ($\bar{h} = 0.923$) implies the transition is effectively

complete before it is observable, which is consequential for the timing of macroprudential intervention. Under this maintained identifying assumption, demand dynamics adjust before supply, credit, asset, and property price responses.

The demand block acts as a common regime clock as $\langle 1|\pi_t \rangle^D$ rises, downstream blocks respond conditional on that shift via the combined emission density in (32). Conversely, the recursive ordering reflects the sequence in which information propagates through the housing system, **not** an assumption of **strict exogeneity**.

2.3 Stage 1: Block-Wise (BVAR) Prior Estimation with the $\mathfrak{J}\mathfrak{A}$ Prior

Prior to estimating the Markov-switching system, a Bayesian VAR (BVAR) is fitted separately to each of the five macroeconomic blocks — demand, supply, credit, assets, and RPPI — under the $\mathfrak{J}\mathfrak{A}$ prior. The block-recursive approach follows the framework for inference in multiple-equation Markov-switching models developed by Sims et al. (2008), with the Bayesian VAR methodology surveyed in Del Negro and Schorfheide (2009). Additionally, this stage carries shrinkage uncertainty forward through the jointly sampled hyperparameters (τ_A, φ, ν_A) .

2.4 Block BVAR Likelihood

Let $\mathbf{y}_t^b$ denote the n_b -dimensional observation vector for block $b \in \mathcal{B} = \{D, S, C, A, R\}$ at time t ,

where $n_b = 4$ for $b \in \{D, S, C, A\}$ and $n_R = 1$. For the four multivariate blocks, the block-specific BVAR(p) is given by

$$\mathbf{y}_t^b = \mathbf{a}_0^b + \sum_{\ell=1}^p \mathbf{A}_\ell^b \mathbf{x}_{t-\ell}^b + \boldsymbol{\varepsilon}_t^b, \quad \boldsymbol{\varepsilon}_t^b \stackrel{\text{iid}}{\sim} \mathcal{T}_{n_b}(\nu^b, \mathbf{0}, \boldsymbol{\Omega}^b), \quad (6)$$

where $\nu^b > 2$ degrees of freedom are imposed. For the RPPI block ($b = R, n_R = 1$), $\boldsymbol{\Omega}^b$ reduces to the scalar variance $(\sigma^R)^2$.

Collecting the regressors into $\mathbf{x}_t^{b'} = (\mathbf{1}, \mathbf{y}_{t-1}^{b'}, \dots, \mathbf{y}_{t-p}^{b'}) \in \mathbb{R}^{1+n_b p}$ and stacking over $T' = T - p$ usable observations gives the compact matrix form

$$\mathbf{Y}^b = \mathbf{X}^b \mathbf{A}^b + \mathbf{E}^b, \quad (7)$$

where $\mathbf{A}^b \in \mathbb{R}^{(1+n_b p) \times n_b}$ collects the intercept and lag coefficients column-wise.

Each block is estimated independently to isolate cross-block transmission and latent regime dependence, which enter only in Stages 2 and 3 through the conditioning structure of the block-recursive framework.

2.5 The $\mathfrak{J}\mathfrak{A}$ Prior

Standard Bayesian VAR applications in macroeconomics often rely on the normal-inverse-Wishart conjugate family and the Minnesota prior of [Litterman \(1986\)](#). Accordingly, this encodes the belief that most variables follow a “*near-random-walk*” with respect to the first lag while subsequent lags become shrunken towards zero.

The variance of off-diagonal elements is shrunk more aggressively than the variance of own lags, improving forecasting performance in small samples and imposing a parsimonious structure on the VAR coefficients ([Litterman, 1986](#); [Kadiyala and Karlsson, 1997](#)).

The challenge of prior specification is more acute within regime-switching models. Excessive shrinkage risks suppressing genuine differences across states, collapsing the posterior toward a single-regime solution. Insufficient shrinkage, conversely, produces unstable inference characterised by divergent transitions and poor chain mixing, a problem compounded by the anisotropic funnel geometry that arises in hierarchical posteriors when hyperparameters govern the scale of lower-level coefficients, as noted by [Neal \(2003\)](#).

This geometry is particularly severe under tightly-centred Gaussian priors in small samples, where posterior correlations between hyperparameters and lower-level parameters intensify as the data become sparse, sharply narrowing the funnel throat and degrading sampler efficiency ([Betancourt and Girolami, 2013](#)).

The $\mathfrak{J}\mathfrak{A}$ prior addresses both of these problems by treating shrinkage intensity, lag decay, and tail thickness as unknown quantities learned jointly with the VAR coefficients, building on the earlier approach of [Giannone et al. \(2015\)](#) for BVARs. The $\mathfrak{J}\mathfrak{A}$ prior’s Student- t structure preserves the regularising role of Minnesota-style shrinkage near zero while permitting occasional large coefficient deviations when supported by the data.

Crisis episodes — abrupt demand shocks, credit tightening, and bubble dynamics — generate shocks poorly captured by tightly centred Gaussian priors; the heavier tails of

the Student- t distribution accommodate these without sacrificing the regularisation that prevents overfitting in the small-to-moderate sample sizes available for the Irish housing market.

Definition 2.1 ($\mathfrak{JA}$ Prior). Let $n_q = 1 + n_b p$ denote the number of regressors per equation. The $\mathfrak{JA}$ prior on the coefficient matrix $\mathbf{A}^b \in \mathbb{R}^{n_q \times n_b}$ is the hierarchical Student- t specification:

$$\tau_A \sim \text{HN}(1), \quad (8a)$$

$$\varphi \sim \text{Exp}(1), \quad (8b)$$

$$\nu_A \sim \Gamma(2, 0.1), \quad (8c)$$

$$\sigma_{r,j} = \begin{cases} 10, & r = 0 \text{ (intercept)}, \\ \tau_A \ell(r)^{-\varphi}, & r \geq 1, \end{cases} \quad (8d)$$

$$A_{r,j}^b \stackrel{\text{iid}}{\sim} \mathcal{T}_1(\nu_A, 0, \sigma_{r,j}), \quad r = 0, \dots, n_q - 1, \quad j = 1, \dots, n_b, \quad (8e)$$

Here, $\ell(r) = \lceil r/n_b \rceil$ maps regressor index r to the corresponding lag number, τ_A is the global shrinkage scale, φ governs lag-order decay, and ν_A controls the tail thickness of the marginal coefficient distribution.

The innovation covariance is parameterised as

$$\mathbf{\Omega}^b = \underbrace{\mathbf{D}^b}_{\text{scale}} \mathbf{R}^b \underbrace{\mathbf{D}^b}_{\text{scale}}, \quad \mathbf{D}^b = \text{diag}(\sigma_1^b, \dots, \sigma_{n_b}^b) \in \mathbb{R}^{n_b \times n_b}, \quad (9)$$

where $\sigma_j^b \sim \text{HN}(1)$, and the Cholesky factor of the correlation matrix $\mathbf{R}^b$ is assigned an LKJCholesky($n_b, \eta = 1$) prior (Lewandowski et al., 2009), where $\eta = 1$ implies a uniform prior over the space of correlation matrices.

Prior convention. Throughout Stages 1–3, positive scale parameters follow $\text{HN}(1)$, degrees-of-freedom parameters follow $\Gamma(2, 0.1)$, and prior or regime-specific correlation Cholesky factors follow LKJCholesky($n_b, \eta = 1$) unless stated otherwise. ***These conventions are***

not restated in each equation.

Hyperparameter Interpretation.

- τ_A determines the overall shrinkage intensity applied to the lag coefficients. Smaller values imply tighter shrinkage and favour more parsimonious dynamics.
- φ governs lag decay, with larger values shrinking more distant lags more aggressively than recent lags.
- ν_A controls the tail thickness of the coefficient prior. As $\nu_A \rightarrow \infty$, the prior approaches a Gaussian form, whereas smaller values permit heavier tails and therefore greater tolerance for large coefficient deviations.

Formally, as $\nu_A \rightarrow \infty$ and $\varphi \rightarrow 1$, the $\mathfrak{JA}$ prior converges approximately to the Minnesota prior of [Litterman \(1986\)](#), and its posterior mode admits a generalised Tikhonov regularisation interpretation. Both results are shown in [Appendix B.4](#). One advantage of this heavy-tailed structure is that the prior also suppresses spurious regimes; see [Remark B.1](#) in [Appendix B](#).

2.6 Stage 2: Block-Specific MS-VAR Estimation and Demand Regime Identification

Stage 2 introduces a two-regime MS-VAR estimated separately for each block $b \in \mathcal{B} = \{D, S, C, A, R\}$.

Definition 2.2 (Markov Chain). A sequence of random variables $\{s_t : t \in \mathbb{N}_0\}$ taking values in a countable state space $\mathcal{S}$ is a *first-order Markov chain* if

$$\Pr(s_t = j \mid s_{t-1} = i, s_{t-2}, \dots) = \Pr(s_t = j \mid s_{t-1} = i) =: p_{ij}, \quad \forall i, j \in \mathcal{S}.$$

The chain is *time-homogeneous* if p_{ij} does not depend on t , and is fully characterised by the transition matrix $\mathbf{P} = [p_{ij}]_{i,j \in \mathcal{S}}$ satisfying $p_{ij} \geq 0$ and $\sum_{j \in \mathcal{S}} p_{ij} = 1$ for all i .

The latent regime process $\{s_t\}$ is modelled as a time-homogeneous first-order Markov chain on $\mathcal{S} = \{0, 1\}$ in the sense of [Definition 2.2](#).

Let the regime-specific parameterisation be centred on the corresponding Stage 1 posterior means and standard deviations $(\hat{\mathbf{A}}^b, \hat{\mathbf{A}}_{\text{sd}}^b, \hat{\boldsymbol{\sigma}}^b)$,

where $\hat{\mathbf{A}}^b$ is the posterior mean coefficient matrix, $\hat{\mathbf{A}}_{\text{sd}}^b$ its elementwise posterior standard deviation, and $\hat{\boldsymbol{\sigma}}^b$ the posterior mean residual standard deviation vector.

Remark 2.3 (Demand Regime Probabilities as the Master Regime Clock). The demand block plays a distinctive role among the five blocks as during Section 2.7 it is treated as the master regime process, such that during Stage 3, the posterior regime log-probabilities serve as the common regime clock linking demand to supply, credit, assets, and nominal residential property prices.

The posterior-mean smoothed regime probabilities from the demand block are therefore given by

$$\bar{\boldsymbol{\rho}}_{\text{D}} = \{\rho_{t|T'}^{(\text{D},k)}\}_{t=1}^{T'},$$

where $\bar{\boldsymbol{\rho}}_{\text{D}}$ collects the full sequence of smoothed regime probabilities across all T' time periods and both regime states $k \in \{0, 1\}$, and does not contain the VAR coefficients or covariance parameters of the demand block, which are handled separately in the centring structure of Stage 2. Moment-matched transition objects and log-probability emissions are computed in parallel and passed to Stage 3.

2.6.1 Regime Process

Let $s_t \in \mathcal{S} = \{0, 1\}$ denote the latent regime, with $K = 2$. Regime $k = 0$ is the low-volatility state and $k = 1$ the high-volatility state.

The transition matrix $\mathbf{P} = [p_{ij}]_{i,j \in \mathcal{S}}$ collects these probabilities, with $\sum_{j=0}^1 p_{ij} = 1$ for each i .

The filtered regime-probability vector is then collected into a ket:

$$|\pi_t\rangle = \begin{pmatrix} \pi_{t,0} \\ \pi_{t,1} \end{pmatrix} \in \Delta^1, \quad \pi_{t,k} := \langle k | \pi_t \rangle, \quad (10)$$

with $\pi_{t,0} + \pi_{t,1} = 1$ and $\pi_{t,k} \geq 0$.

The proportionality update is $|\pi_t\rangle \propto \mathbf{Z}_t^b \mathbf{P}' |\pi_{t-1}\rangle$, where $\mathbf{Z} \in \mathbb{R}^{K \times K}$ is a matrix of

latent transition parameters defined formally in Remark 2.5, normalised so that $|\pi_t\rangle \in \Delta^1$ at every t .

Conditional on regime state $|k\rangle \in \mathcal{S}$, the observation equation is given by

$$\mathbf{y}_t^b = \mathbf{A}_k^{b'} \mathbf{x}_t^b + \boldsymbol{\varepsilon}_{t,k}^b, \quad \boldsymbol{\varepsilon}_{t,k}^b \stackrel{\text{iid}}{\sim} \mathcal{T}_{n_b}(\nu^b, \mathbf{0}, \boldsymbol{\Omega}_k^b), \quad (11)$$

where $\mathbf{x}_t^{b'} = (\mathbf{1}, \mathbf{y}_{t-1}^{b'}, \dots, \mathbf{y}_{t-p}^{b'}) \in \mathbb{R}^{1+n_b p}$ stacks the intercept and own-block lags, and $\mathbf{A}_k^b \in \mathbb{R}^{(1+n_b p) \times n_b}$ collects the regime-specific intercept and lag coefficients.

Since each block is estimated independently (a multivariate MS-VAR for blocks $b \in \{\text{D}, \text{S}, \text{C}, \text{A}\}$ and a univariate MS-AR for RPPI), the regime-conditional mean and log-likelihood are defined as

$$\boldsymbol{\mu}_{t,k}^b = \mathbf{A}_k^{b'} \mathbf{x}_t^b, \quad (12)$$

$$e_{t,k}^b = \log f_{\mathcal{T}_{n_b}}(\mathbf{y}_t^b \mid \nu^b, \boldsymbol{\mu}_{t,k}^b, \boldsymbol{\Omega}_k^b). \quad (13)$$

Here $e_{t,k}^b : \mathcal{Y} \times \mathcal{S} \rightarrow \mathbb{R}$ denotes the regime-conditional log-emission probability, that is, the log-density of $\mathbf{y}_t^b$ evaluated under regime k , summarising how likely the current observation is given that the system occupies state k .

These contributions enter the Hamilton filter $\mathcal{H}$, defined as the nonlinear operator

$$\mathcal{H}(\mathbf{P}, |\mathbf{e}\rangle) |\pi\rangle := \frac{\underbrace{\text{diag}(\exp \mathbf{e}) \mathbf{P}' |\pi\rangle}_{\text{emission-reweighted prediction}}}{\underbrace{\langle \mathbf{1} | \text{diag}(\exp \mathbf{e}) \mathbf{P}' |\pi\rangle}_{\text{normalisation}}}, \quad \exp \mathbf{e} := (\exp e_0, \dots, \exp e_{K-1})', \quad (14)$$

Stacking (11) over $T' = T - p$ observations yields the regime-conditional matrix form:

$$\mathbf{Y}^b = \mathbf{X}^b \mathbf{A}_k^b + \mathbf{E}_k^b. \quad (15)$$

The mapping $\mathcal{H} : \Delta^1 \times \mathbb{R}^K \times \mathbb{R}^{K \times K} \rightarrow \Delta^1$ accumulates log emission-reweighted probabilities and normalises to the simplex; filtered probabilities are recovered from the

forward recursion and smoothed probabilities via the backward smoother.

The expected regime duration follows directly from the smoothed probability:

$$\bar{d}_0 = \frac{1}{p_{01}}, \quad \bar{d}_1 = \frac{1}{p_{10}}. \quad (16)$$

where asymmetry arises if $p_{01} \neq p_{10}$, implying different expected regime durations across states.

2.6.2 Coefficient Prior

The regime-specific coefficient matrices are centred on the Stage 1 posterior mean via a scaled perturbation:

$$[\Delta_k^b]_{ij} \stackrel{\text{iid}}{\sim} \mathcal{T}_1(\nu_\delta, 0, 1), \quad i = 1, \dots, n_q, \quad j = 1, \dots, n_b, \quad k \in \{0, 1\}, \quad (17)$$

$$\mathbf{A}_k^b = \underbrace{\hat{\mathbf{A}}^b}_{\text{Stage 1 baseline}} + \underbrace{x_A \hat{\mathbf{A}}_{\text{sd}}^b \odot \Delta_k^b}_{\text{regime deviation}}. \quad (18)$$

where $\odot$ denotes element-wise multiplication and the parameter x_A dictates the magnitude of the deviation, while $\hat{\mathbf{A}}_{\text{sd}}^b$ acts as an elementwise scalar factor for its posterior uncertainty.

The tail thickness in Δ_k^b is governed by ν_δ under the $\mathfrak{JA}$ prior's Student- t distribution, allowing the data to support sharper cross-regime differences when large deviations warrant them, while still regularising toward the Stage 1 baseline when the data do not strongly support distinct regime separation.

To address label-switching, a hard volatility ordering is imposed on the regime-specific log-volatilities. For each variable j in block b , latent standard normal draws are generated for each regime:

$$u_j^{(k)} \sim \mathcal{N}(0, 1), \quad k \in \{0, 1\}, \quad j = 1, \dots, n_b. \quad (19)$$

The latent draws are sorted across regimes for each variable separately, imposing the ordering:

$$u_{j,(1)} := \min(u_j^{(0)}, u_j^{(1)}), \quad u_{j,(2)} := \max(u_j^{(0)}, u_j^{(1)}), \quad j = 1, \dots, n_b, \quad (20)$$

which ensures $u_{j,(1)} \leq u_{j,(2)}$ for all j . The regime-specific log-volatilities are then constructed as

$$\log \sigma_{j,0}^b = \log \hat{\sigma}_j^b + \tau_\sigma u_{j,(1)}, \quad \log \sigma_{j,1}^b = \log \hat{\sigma}_j^b + \tau_\sigma u_{j,(2)}, \quad (21)$$

where $\hat{\sigma}_j^b$ denotes the Stage 1 posterior mean residual standard deviation for variable j in block b , ensuring that regime $k = 0$ is identified as the low-volatility state and regime $k = 1$ as the high-volatility state, without any ex-post relabelling.

The regime-conditional covariance matrix $\mathbf{\Omega}_k^b$ is parameterised as

$$\mathbf{\Omega}_k^b = \mathbf{D}_k^b \mathbf{R}_k^b \mathbf{D}_k^b, \quad \mathbf{D}_k^b = \text{diag}(\sigma_{1,k}^b, \dots, \sigma_{n_b,k}^b), \quad (22)$$

where $\sigma_{j,k}^b$ denotes the regime-specific standard deviation for variable j in block b , and $\mathbf{R}_k^b$ is the corresponding correlation matrix. The diagonal elements are assigned the hard-ordered log-normal prior described in Section 2.6.2, ensuring that $\sigma_{j,0}^b \leq \sigma_{j,1}^b$ for all $j = 1, \dots, n_b$.

Following the prior conventions in Section 2.5, the Cholesky factor $\mathbf{L}_{\mathbf{R},k}^b$ of $\mathbf{R}_k^b$ is assigned an LKJCholesky($n_b, \eta = 1$) prior for every regime k and block b , ensuring positive-definiteness and a uniform prior over the space of correlation matrices. The full Cholesky factor of $\mathbf{\Omega}_k^b$ is then

$$\mathbf{L}_{\mathbf{\Omega},k}^b = \text{diag}(\sigma_k^b) \mathbf{L}_{\mathbf{R},k}^b. \quad (23)$$

For the univariate RPPI block ($b = \text{R}, n_{\text{R}} = 1$), the covariance $\mathbf{\Omega}_k^{\text{R}}$ reduces to the scalar variance $(\sigma_k^{\text{R}})^2$.

Remark 2.4 (Block-Diagonal Joint Covariance). Since blocks are estimated independently, the joint covariance across all blocks in $\mathcal{B}$ is block-diagonal. Denoting the direct sum over blocks as

$$\mathbf{\Omega}^{\mathcal{B}} = \bigoplus_{b \in \mathcal{B}} \mathbf{\Omega}^b = \text{diag}(\mathbf{\Omega}^{\text{D}}, \mathbf{\Omega}^{\text{S}}, \mathbf{\Omega}^{\text{C}}, \mathbf{\Omega}^{\text{A}}, \mathbf{\Omega}^{\text{R}}), \quad (24)$$

the off-diagonal blocks are zero by construction, reflecting the identifying assumption that cross-block residual dependence enters only through the shared regime state s_t rather than through contemporaneous error correlation.

Since blocks differ in dimension (n_b varies across $b \in \mathcal{B}$), the individual block covariance is recovered via the selection matrix $\mathbf{S}_b \in \{0, 1\}^{n_b \times \sum_{b'} n_{b'}}$ that selects the rows and columns corresponding to block b :

$$\boldsymbol{\Omega}^b = \mathbf{S}_b \boldsymbol{\Omega}^{\mathcal{B}} \mathbf{S}_b^\top. \quad (25)$$

The posterior-mean correlation matrices $\bar{\mathbf{R}}_0^b$ and $\bar{\mathbf{R}}_1^b$ yield the cross-regime difference

$$\Delta \mathbf{R}^b = \bar{\mathbf{R}}_1^b - \bar{\mathbf{R}}_0^b. \quad (26)$$

2.6.3 Transition Matrix

Housing-market regimes are expected to be persistent rather than transitory episodes. Accordingly, transition probabilities are parameterised using a κ -inflated softmax, where larger values of κ place greater prior mass on diagonal elements while leaving the realised degree of persistence to be data-driven through the posterior.

†

The transition probabilities are given by

$$\kappa \sim \Gamma(2, 0.1), \quad (27)$$

$$\zeta_{ij} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1), \quad i, j \in \{0, 1\}, \quad (28)$$

$$p_{ij} = \frac{\exp(\zeta_{ij} + \kappa \delta_{ij})}{\sum_{m=1}^K \exp(\zeta_{im} + \kappa \delta_{im})}, \quad i, j \in \{0, 1\}, \quad (29)$$

Remark 2.5. Here, $\mathbf{Z} \in \mathbb{R}^{K \times K}$ denotes a matrix of latent parameters estimated via NUTS, with every element ζ_{ij} assigned an independent normal prior.

†This reparameterisation avoids constrained sampling because the row-wise normalisation in (29) enforces row-stochasticity by construction. The latent transition parameters $(\mathbf{Z}, \kappa)$ are sampled jointly with the remaining model parameters under NUTS, and the implied transition matrix $\mathbf{P}$ is recovered deterministically from (29) at each iteration.

satisfying the stationary distribution $\pi(i)$, $\sum_{i \in \mathcal{S}} \pi(i) p_{ij} = \pi(j)$ and $\sum_i \pi(i) = 1$.

Remark 2.6 (Spectral Gap for $K = 2$). For a two-state transition matrix the second eigenvalue of $\mathbf{P}$ is $\lambda_2 = p_{00} + p_{11} - 1$, such that the spectral gap is

$$\text{gap}(\mathbf{P}) = 1 - |\lambda_2| = 1 - |p_{00} + p_{11} - 1|. \quad (30)$$

The spectral gap specifies the rate at which the chain forgets its initial state: a gap close to zero implies slow mixing and persistent regimes; a gap close to one implies rapid mixing and transient regimes. The values reported in Table 2 are posterior medians of $\text{gap}(\mathbf{P}^{(s)})$ computed from each MCMC draw $\mathbf{P}^{(s)}$; because $\text{gap}(\cdot)$ is nonlinear, these differ in general from the gap evaluated at the posterior-median transition probabilities. For the assets block, the reported posterior median gap is 0.195, the lowest across all blocks. The RPPI block reports a value of approximately 0.628, reflecting rapid regime turnover, consistent with the credible intervals for the diffuse transition matrix in Table 14.

2.6.4 Likelihood and Hamilton Filter

Remark 2.7 (Kim Smoother and NUTS Compatibility). The regime-conditional emission log-probability (see Appendix B.1 for the explicit form) is evaluated at each observation. The Hamilton (1989) forward filter then accumulates these terms in log-space to construct the full sample log-likelihood, initialised at the stationary distribution $\pi(i)$. Filtered probabilities are recovered from the forward recursion, while smoothed probabilities are obtained via the Kim–Hamilton backward pass. For the demand block, posterior-mean smoothed probabilities serve as the regime-dating object propagated into Stage 3. The smoothed probabilities reported in this paper are obtained via the Kim (1994) backwards smoother applied to the posterior-mean filtered probabilities from the Hamilton forward filter. This approach is valid within the NUTS framework employed here, but would be inappropriate within a Gibbs sampler. In Gibbs sampling, the latent states must be sampled conditional on all other states and parameters at each iteration, as detailed in Chib (1996).

The Kim smoother, by contrast, computes smoothed probabilities conditional only on

the data and parameters, treating each time period independently given those conditioning variables. This independence property is precisely what makes the Kim smoother suitable for post-processing the output of gradient-based samplers such as NUTS, where the latent regime path is integrated out analytically through the Hamilton filter rather than being sampled directly.

The NUTS algorithm samples from the marginal posterior over continuous parameters $(\mathbf{A}_k^b, \mathbf{\Omega}_k^b, \mathbf{P}, \kappa)$, with the discrete regime sequence already marginalised via the forward recursion. Smoothed probabilities are then recovered as a deterministic post-processing step, conditional on posterior-mean parameter estimates. This contrasts sharply with strict Gibbs-based approaches, where the regime sequence is treated as an explicit latent variable to be sampled at each iteration. For further discussion of smoothing versus sampling in Markov-switching models, see [Chib \(1996\)](#) and [Frühwirth-Schnatter \(2006\)](#) (Chapter 11).

Remark 2.8 (Demand Block as Master Regime Process). The demand block provides a moment-matched handoff for Stage 3 consisting of precomputed demand emission log-probabilities together with summary objects for the transition process, notably: $(\boldsymbol{\mu}_Z, \boldsymbol{\sigma}_Z, \mu_{\log \kappa}, \sigma_{\log \kappa})$.

These objects allow the uncertainty in the demand regime process to be reintroduced in Stage 3 without conditioning on a fixed regime path. Economically, this reflects the identifying assumption that housing demand conditions, such as income, employment, and financing costs, are determined prior to supply, credit, asset, and property-price adjustment. The framework, therefore, imposes a common demand-driven regime clock while still allowing each downstream block to retain its own regime-specific dynamics.

The moment-matched handoff objects and their derivation are given in [Appendix B.5](#).

Posterior Simulation. Inference proceeds via the No-U-Turn Sampler ([Hoffman and Gelman, 2014](#)) (NUTS), a gradient-based MCMC algorithm that adaptively sets leapfrog step sizes using Hamiltonian dynamics; see [Appendix C](#) for a self-contained description of the algorithm and its implementation details. The sampler targets the continuous parameter vector.

Specifically, it targets $\theta^b = \{\mathbf{A}^b, \mathbf{\Omega}^b, \nu^b, \tau_A, \varphi, \nu_A\}$, with the discrete regime sequence analytically marginalised at each leapfrog step via the Hamilton forward filter, such that no discrete sampling or Metropolis–Hastings step is required.

The sampler is implemented in NumPyro 0.20.0 (Phan et al., 2019) with JAX 0.9.0.1 (Bradbury et al., 2018) as the computational backend.

‡

2.7 Stage 3: Block-Specific MS-VARX Estimation with Demand Regime Conditioning

2.7.1 Joint Regime Identification

Rather than treating the demand-regime path from Stage 2 as fixed, Stage 3 allows demand’s uncertainty to enter each downstream block via the likelihood itself. Under the maintained assumption that demand and block residuals are conditionally independent given s_t , the joint emission density factorises as

$$p(\mathbf{y}_t^D, \mathbf{y}_t^b \mid s_t = k) = p(\mathbf{y}_t^D \mid s_t = k) p(\mathbf{y}_t^b \mid s_t = k), \quad (31)$$

The Hamilton filter then operates on the combined log-emission,

$$e_{t,k,b}^\circ = \underbrace{\bar{e}_{t,k}^D}_{\text{demand signal}} + \underbrace{\alpha_b \bar{e}_{t,k}^b}_{\text{block signal}}. \quad (32)$$

Equation (32) introduces a block-specific decoupling weight $\alpha_b \in (0, 1)$, which governs the contribution of each block’s log-emission probability within the Hamilton filter. This weight addresses a trade-off at the centre of multi-block regime identification: each downstream block (supply, credit, assets, and RPPI, with the demand block serving as the conditioning input rather than a downstream recipient) may carry its own signals that are either coupled to or decoupled from the demand regime classification. Setting $\alpha_b = 0$

‡Stage 1 uses 128 parallel chains across two CUDA devices (*NVIDIA RTX 5090* and *NVIDIA RTX 4090*). Stages 2 and 3 use 32 CPU chains on an *AMD Ryzen 7950X3D*, with 60,000 warmup and 60,000 sampling iterations per block, a target acceptance probability of 0.90, and dense mass-matrix adaptation. Convergence is assessed using the rank-normalised split- $\hat{R}$ of Vehtari et al. (2021) and the effective sample size (ESS); all reported results satisfy $\hat{R} < 1.01$ and $\text{ESS} > 400$ per parameter. For discrete latent variables subject to label-switching (e.g. π_t, π_{stat}), elevated $\hat{R}$ values reflect permutation invariance across chains rather than a failure of convergence; see Appendix D for full diagnostics.

reduces to pure demand-driven conditioning, in which regime classification is determined entirely by demand dynamics and downstream blocks respond passively.

Setting $\alpha_b = 1$ recovers full joint identification, in which both demand and block-specific likelihood contributions determine regime probabilities at each iteration. The posterior of α_b is therefore economically informative: a posterior mean near unity indicates that the block's own emission density is informative about the common regime state, while values near zero indicate that regime classification is primarily driven by demand, with limited independent information from the block's own dynamics.

The learnable weight lets the data determine how much each block should influence the shared regime process, balancing under-identification (true regime signals ignored) against over-identification (spurious block-specific variation mistaken for genuine regime shifts). The coupling weight is assigned a flat prior on $(0, 1)$ and reparameterised for unconstrained NUTS sampling:

$$\tilde{\alpha}_b \sim \mathcal{N}(0, 1), \quad \alpha_b = \frac{1}{1 + \exp(-\tilde{\alpha}_b)}, \quad (33)$$

where the mapping $\tilde{\alpha}_b \mapsto (1 + \exp(-\tilde{\alpha}_b))^{-1}$ is the standard logistic transform. In implementation, $\tilde{\alpha}_b$ is sampled on the real line and mapped into $(0, 1)$ via this transform, so the posterior over α_b is learned jointly with the remaining model parameters.

The regime-conditional demand and block-specific emission log-probabilities are

$$\bar{\boldsymbol{\mu}}_{t,k}^D = \mathbf{x}_t^D \bar{\mathbf{A}}_k^D, \quad \bar{\boldsymbol{\mu}}_{t,k}^b = \mathbf{x}_t^b \bar{\mathbf{A}}_k^b, \quad (34)$$

$$\bar{e}_{t,k}^D = \log f_{\mathcal{T}_{n_D}}(\mathbf{y}_t^D \mid \bar{\nu}^D, \bar{\boldsymbol{\mu}}_{t,k}^D, \bar{\mathbf{L}}_{\Omega,k}^D), \quad (35)$$

$$\bar{e}_{t,k}^b = \log f_{\mathcal{T}_{n_b}}(\mathbf{y}_t^b \mid \bar{\nu}^b, \bar{\boldsymbol{\mu}}_{t,k}^b, \bar{\mathbf{L}}_{\Omega,k}^b). \quad (36)$$

Both are evaluated at Stage 2 posterior-mean parameters and precomputed in the reparameterisation handoff. These objects summarise the likelihood of each observation under each regime in the demand block and the downstream block, respectively. However, to preserve uncertainty in the demand regime process, Stage 3 does not condition on fixed

smoothed probabilities from Stage 2.

Instead, the demand regime path is integrated over at each NUTS iteration via a differentiable auxiliary reparameterisation and Hamilton forward filter; full details are in Appendix B.5.

The resulting filtered probabilities $|\pi_t\rangle$ depend jointly on the reconstructed demand transition process and on the regime-specific likelihood contributions of both demand and the downstream block. The explicit closed-form recursion for $|\pi_t\rangle$ is given in Appendix B.6.

2.7.2 MS-VARX Specification

For multivariate blocks $b \in \{S, C, A\}$, Stage 3 marginalises over the demand regime state $|\pi_t\rangle$, yielding the observation equation as a filtered-probability-weighted mixture:

$$\begin{aligned} \mathbf{y}_t^b &= \sum_{k=0}^1 \pi_{t,k} [\mathbf{A}_k^b \mathbf{x}_t^b + \mathbf{B}_k^b \mathbf{x}_t^D] + \boldsymbol{\varepsilon}_t^b, & \pi_{t,k} &= \langle k | \pi_t \rangle, \\ \boldsymbol{\varepsilon}_t^b &\sim \mathcal{T}_{n_b} \left(\nu^b, \mathbf{0}, \sum_{k=0}^1 \pi_{t,k} \boldsymbol{\Omega}_k^b \right). \end{aligned} \quad (37)$$

where $\mathbf{x}_t^b \in \mathbb{R}^{1+n_b p}$ stacks the intercept and own-block lags, $\mathbf{x}_t^D \in \mathbb{R}^{n_D(p+1)}$ stacks contemporaneous and lagged demand variables, $\mathbf{A}_k^b \in \mathbb{R}^{n_b \times (1+n_b p)}$ contains regime-conditional own-lag coefficients, and $\mathbf{B}_k^b \in \mathbb{R}^{n_b \times n_D(p+1)}$ contains regime-conditional demand-transmission coefficients. The mixture weights $\pi_{t,k} = \langle k | \pi_t \rangle$ are the filtered demand-regime probabilities delivered by $\mathcal{H}$. The corresponding log-likelihood is formed as a finite mixture weighted by the filtered demand probabilities $|\pi_t\rangle$:

$$\mathcal{L}(\boldsymbol{\theta}^b) = \sum_{t=p+1}^T \log \left(\sum_{k \in \mathcal{S}} \langle k | \pi_t \rangle f_{\mathcal{T}_{n_b}}(\mathbf{y}_t^b \mid \nu^b, \mathbf{A}_k^b \mathbf{x}_t^b + \mathbf{B}_k^b \mathbf{x}_t^D, \mathbf{L}_{\Omega,k}^b) \right). \quad (38)$$

Economically, this allows each downstream block to respond to demand through both observed demand covariates and the latent demand regime.

Remark 2.9 (Uncertainty Propagation Without Double Counting). Two distinct channels link demand to the downstream blocks. The demand variables $\mathbf{x}_t^D$ enter directly through $\mathbf{B}_k^b$, capturing level effects within a regime. The latent regime process enters through $|\pi_t\rangle$, which serves as mixture weights that determine which regime-specific parameter

vector is relevant at each date. These channels are econometrically distinct and do not double-count, since the Hamilton filter operates on precomputed Stage 2 emission objects rather than reusing the Stage 3 structural equation itself for regime construction.

2.7.3 MS-ARX Specification

RPPI's block ($b = R$, $n_R = 1$), reduces to a scalar MS-ARX[†]:

$$\begin{aligned} y_t^R &= \sum_{k=0}^1 \pi_{t,k} [\mathbf{a}_k^{R'} \mathbf{x}_t^R + \mathbf{b}_k^{R'} \mathbf{x}_t^D] + \varepsilon_t^R, & \pi_{t,k} &= \langle k | \pi_t \rangle, \\ \varepsilon_t^R &\sim \mathcal{T}_1 \left(\nu^R, 0, \sum_{k=0}^1 \pi_{t,k} \sigma_k^R \right). \end{aligned} \quad (39)$$

2.7.4 Priors on Exogenous Coefficients

The own-lag matrices $\mathbf{A}_k^b$ inherit the Stage 2 centring structure from (18). The exogenous coefficient matrices $\mathbf{B}_k^b$ receive a zero-centred shrinkage prior:

$$[\tilde{\mathbf{B}}_k^b]_{ij} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1), \quad i = 1, \dots, n_b, \quad j = 1, \dots, n_D(p+1), \quad (40)$$

$$\mathbf{B}_k^b = x_B \tilde{\mathbf{B}}_k^b. \quad (41)$$

The parameter x_B is learned from the data, shrinking demand-transmission coefficients toward zero when cross-block effects are weak.

Remark 2.10 (Information Flow and Identification). The conditioning structure in (37)–(39) imposes a one-directional flow of information where demand conditions drive supply, credit, asset, and property-price dynamics, but not the reverse. This is consistent with the identifying assumption that housing demand is determined by income, financing costs, and household formation prior to supply and price adjustments. Feedback from property prices to demand is therefore excluded by construction. Although this restriction could be relaxed in a fully simultaneous extension, doing so would require an alternative identification strategy for the demand block itself, and is left for future work.

[†]The Univariate MS-ARX case again uses the mixture log-likelihood analogous to (38) and the $|\pi_t\rangle$ from (73).

2.8 Identification and Regime Labelling

Label-switching is a well-known phenomenon and often a challenge in mixture and Markov-switching models: the likelihood is invariant to permutations of the regime labels. Such that, posterior samplers may assign probability mass to multiple orderings of the same solution (see [Frühwirth-Schnatter \(2006\)](#), Chapter 3). This paper addresses label-switching through a hard volatility ordering imposed via the sorted prior in Section 2.6.2: by construction:

$$\sigma_{j,0}^b \leq \sigma_{j,1}^b \quad (42)$$

for every variable j and block b , such that regime $k = 0$ is always the low-volatility state and regime $k = 1$ the high-volatility state. This constraint is differentiable and compatible with NUTS, requiring no ex-post relabelling of posterior draws.

The regime labels *low-volatility* (S_0) and *high-volatility* (S_1) are therefore economically meaningful by construction: S_1 denotes the state in which every variable in the block exhibits weakly greater posterior volatility than in S_0 . The identification does not restrict the coefficient matrices $\mathbf{A}_k^b$, which are free to differ across regimes in both sign and magnitude. Regime comparisons (impulse responses, transmission coefficients) are therefore comparisons between the low- and high-volatility states of the system, as defined by the sorted-prior constraint.

2.9 Robustness

The share of posterior draws requiring companion-matrix stabilisation at IRF computation is small across the Demand, Supply, Credit, and Assets blocks, consistent with the identified regimes reflecting genuine shifts in volatility and dependence structure rather than artefacts of explosive dynamics. The RPPI block exhibits a modestly higher stabilisation rate, reflecting the known persistence of real residential price growth at business-cycle frequencies. Stabilisation is applied per draw via Schur-based eigenvalue clipping at IRF computation time, leaving the estimated posterior unchanged; see Appendix B.7 for the explicit form.

3. Results

3.1 Regime Identification

Table 2 and Figures 2–3 characterise the identified regimes. The headline pattern is that regime non-linearity is concentrated in the asset and demand blocks rather than house prices.

Table 2: Regime characterisation. T' : usable observations after lag removal. N_1 : quarters classified as S_1 . $\bar{d}_k = 1/(1 - p_{kk})$ in quarters.

Block	T'	N_1	%	$\bar{d}_0$	$\bar{d}_1$	$\bar{\pi}_0$	$\bar{\pi}_1$	Hardness
Demand	96	11	11.5	22.4	4.2	0.818	0.182	0.961
Supply	96	17	17.7	9.3	3.1	0.738	0.262	0.870
Credit	97	8	8.2	15.1	2.8	0.832	0.168	0.922
Assets	96	37	38.5	13.9	8.2	0.634	0.366	0.923
RPPI	96	12	12.5	2.9	3.0	0.482	0.518	0.874

Notes. Smoothed probabilities from the Kim (1994) backward smoother applied to posterior-mean filtered probabilities. Hardness = $\mathbb{E}[\max_k \hat{\pi}_{t|T}^{(k)}]$; values exceeding 0.75 indicate sharp regime separation.

The demand block identifies three S_1 episodes: four quarters in 2008Q3–2009Q2 (GFC income shock), six quarters in 2020Q2–2021Q3 (pandemic), and a single quarter in 2022Q2, with a regime hardness of 0.961 indicating sharp, clean transitions. The assets block classifies 2008Q2–2015Q1 as a single uninterrupted 28-quarter S_1 episode, five times longer than the 2008Q3–2009Q2 demand episode.

This duration gap is consistent with a financial accelerator operating independently of demand: the sovereign spread and systemic risk (sovCISS) appear to have sustained the high-volatility state well beyond the demand shock that triggered it. The assets transition matrix (Table 5) is consistent with this: the S_1 -to- S_1 persistence probability is 0.887 (95% CI: [0.720, 0.983]).

Supply exhibits seven fragmented S_1 episodes totalling 17 quarters. The fragmentation confirms that supply volatility is episodic and primarily cost-driven, as the construction cost’s volatility amplification ratio reached approximately 5.09 times (Table 15). This

Table 3: High-volatility (S_1) episodes

Block	Episode	Start	End	Quarters
Demand	1	2008Q3	2009Q2	4
	2	2020Q2	2021Q3	6
	3	2022Q2	2022Q2	1
Supply	1	2003Q1	2003Q1	1
	2	2004Q1	2004Q2	2
	3	2006Q2	2006Q2	1
	4	2010Q1	2010Q1	1
	5	2015Q1	2015Q1	1
	6	2018Q1	2018Q1	1
	7	2020Q2	2022Q3	10
Credit	1	2001Q4	2001Q4	1
	2	2008Q3	2009Q1	3
	3	2012Q4	2012Q4	1
	4	2022Q2	2022Q4	3
Assets	1	2008Q2	2015Q1	28
	2	2016Q3	2017Q1	3
	3	2019Q4	2020Q2	3
	4	2022Q2	2022Q4	3
RPPI	1	2000Q4	2000Q4	1
	2	2008Q2	2009Q1	4
	3	2020Q1	2021Q2	6
	4	2022Q1	2022Q1	1

Notes. Contiguous quarters with $\Pr(s_t = 1 \mid \mathcal{Y}_T) > 0.5$.

Table 4: Posterior transition matrix and regime statistics: Demand block.

Panel A: Transition matrix $\hat{\mathbf{P}}$

Median	95 % CI
$\begin{bmatrix} 0.962 & 0.038 \\ 0.210 & 0.790 \end{bmatrix}$	$\begin{bmatrix} [0.884, 0.992] & [0.008, 0.116] \\ [0.042, 0.587] & [0.413, 0.958] \end{bmatrix}$

Panel B: Ergodic distribution and mixing

	$\bar{\pi}_0$		$\bar{\pi}_1$		Spectral gap		κ		
	Med.	95 % CI	Med.	95 % CI	Med.	95 % CI	Med.	95 % CI	
Demand	0.843	[0.533, 0.961]	0.157	[0.039, 0.467]	0.252	[0.066, 0.662]	1.940	[0.448, 4.017]	<i>Notes.</i> Posterior

medians with 95 % equal-tailed credible intervals in brackets. *Panel A:* left matrix = posterior medians; right matrix = 95 % CIs; rows index s_{t-1} , columns s_t . *Panel B:* $\bar{\pi}_k$ = ergodic probability; spectral gap = $1 - |\lambda_2(\hat{\mathbf{P}})|$; κ = softmax persistence concentration.

Table 5: Posterior transition matrix and regime statistics: Assets block.

Panel A: Transition matrix $\hat{\mathbf{P}}$

Median	95 % CI
$\begin{bmatrix} 0.936 & 0.064 \\ 0.113 & 0.887 \end{bmatrix}$	$\begin{bmatrix} [0.815, 0.994] & [0.006, 0.185] \\ [0.017, 0.280] & [0.720, 0.983] \end{bmatrix}$

Panel B: Ergodic distribution and mixing

	$\bar{\pi}_0$		$\bar{\pi}_1$		Spectral gap		κ		
	Med.	95 % CI	Med.	95 % CI	Med.	95 % CI	Med.	95 % CI	
Assets	0.640	[0.331, 0.902]	0.360	[0.098, 0.669]	0.185	[0.030, 0.418]	2.063	[0.495, 4.419]	

Notes. Posterior medians with 95 % equal-tailed credible intervals in brackets. *Panel A:* left matrix = posterior medians; right matrix = 95 % CIs; rows index s_{t-1} , columns s_t . *Panel B:* $\bar{\pi}_k$ = ergodic probability; spectral gap = $1 - |\lambda_2(\hat{\mathbf{P}})|$; κ = softmax persistence concentration.

finding suggests that supply is slow-moving and demand-response invariant.

The RPPI transition matrix (Table 14) is nearly uninformative: $p_{00} = 0.660$ and $p_{11} = 0.727$, with 95% credible intervals spanning almost the full unit interval. The model cannot separate RPPI regimes from price data alone, suggesting that volatility shifts in house-price growth are insufficiently persistent or distinct to be identified as separate regimes. This is consistent with the interpretation that prices are a lagging indicator of the cycle rather than a driver.

The diffuse transition matrix is corroborated by the smoothed probabilities in Figure 2, which show persistent uncertainty between S_0 and S_1 , further supporting the conclusion that price volatility shifts do not form distinct regimes. House prices respond to underlying demand and supply conditions rather than driving them; volatility shifts in prices reflect changes in the underlying economic environment rather than independent regime shifts in price dynamics. Within the Irish housing market, prices are primarily a lagging indicator of the cycle, consistent with interpreting them as an outcome variable rather than a source of volatility shifts in the system.

The symmetric Kullback-Leibler (Jeffreys) divergence J_b between the two regime-conditional innovation distributions $\mathcal{N}(\mathbf{0}, \Sigma_k^b)$, reported in the final column of Table 2 and in Table 7, corroborates the smoothed-probability hardness on a distributional scale. Supply yields $J_b = 9.33$ nats (where nats denote natural-logarithm units of information; 95% CI: [3.42, 23.12]), while assets record 9.20 nats ([3.78, 21.02]), the widest separation overall. This is consistent with the fragmented supply-cost episodes and the sustained financial-accelerator dynamics already identified. Demand yields 5.25 nats (95% CI: [0.07, 0.41]) and credit 5.89 nats.

RPPI registers approximately two orders of magnitude less at 0.12 nats ([0.07, 0.41]), indicating that the regime-conditional innovation distributions for house-price growth are statistically indistinguishable, consistent with the diffuse transition matrix and the interpretation of prices as a lagging indicator. Across all multivariate blocks, $\text{KL}(P_1 \parallel P_0)$ exceeds $\text{KL}(P_0 \parallel P_1)$ by a factor of three to eight: the crisis regime is the distinctive one, with the tranquil-period innovation distribution substantially lighter-tailed than its crisis

counterpart.

The regime-conditional correlation matrices reveal a further dimension of asymmetry. In the demand block (Table 8), the MDD–CPI correlation shifts from approximately -0.055 in S_0 to $+0.544$ in S_1 (95% CI: $[0.074, 0.810]$): demand growth and inflation co-move during crisis periods, consistent with conventional stagflationary dynamics.

The MDD–RTI correlation becomes more negative, shifting from -0.060 to -0.489 (95% CI: $[-0.796, -0.011]$), indicating that affordability deteriorates more sharply when incomes are under stress. In the assets block (Table 11), the spread–sovCISS correlation rises from -0.172 to $+0.464$ (CI: $[0.141, 0.705]$), demonstrating that sovereign risk and systemic stress co-amplify only in S_1 , consistent with a self-reinforcing financial accelerator mechanism.

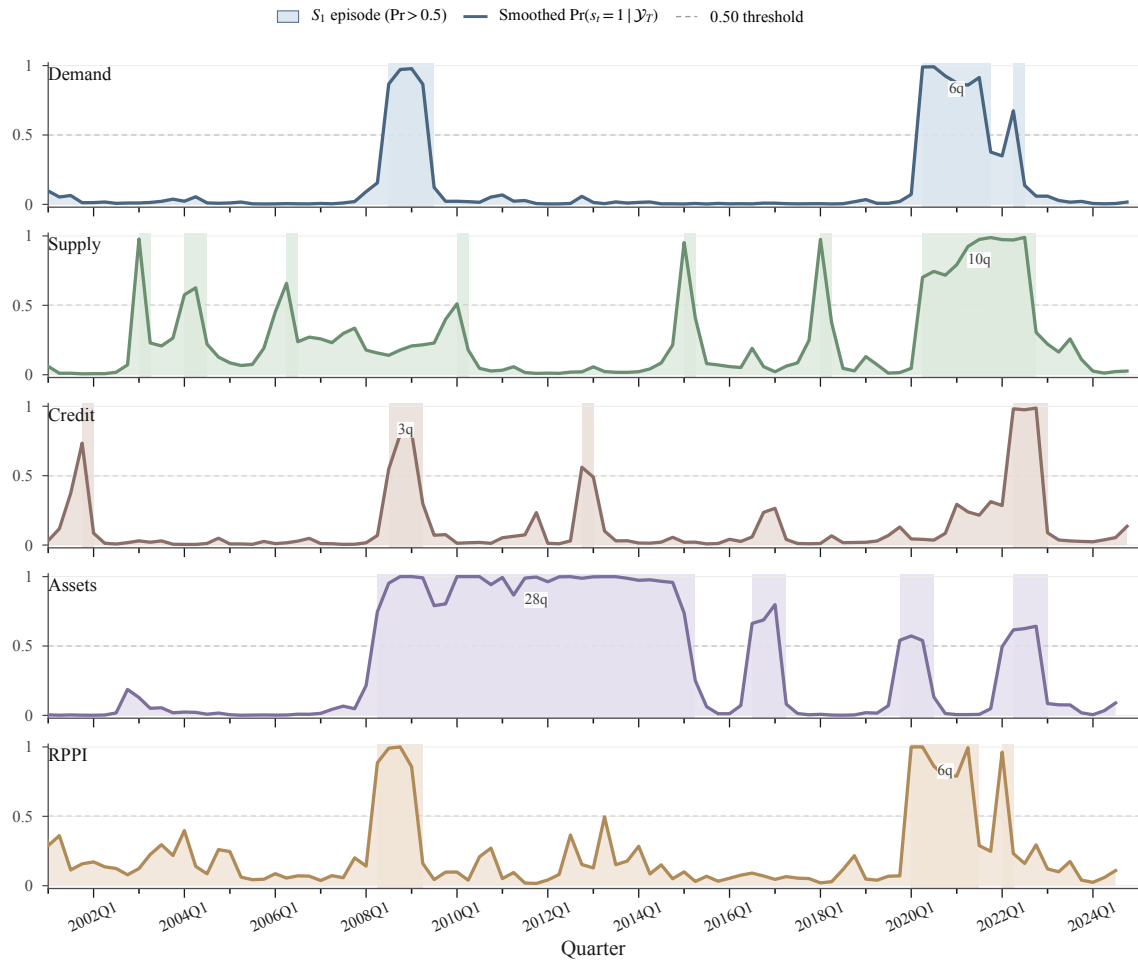
3.2 Impulse Responses

Full GIRF plots are reported in Figures 9–16 (Appendix A.2). GIRFs are computed using the ordering-invariant Pesaran–Shin estimator (Pesaran and Shin, 1998) and displayed as S_0/S_1 overlays with 68% and 90% credible bands.

Demand (Figure 9). Own-block GIRFs show moderate cross-regime differentiation. The MDD own-shock decays within two quarters regardless of regime. The clearest asymmetry is the MDD-to-CPI cross-response: approximately 0.8 standard deviations in S_1 against 0.4 in S_0 , indicating demand-pull inflation is amplified in high-volatility states. Moderate differentiation is by design: the demand block provides the regime clock rather than the primary amplification channel.

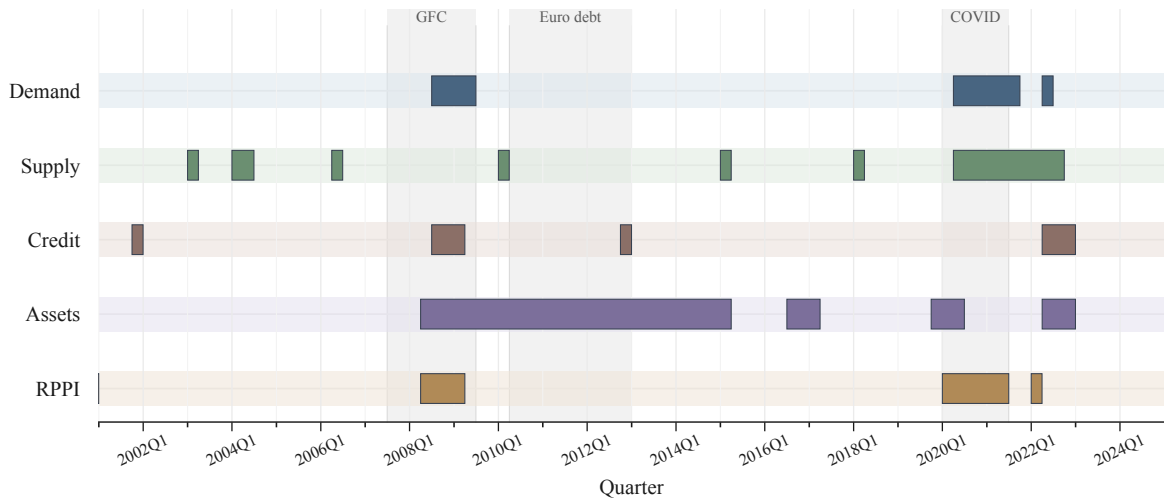
Supply (Figures 10 and 13). S_0 and S_1 profiles are nearly identical across both stages. Construction cost own-shock is larger in S_1 (approximately 1.5 standard deviations), but quantity variables (construction volume, planning permissions) show no corresponding amplification. The decoupling of cost volatility from quantity response is consistent with structural supply inelasticity, in which non-price constraints limit adjustment regardless of the prevailing regime. Stage 3 conditioning on the demand regime path does not alter this conclusion.

Credit (Figures 11 and 14). The shadow rate-to-LDR response under S_1 reaches



Shaded windows denote contiguous quarters with $\Pr(s_t = 1 | \mathcal{Y}_T) > 0.5$; the longest episode in each panel is annotated.

Figure 2: Smoothed regime probabilities and high-volatility episodes for all five blocks. Grey episode bands denote contiguous quarters where $\mathbb{P}(s_t = 1 | \mathcal{Y}_T) > 0.5$; duration labels and panel summaries make cross-block asymmetry explicit. Demand identifies two substantive S_1 episodes (GFC: 2008Q3–2009Q2; pandemic: 2020Q2–2021Q3). Assets exhibit a single 28-quarter episode (2008Q2–2015Q1), nearly three times the length of the demand GFC episode, establishing that financial stress persisted well beyond the initial demand shock. Supply episodes are brief and fragmented. RPPI probabilities are diffuse throughout, consistent with the nearly uninformative transition matrix estimated for that block (Table 14).



Bars denote contiguous quarters with $\Pr(s_t = 1 | \mathcal{Y}_T) > 0.5$; shaded columns mark major global stress periods.

Figure 3: Calendar strip plot of S_1 classification by block and quarter. Bars mark quarters for which $\mathbb{P}(s_t = 1 | \mathcal{Y}_T) > 0.5$, aligned to quarter-end. Light grey background bands indicate major global stress periods. The figure illustrates the temporal cascade: demand enters S_1 in 2008Q3; assets follow in 2008Q2 and remain there through 2015Q1; credit has discrete episodes in 2008–2009, 2012, and 2022. Supply and RPPI S_1 classifications are scattered and do not align with the GFC demand episode, confirming that these blocks respond to different drivers.

approximately 1.5 standard deviations at Stage 2, compared with near zero under S_0 , suggesting that unconventional monetary policy had substantially greater pass-through during periods of financial stress. At Stage 3, the shadow rate own-shock reaches approximately two posterior standard deviations under S_1 , with non-overlapping S_0/S_1 bands through the first three quarters. Net lending response to a cost-of-borrowing shock is near zero in S_0 but approximately -0.5 standard deviations in S_1 , consistent with credit rationing during stress. These patterns establish that credit transmission is state-dependent in both direction and magnitude.

Assets (Figures 12 and 15). The largest cross-regime differences occur here. The spread-to-sovCISS cross-response reaches approximately 0.8 standard deviations in S_1 but near zero in S_0 : sovereign risk and systemic stress co-amplify only during financial stress, with non-overlapping credible bands through the first four quarters. A CB Assets shock produces a positive spread response in S_1 but not S_0 , consistent with the ECB balance sheet expansion during the sovereign crisis period. Stage 3 conditioning sharpens these differences further, consistent with the high own-block signal weight $\hat{\alpha}_A = 0.885$: combining demand and own-block likelihood contributions appears to improve regime

separation relative to demand-only conditioning.

RPPI (Figure 16). The S_1 impact response (approximately 0.65 standard deviations) exceeds the S_0 impact (approximately 0.40), but the 90% posterior bands overlap substantially from quarter two onward. Residential property prices respond more vigorously to demand shocks in high-volatility periods, but the model cannot confidently attribute this difference to a discrete state change rather than continuous volatility variation. Regime differentiation in prices is real but imprecise.

3.3 Own-Block Signal Weights

Table 6 reports posterior estimates of the own-block signal weight $\alpha_b \in (0, 1)$, which governs the contribution of each block's own emission log-probabilities to the combined Hamilton filter (32); a higher α_b indicates that the block's own dynamics contribute more to regime classification beyond the demand signal.

The assets block has the highest weight ($\hat{\alpha}_A = 0.885$, $\mathbb{P}(\alpha_b > 0.5) = 1.000$, 95% CI: $[0.731, 0.966]$). Asset market dynamics are sufficiently informative about the regime state that they sustain the S_1 classification for 24 quarters beyond the four-quarter demand S_1 episode that initiated it. This pattern is consistent with the quantitative mechanism of the financial accelerator: the own-block signal in the assets block appears to maintain the high-volatility classification through sovereign spread and systemic risk dynamics beyond the initial demand episode.

Supply ($\hat{\alpha}_S = 0.819$, $\mathbb{P}(\alpha_b > 0.5) = 0.995$) and credit ($\hat{\alpha}_C = 0.805$, $\mathbb{P}(\alpha_b > 0.5) = 0.983$) both carry substantial own-block signal. For supply, this captures cost-driven S_1 episodes misaligned with demand timing. For credit, own stress episodes in 2008–2009 and 2022 are driven by borrowing cost and lending dynamics rather than demand conditions.

The RPPI block has the lowest weight ($\hat{\alpha}_R = 0.701$, $\mathbb{P}(\alpha_b > 0.5) = 0.813$, 95% CI: $[0.237, 0.929]$) and the widest credible interval. House prices carry a limited own-block signal: regime classification for RPPI is predominantly driven by demand rather than price dynamics. This is consistent with the view that price-threshold-based macroprudential triggers react to information already incorporated into demand and asset regime signals several quarters earlier.

Table 6: Demand-signal mixing weight α_b , Stage 3

Block	$\tilde{\alpha}_b$	90 % CI	95 % CI	$\Pr(\alpha_b > 0.5)$
Supply	0.819	[0.623, 0.936]	[0.582, 0.949]	0.995
Credit	0.805	[0.580, 0.933]	[0.529, 0.947]	0.983
Assets	0.885	[0.760, 0.957]	[0.731, 0.966]	1.000
RPPI	0.701	[0.308, 0.908]	[0.237, 0.929]	0.813

Notes. α_b weights the block-specific emission log-probability relative to the demand-regime signal in the Stage 3 combined Hamilton filter; prior Beta(1, 1). Columns report posterior median, 90 % and 95 % credible intervals, and $\Pr(\alpha_b > 0.5)$, the posterior probability that the block contributes regime information beyond demand. Higher α_b indicates a more informative own-block signal.

4. Discussion and Macroprudential Policy Implications

Three features of the estimates shape the macroprudential discussion that follows, each pointing to a candidate mechanism with direct implications for how policy instruments are specified and timed.

Asset markets are the main locus of regime non-linearity and stress persistence. The assets block exhibits a single 28-quarter high-volatility episode spanning 2008Q2 to 2015Q1, a regime hardness of 0.923, and an expected S_1 duration of 8.2 quarters against 15.6 for S_0 . The sovCISS volatility amplification ratio of 4.76 and the sovereign spread ratio of 2.64 are the largest of the financial variables. The regime-conditional GIRFs (Figures 12 and 15) show that spread and systemic-risk responses are larger and more persistent under S_1 , and the own-block signal weight $\hat{\alpha}_A = 0.885$ ($\mathbb{P}(\alpha_b > 0.5) = 1.000$) indicates that asset dynamics carry substantial regime information beyond the demand signal. These features are consistent with a financial-accelerator interpretation: once a demand shock is absorbed, sovereign spread and systemic-risk indicators appear to sustain the high-volatility state beyond the duration of the initiating shock through co-amplification of risk premia. The ergodic distribution assigns 36.6 per cent probability to the asset S_1 state. The spectral gap of 0.195 is the lowest across all blocks (Remark 2.6), so financial-stress episodes, when they occur, tend to persist. Policy frameworks that assume short, symmetric boom-and-bust cycles are likely to understate this persistence

relative to the transition matrix estimates.

Credit transmission is state-dependent, so linear models are likely to misestimate policy elasticities in stress periods. Credit stress is episodic rather than sustained: the expected S_1 duration is 2.8 quarters, and S_1 episodes are fragmented across the sample. The Stage 3 GIRFs (Figures 11 and 14) show that the shadow-rate response to a cost-of-borrowing shock under S_1 reaches approximately two posterior standard deviations before decaying, roughly twice the S_0 amplitude. The coupling weight $\hat{\alpha}_C = 0.805$ indicates that credit carries a substantial own-block signal independent of the demand signal. The pattern is consistent with asymmetric transmission in the high-volatility state: tighter credit conditions appear to propagate into borrowing costs and loan volumes more rapidly and with greater amplitude, while the stress episode itself is short-lived. Following Gatt (2023), the policy-relevant window between the onset of credit stress and its observability in the data is short, on the order of two to three quarters, which suggests that reactive macroprudential tightening may arrive too late to mitigate systemic risk once the window closes. Buffer-based rules that accumulate capital during credit expansions act before the transmission asymmetry materialises and are therefore better matched to the estimated stress dynamics than symmetric reactive rules.

The supply block is largely regime-invariant and the estimates point to binding non-price constraints. Construction cost volatility amplification of 5.09 is the largest in the dataset, yet the Stage 2 and Stage 3 GIRFs are near-identical across regimes. Quantity variables show little response to demand shocks even as costs amplify sharply under S_1 . This decoupling of cost volatility from quantity adjustment is the key identifying feature for what follows, and three candidate mechanisms are consistent with it.

The first is institutional rigidity. Planning constraints, labour-market frictions, and the post-2008 contraction in construction capacity mean that the supply sector adjusts little to demand regardless of the prevailing regime, consistent with the low supply-elasticity estimates reported by Günnewig-Mönert and Lyons (2024).

The second is a cost-side information problem that the volatility amplification makes

visible. When construction cost volatility amplifies by a factor of 5.09 in S_1 , input prices behave less like stable production costs and more like a financial asset with its own regime dynamics. A developer pricing a project in the high-volatility state therefore faces substantial forward cost uncertainty that is not readily hedged: materials, labour, and financing costs amplify together and their co-movement strengthens under S_1 (Table 9). Search and procurement costs (assembling reliable quotes, locking in materials prices across a multi-month procurement window, and monitoring input markets) also rise with input volatility. In a standard real-options framework, elevated uncertainty raises the value of the option to wait, so deferral of project initiation is a rational response even when current demand conditions appear favourable.

A further source of uncertainty compounds the first two. A developer commits capital today against a sale price that will be realised eighteen to thirty-six months from now. In tranquil periods, this forward sale price can be projected with some confidence from current trends; in S_1 it cannot, because the RPPI transition matrix is nearly uninformative (Table 14) and the own-block signal weight is the lowest of all blocks ($\hat{\alpha}_R = 0.701$).

The model cannot separate RPPI regimes sharply from price data alone, and in that sense, the information available to a developer about the likely realised sale price is also limited. Both legs of the project are therefore uncertain simultaneously (input costs in, sale prices out), and they co-move in ways that make natural hedging between them unreliable. A reduction in exposure is one rational response to this bilateral uncertainty, consistent with the near-zero GIRF quantity responses.

The implication for policy is that supply-side interventions confined to planning permissions or land availability are unlikely to resolve the inelasticity documented here. Instruments that stabilise input costs or reduce forward price uncertainty for developers address the binding constraint identified by the model more directly.

Residential property prices exhibit the weakest regime signal of the five blocks. The RPPI regime hardness of 0.874 is the lowest among the multivariate blocks, and the transition-matrix credible intervals are wide and nearly diffuse (Table 14).

The univariate GIRF (Figure 16) shows S_0 and S_1 impact estimates with substantially

overlapping posterior bands, and the coupling weight $\hat{\alpha}_R = 0.701$ ($\mathbb{P}(\alpha_b > 0.5) = 0.813$) is the lowest of all blocks. Under the block-recursive identification maintained throughout, prices aggregate the outcomes of demand, credit, and asset dynamics rather than feeding back into them within period, so price-based regime signals are, by construction, delayed relative to the upstream blocks.

A practical consequence follows for any policy instrument that relies on house prices as its state variable. Because the RPPI contains little independent regime information, and because the upstream demand and asset regimes can shift without a correspondingly sharp adjustment in the price series, a price-level indicator that is informative in real time about whether the system is in S_0 or S_1 is not identified from these estimates. Instruments that trigger on house-price thresholds therefore act on a signal that is, under the model, among the least informative about the regime state.

The Stage 3 GIRF in Figure 16 does not show a clean downward adjustment in house prices under S_1 : the posterior impact response is larger than in S_0 (approximately 0.65 versus 0.40 standard deviations) and the 90% credible bands overlap from quarter two onward. Standard housing-demand theory predicts that elevated financial stress and tighter credit should reduce demand and place downward pressure on prices; the estimated S_1 response is not obviously consistent with that prediction, though the wide posterior bands mean the data do not decisively reject it either.

One interpretation of this pattern, which the evidence is consistent with but does not identify, runs as follows. The diffuse RPPI transition matrix (Table 14) and the low own-block signal weight $\hat{\alpha}_R = 0.701$ imply that prices contain little independent information about the underlying regime state. The wavelet Hurst estimate for $\Delta \log$ RPPI ($H = -0.173$) is consistent with mean reversion at business-cycle frequencies, while the sovCISS series has the highest long-memory estimate in the dataset (R/S $H = 0.933$).

Taken together, these diagnostics describe an information environment in which prices do eventually correct but no price-based signal is informative about when. Under loss aversion (Kahneman and Tversky, 1979), a household facing this information set may find it individually rational to transact rather than wait, since the downside of continued

rental exposure is observed, while the upside of waiting is uncertain and unscheduled. This is the mechanism labelled “fear of being priced out” in the behavioural housing literature.

This interpretation is offered as a candidate explanation, not as an identification result. The block-recursive MS-VARX framework is not set up to test behavioural hypotheses directly: the GIRFs measure how prices respond in each regime, not why households transact. What the estimates do establish is that, given the regime structure of the data, any plausible account of demand in S_1 must reconcile three features: high cost and asset-market volatility, persistent high-stress episodes in the assets block, and a price series that carries little own-block signal about the regime state. Whether the reconciliation runs through loss aversion, safe-haven reframing of residential property relative to financial assets, or a combination of the two, is beyond what the present framework can identify, and is left to future work using household-level data.

A related limitation should be flagged. The block-recursive cascade precludes feedback from prices and the asset block into demand by construction. If persistent financial stress does, in practice, intensify housing demand for instance through a safe-haven channel the demand block’s own-block signal will understate the long-run coupling between financial-market stress and housing demand, and Stage 3 estimates will be conditional on this restriction. Relaxing the one-directional structure is a priority for future work.

5. Conclusion

This paper makes three contributions to the empirical literature on housing-market dynamics. Regimes are identified from demand-block dynamics rather than from house prices, separating the identification variable from the outcomes whose behaviour is being explained.

A hierarchical Student- t shrinkage prior (the $\mathfrak{JA}$ prior) is proposed for Markov-switching VARs, with lag decay and tail thickness estimated jointly with the VAR coefficients. Regime uncertainty is propagated across stages by embedding a differentiable Hamilton filter inside NUTS, which avoids the generated-regressor bias that would arise from treating Stage 2 smoothed probabilities as fixed inputs.

The empirical results indicate that housing-market transmission in Ireland is state-dependent and that the asset block is the primary locus of regime non-linearity rather than house prices themselves. Asset-market stress coincides with a 28-quarter high-volatility episode from 2008Q2 to 2015Q1, five times longer than the corresponding demand episode, with an ergodic S_1 probability of 36.6 per cent.

Credit transmission is episodic but asymmetric across regimes: the shadow-rate response in the credit block is roughly twice as large under S_1 as under S_0 , with a policy-relevant window of two to three quarters before the stress episode decays. Supply is largely regime-invariant: construction cost volatility amplifies by a factor of 5.09 in S_1 while quantity variables show no corresponding amplification in the GIRFs, consistent with binding non-price constraints on construction activity.

Residential property prices exhibit only a modest and imprecise regime separation: the Stage 3 GIRF shows a larger S_1 impact response than S_0 , but the 90% posterior bands overlap from quarter two onward, so the model cannot attribute the differential to a discrete state change, rather than to continuous volatility variation.

Taken together, the estimates are consistent with prices lagging the demand and asset regime signals by several quarters; any behavioural interpretation of this lag must be read as suggestive rather than identified. The policy implication is that macroprudential instruments calibrated on single-regime estimates, or triggered by price-level thresholds, risk being miscalibrated in precisely the periods where accurate calibration matters most. Countercyclical instruments anchored to demand and asset-market leading indicators, and designed to accumulate buffers during credit expansions, are better matched to the regime dynamics estimated here than reactive rules triggered after stress becomes visible in the price series.

Two limitations apply. The block-recursive structure excludes feedback from prices-to-demand by construction. The 98-quarter sample limits the number of S_1 episodes available for transition matrix inference, particularly in the credit and supply blocks. Allowing the own-block signal weight α_b to vary over time and incorporating Central Bank of Ireland household balance-sheet data are the natural extensions.

Tables 16–19 report the contemporaneous demand transmission coefficients $B_{b,k}$ for every downstream block and regime. The headline finding is that direct demand transmission into downstream blocks is weak and statistically diffuse in the posterior, with near-zero median estimates and wide credible intervals across cross-regime inference. No single demand variable reported a posterior probability of positive sign exceeding 0.90 for any downstream block variable in either regime, with marginal exceptions for MDD-to-construction-volume in S_0 ($\mathbb{P}(B_{b,k} > 0) = 0.69$) and CPI-to-construction-cost in S_0 with a $\mathbb{P}(B_{b,k} > 0) = 0.84$.

This reflects the identification structure of the modelling framework. Demand does not enter Stage 3 as a fixed exogenous input in the conventional VARX sense; rather, demand uncertainty propagates into each downstream block through the joint likelihood via the precomputed emission log-probabilities. These log-probabilities encode the full probabilistic content of the demand regime path, including its uncertainty, at each quarter t .

Once the Hamilton filter conditions on this signal, the residual variation available to $B_{b,k}$ is the component of demand orthogonal to regime membership; demand-level fluctuations exist within regimes but not across them. Future work may therefore focus on time-varying spatial modelling of demand spillovers, rather than on regime shifts in the spillover coefficients themselves, to capture a complementary dimension of cross-block transmission.

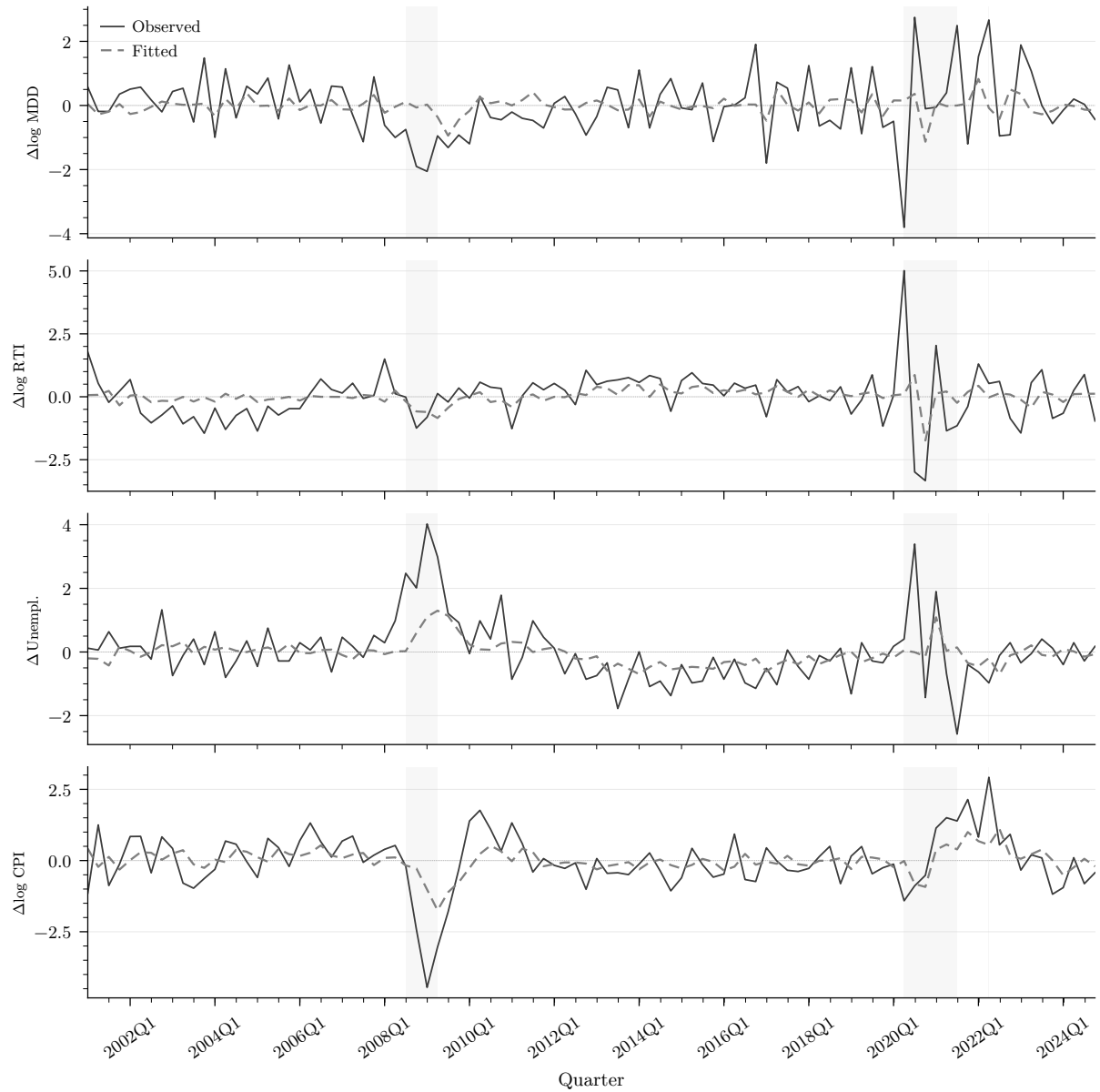
The finding of near-zero $B_{b,k}$ coefficients is consistent with the Irish housing literature, which attributes price dynamics primarily to lagged credit and supply conditions rather than to contemporaneous demand levels (Egan and McQuinn, 2024b): demand conditions shape the regime within which transmission occurs, but their direct level effects are largely absorbed by the Hamilton filter’s emission signal.

A. Figures and Tables

This appendix collects the figures, posterior predictive checks, and supporting tables underlying the discussion in Section 3.

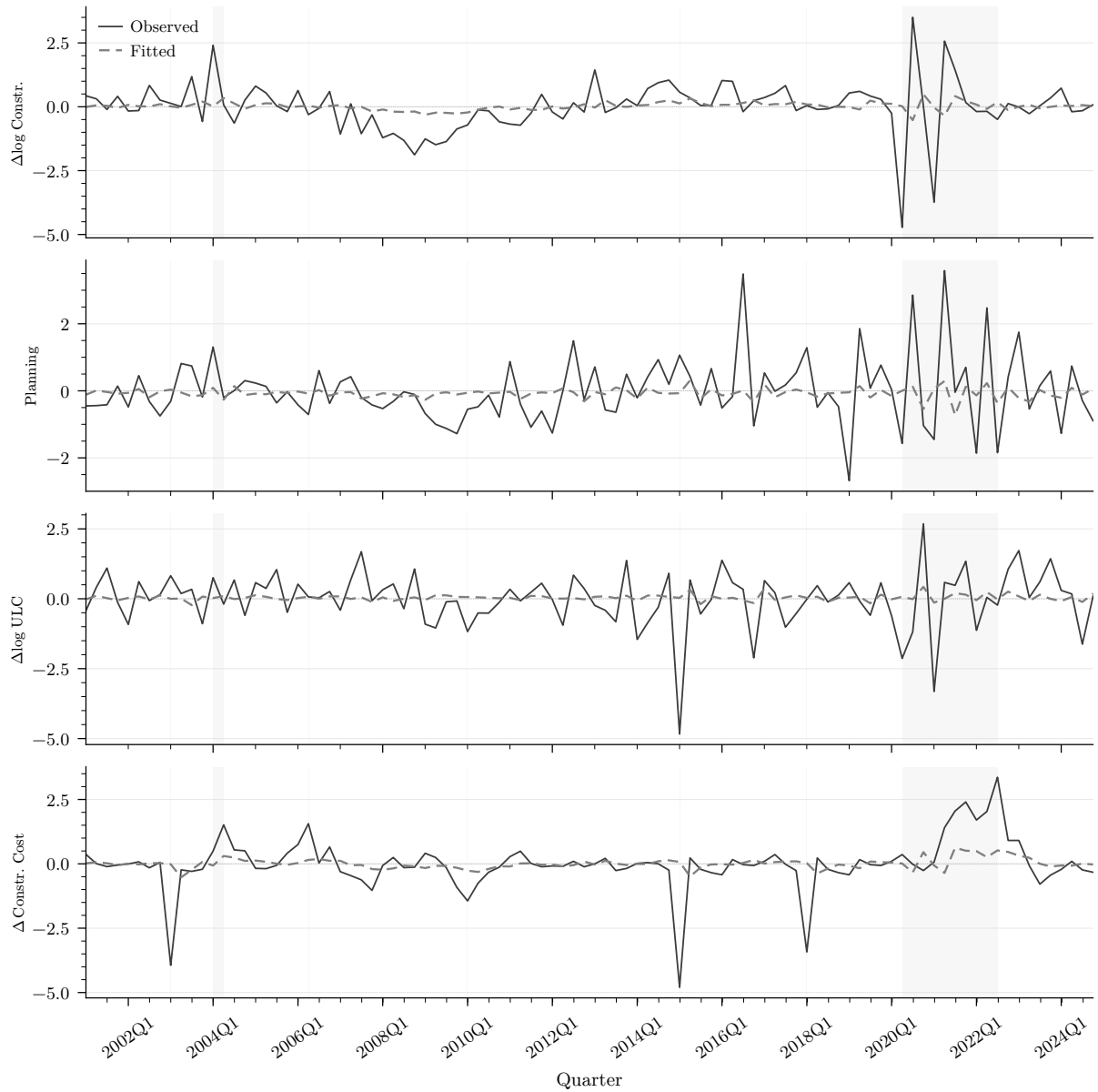
A.1 Smoothed Regime Probabilities and Posterior Predictive Checks

Figures 2 and 3 appear in Section 3.1 of the main text. Figures 4–8 below report block-wise posterior predictive checks overlaid on the smoothed S_1 probabilities, providing a visual diagnostic of model fit against the regime classification discussed in Section 3.



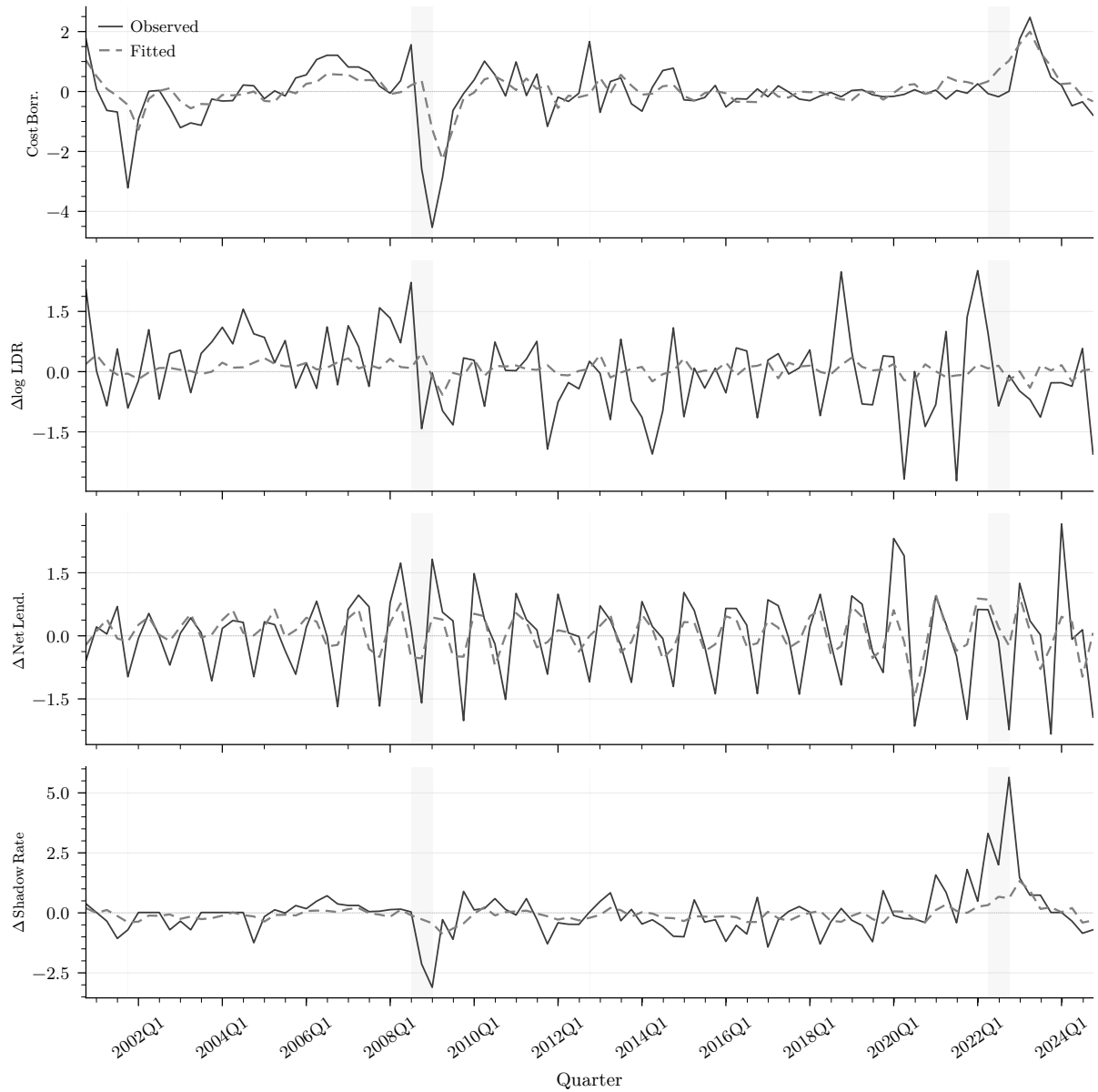
Demand block: posterior predictive check, Stage-2. Fitted values $\hat{y}_{t,j} = \sum_k \pi_{t|T}^{(k)} \mathbf{x}_t \hat{\mathbf{A}}^{(k)}$. Grey bands: S_1 episodes.

Figure 4: Demand block: posterior predictive check (Stage 2) and smoothed regime probabilities. Fitted values $\hat{y}_{t,j} = \sum_k \hat{\rho}_{t|T}^k \mathbf{x}_t \hat{\mathbf{A}}^{(k)}$. Grey bands: S_1 episodes. The model captures both the GFC demand collapse (sharp 2008Q3–2009Q2 unemployment and MDD spike) and the 2020 pandemic income shock. RTI rises sharply in 2020–2021 as rents continued to increase while incomes temporarily fell, a pattern the S_1 band correctly identifies as distinct from the MDD-led GFC episode.



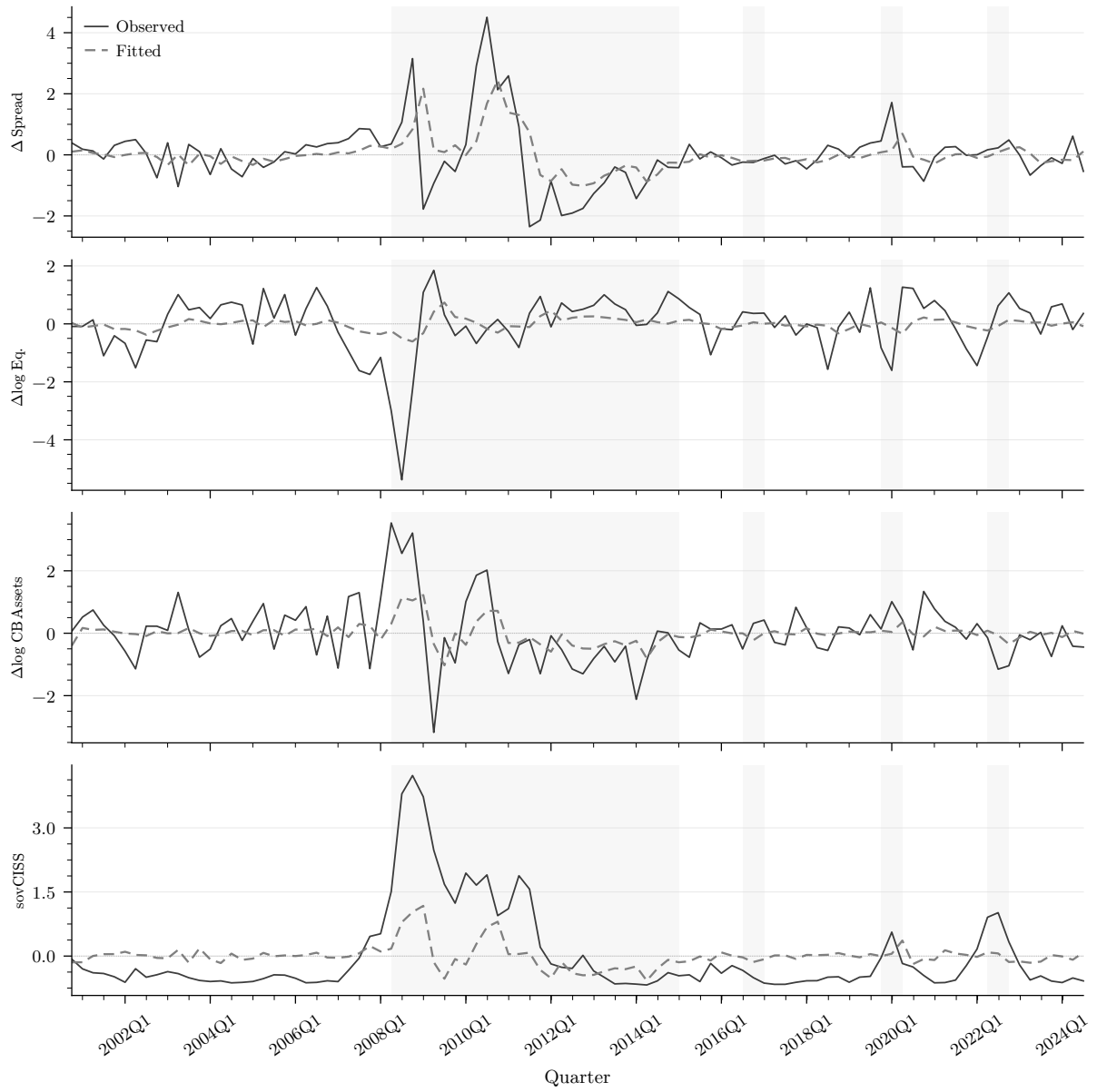
Supply block: posterior predictive check, Stage-2. Fitted values $\hat{y}_{t,j} = \sum_k \pi_{t|T}^{(k)} \mathbf{X}_t \hat{\mathbf{A}}^{(k)}$. Grey bands: S_1 episodes.

Figure 5: Supply block: posterior predictive check (Stage 2) and smoothed regime probabilities. Grey bands: S_1 episodes. Construction cost (volatility amplification ratio 5.09) shows the most pronounced S_1 amplitude expansion. Quantity variables (construction volume, planning permissions) exhibit substantially smaller regime amplification throughout the sample, consistent with the supply inelasticity result from the GIRFs.



Credit block: posterior predictive check, Stage~2. Fitted values $\hat{y}_{t,j} = \sum_k \pi_{t|T}^{(k)} \mathbf{X}_t \hat{\mathbf{A}}^{(k)}$. Grey bands: S_1 episodes.

Figure 6: Credit block: posterior predictive check (Stage 2) and smoothed regime probabilities. Grey bands: S_1 episodes. Cost of borrowing shows a large negative spike in 2008Q3; net lending exhibits the largest amplitude swings, consistent with rapid household deleveraging from 2008 to 2013. The shadow rate remains at its lower bound from 2012 to 2022; the 2022Q2–2022Q4 S_1 episode reflects the ECB tightening cycle identified through shadow rate and borrowing cost channels.



Assets block: posterior predictive check, Stage-2. Fitted values $\hat{y}_{t,j} = \sum_k \pi_{t|T}^{(k)} \mathbf{X}_t \hat{\mathbf{A}}^{(k)}$. Grey bands: S_1 episodes.

Figure 7: Assets block: posterior predictive check (Stage 2) and smoothed regime probabilities. Grey bands: S_1 episodes. The 2008Q2–2015Q1 S_1 period is driven by the sovereign spread (large positive deviations during the sovereign crisis) and sovCISS (elevated systemic stress). CB assets expand monotonically through the QE period, co-moving positively with spread and sovCISS in S_1 , consistent with the regime-conditional correlation shift identified in Table 11.

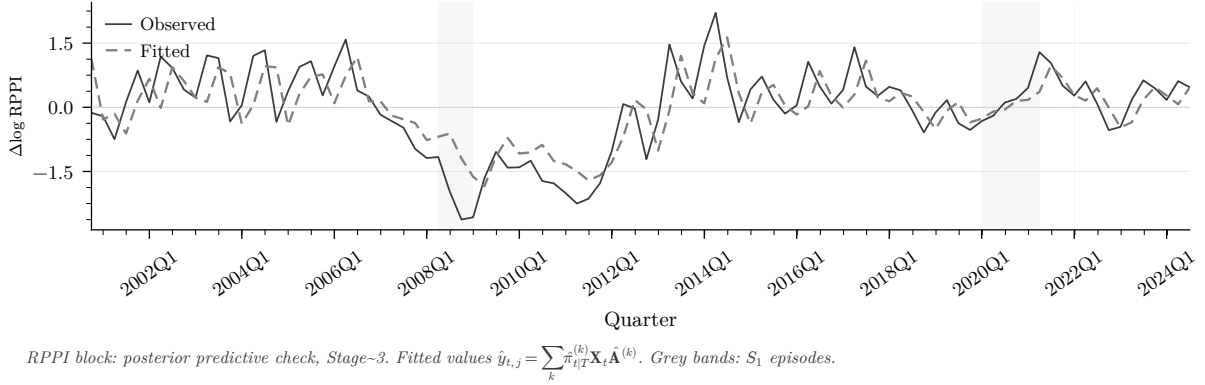


Figure 8: RPPI block: posterior predictive check (Stage 3) and smoothed regime probabilities. Grey bands: S_1 episodes. The model fits house price growth well through both the GFC collapse and the post-2012 recovery period. However, regime probabilities are diffuse with smoothed probability $\mathbb{P}(s_t = 1)$ rarely exceeding 0.75, and S_1 episodes are short-lived. The model also captures the 2020Q1–2021Q2 identification, which reflects the pandemic-era price acceleration, Although it is notable that the posterior probability does not reach the high levels seen in the assets block, confirming that price volatility is less sharply regime-separated than financial stress.

A.2 Regime-Conditional Impulse Response Functions

The impulse responses summarised in Section 3 are computed using the ordering-invariant (Pesaran and Shin, 1998):

$$\text{GIRF}_y(h, \boldsymbol{\delta}, \mathcal{I}_{t-1}) = \mathbb{E}[\mathbf{y}_{t+h} | \boldsymbol{\varepsilon}_t = \boldsymbol{\delta}, \mathcal{I}_{t-1}] - \mathbb{E}[\mathbf{y}_{t+h} | \mathcal{I}_{t-1}]. \quad (43)$$

Applying the law of iterated expectations with respect to the regime sequence $\{s_\tau\}_{\tau \geq t}$ and the regressor history $\mathbf{x}_{t-1}^b$, each conditional expectation expands as

$$\mathbb{E}[\mathbf{y}_{t+h}^{b,j} | S_{t-1}] = \mathbb{E}\left[\mathbb{E}[\mathbf{y}_{t+h}^{b,j} | S_{t-1}, \mathbf{x}_{t-1}^b] | S_{t-1}\right], \quad (44)$$

where the inner expectation conditions on the full regressor history and the outer expectation marginalises over it. The regime-conditional GIRFs reported here fix $S_{t-1} = k$ and average over $\mathbf{x}_{t-1}^b$ using the empirical distribution within regime k , following Pesaran and Shin (1998).

GIRFs are computed separately for each regime and displayed as S_0/S_1 overlays with 68% and 90% credible bands in Figures 9–16 below.

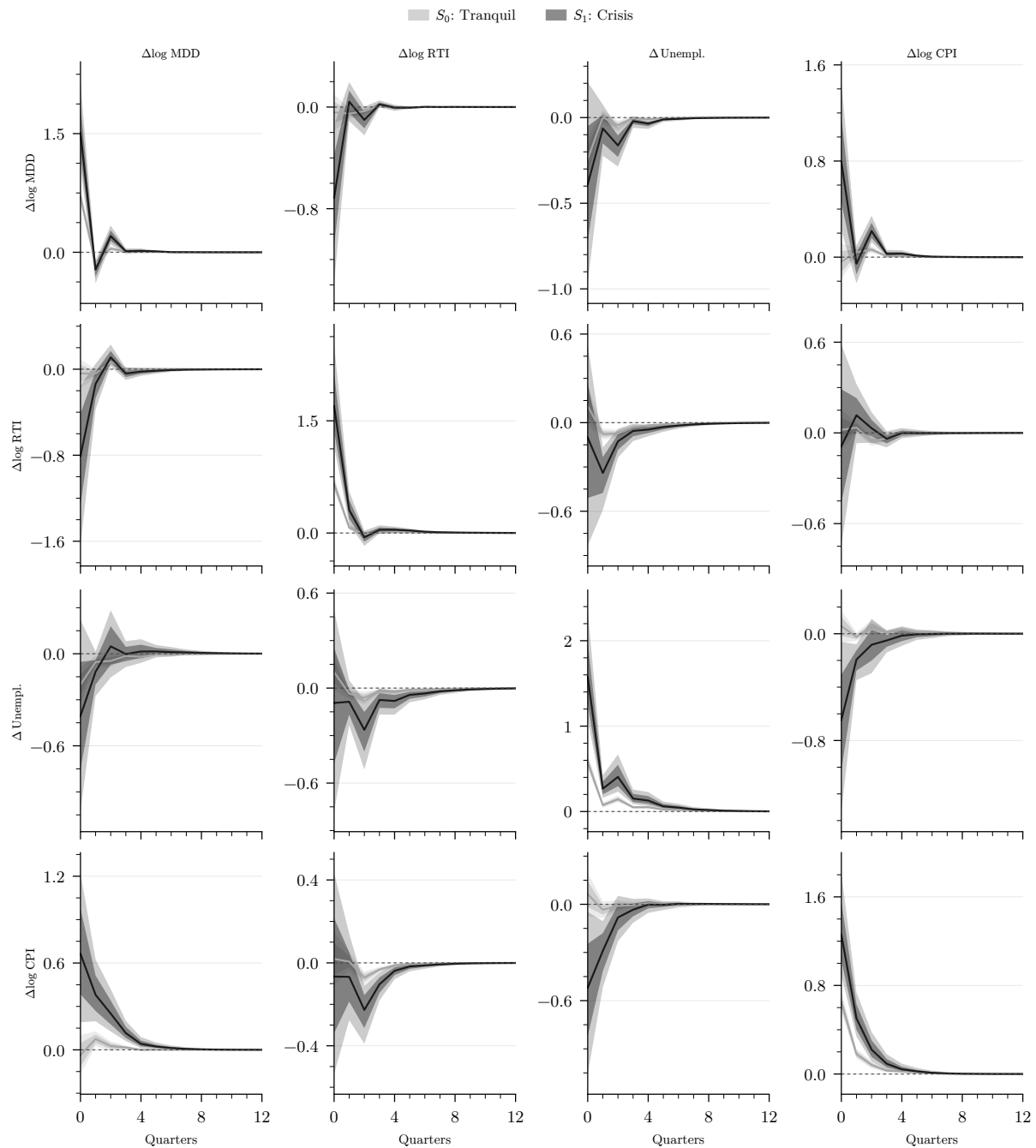


Figure 9: Stage 2 demand block GIRFs by regime (S_0 : tranquil; S_1 : crisis). Bands: 68% and 90% credible intervals. Cross-regime differences are moderate, with MDD exhibiting own-shock decay within two quarters in both regimes. The unemployment own-shock is more persistent in S_1 , though posterior bands overlap. The MDD-to-CPI cross-response is approximately twice as large in S_1 (0.8 vs. 0.4 standard deviations), indicating that demand-pull inflation is amplified in high-volatility states, consistent with conventional macroeconomic findings. Still, the modest regime differentiation is a feature by design: the demand block is the regime clock, not the primary amplification channel.

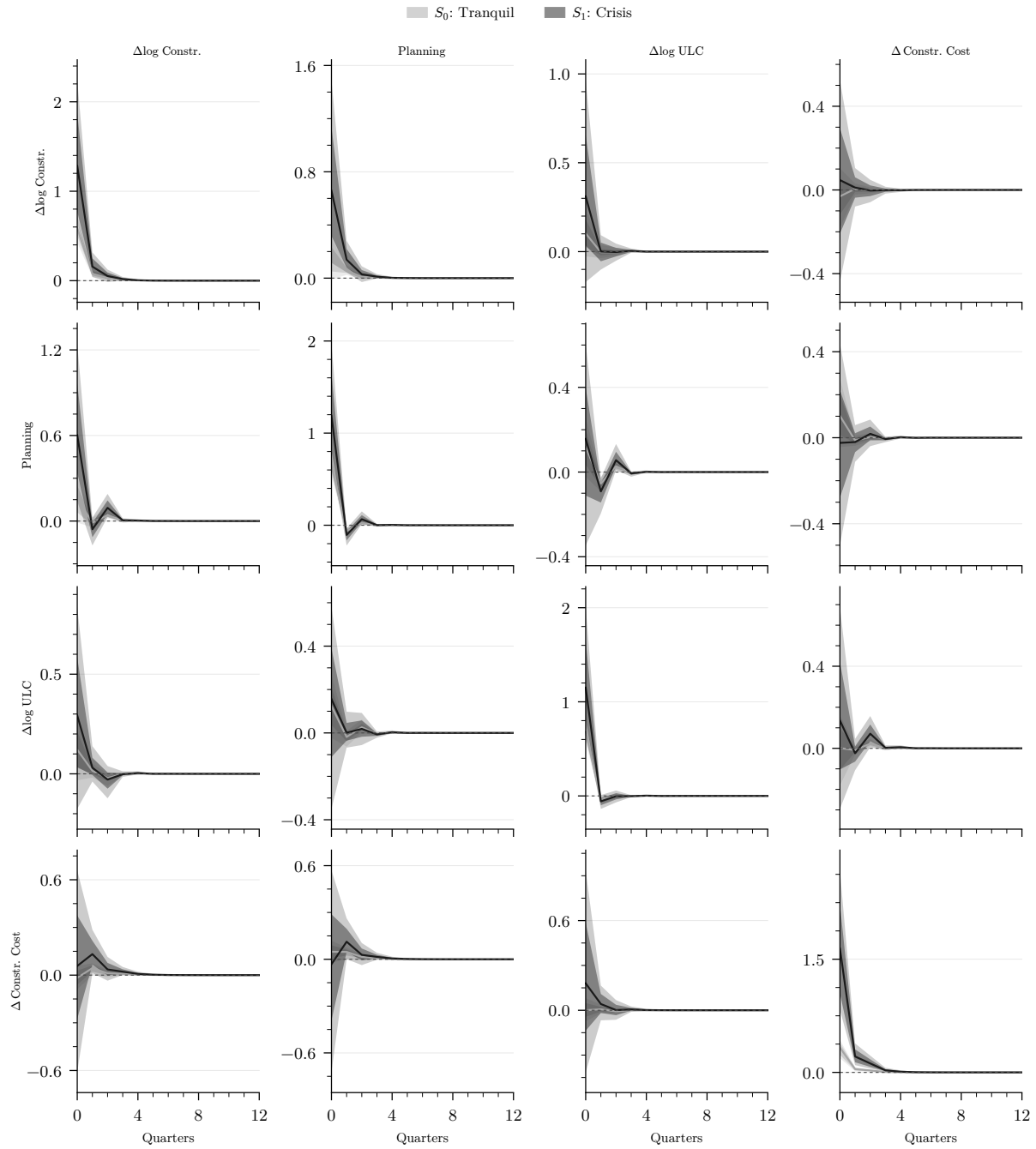


Figure 10: Stage 2 supply block GIRFs by regime. The S_0 and S_1 supply profiles are nearly identical for all variable pairs. Construction cost own-shock is larger in S_1 (approximately 1.5 standard deviations), but quantity variables (construction volume, planning permissions) show no corresponding amplification. The decoupling of cost volatility from quantity response is the quantitative evidence for supply inelasticity: non-price constraints prevent adjustment regardless of regime.

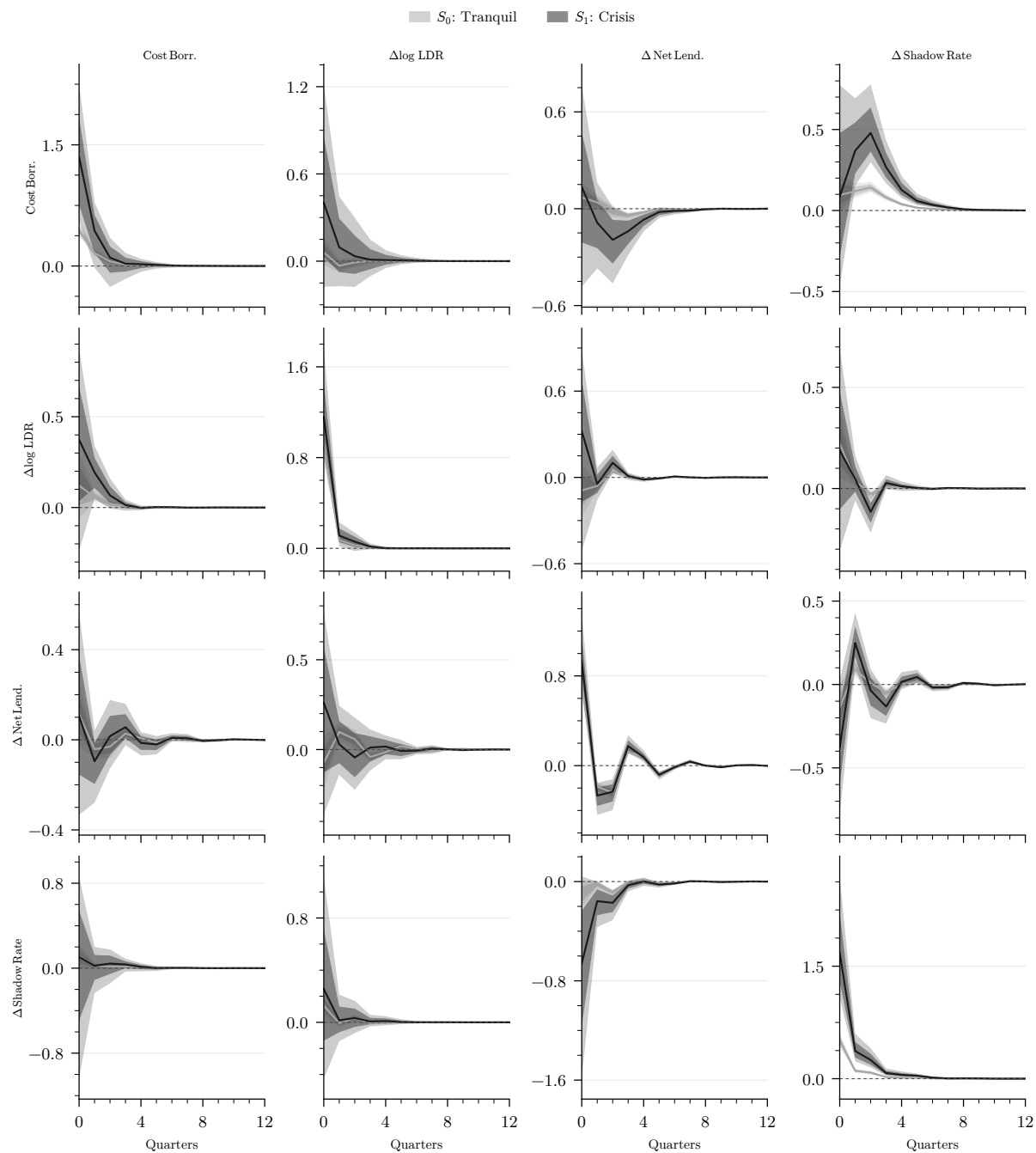


Figure 11: Stage 2 credit block GIRFs by regime. Cost of borrowing own-shock: similar across regimes. Shadow rate shock into loan-to-deposit ratio: approximately 1.5 standard deviations in S_1 versus near zero in S_0 , indicating strongly regime-dependent monetary transmission. Net lending response to a borrowing cost shock is negative and significant only in S_1 , consistent with credit rationing during stress.

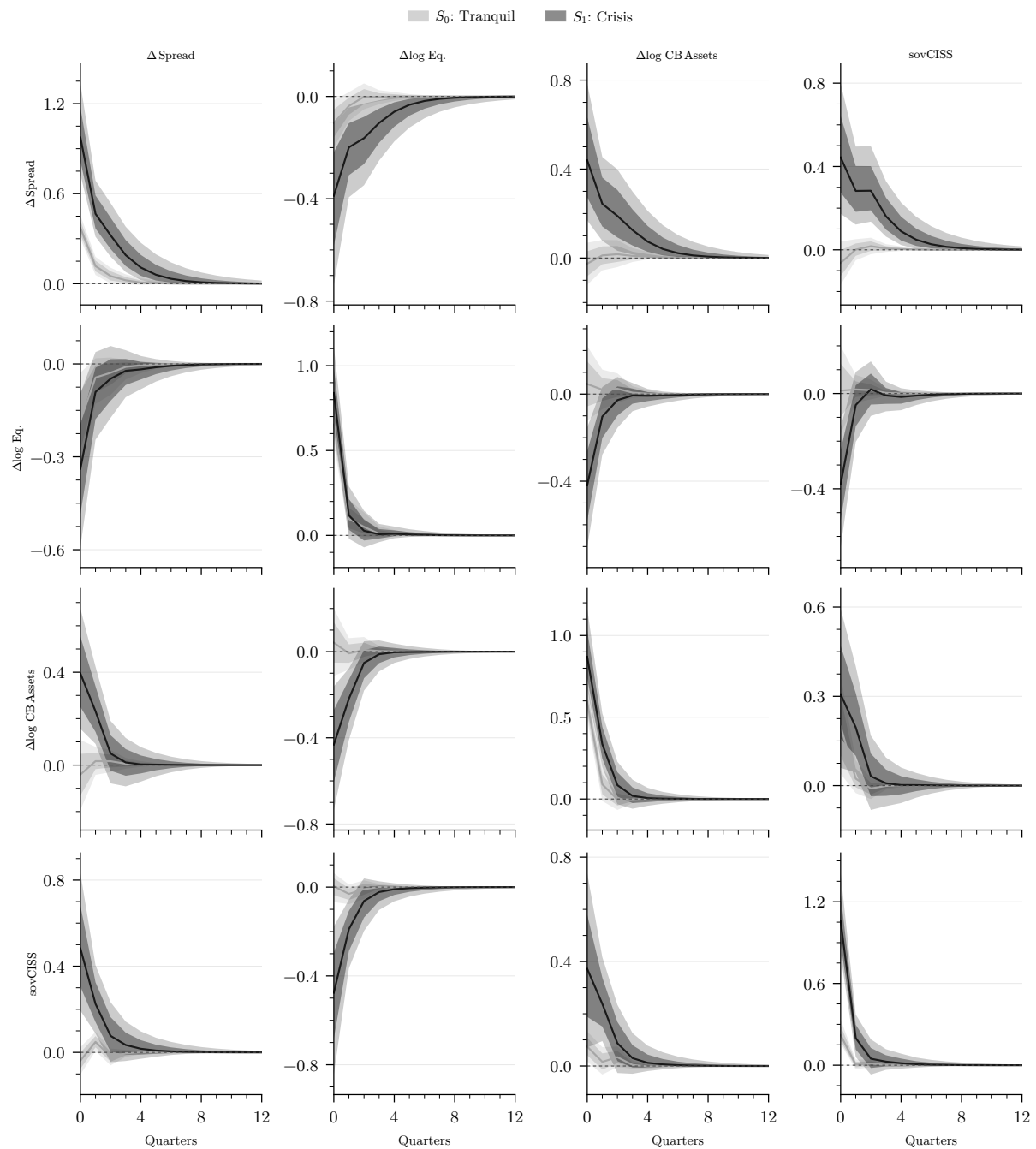


Figure 12: Stage 2 assets block GIRFs by regime. Largest cross-regime differences of all blocks. SovCISS own-shock: non-overlapping S_0/S_1 bands through the first four quarters. Spread shock into sovCISS: approximately 0.8 standard deviations in S_1 versus near zero in S_0 ; sovereign risk and systemic stress co-amplify only in the high-volatility state. CB Assets shock into spread: positive in S_1 , near zero in S_0 , reflecting ECB balance sheet expansion during the sovereign stress period.

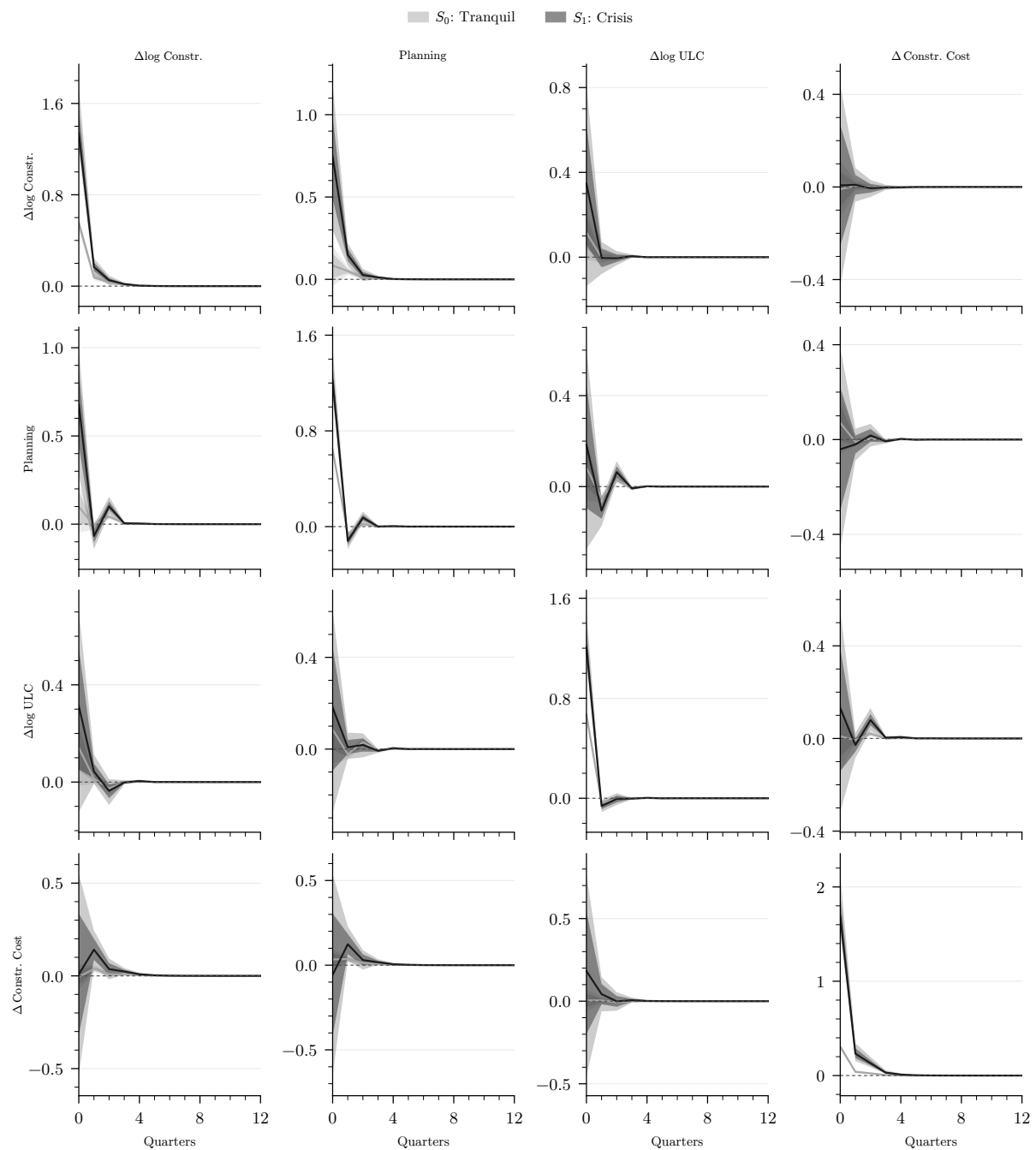


Figure 13: Stage 3 supply block GIRFs conditional on demand regime path. Supply profiles are nearly identical to Stage 2, with no meaningful S_0/S_1 differentiation in quantity variables, which confirms that supply inelasticity is robust to the fuller conditioning structure of Stage 3 and is not an artefact of Stage 2 specification.

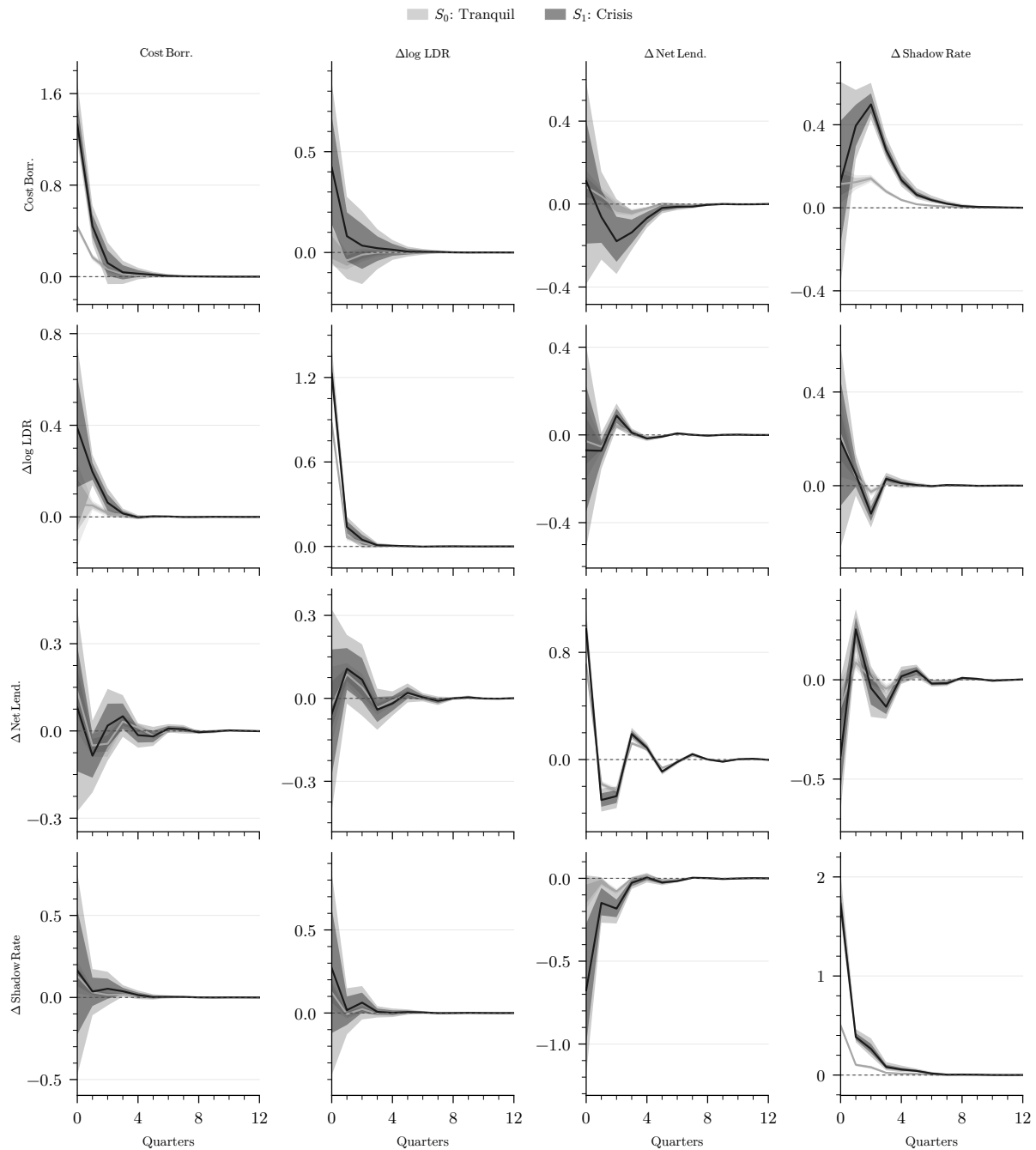


Figure 14: Stage 3 credit block GIRFs conditional on demand regime path. The MRO Shadow rate's own-shock reaches approximately 2 standard deviations in S_1 versus approximately 1 in S_0 , with non-overlapping bands through the first three quarters. Net lending's response to borrowing cost shock reaches near zero in S_0 and approximately -0.5 standard deviations in S_1 . These cross-regime differences confirm that credit transmission is state-dependent in both direction and magnitude. Linear models will therefore systematically understate the policy multiplier during stress.

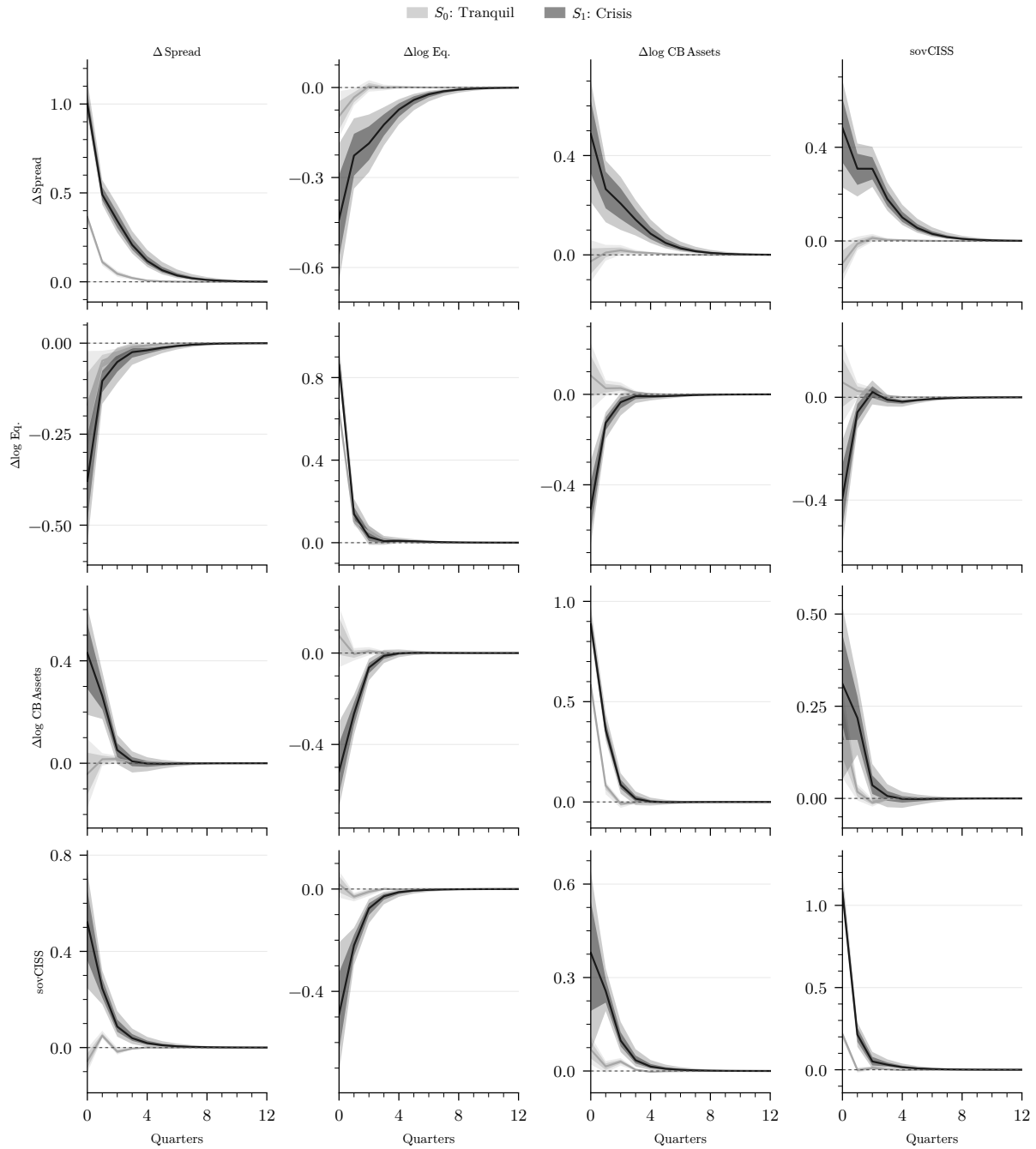
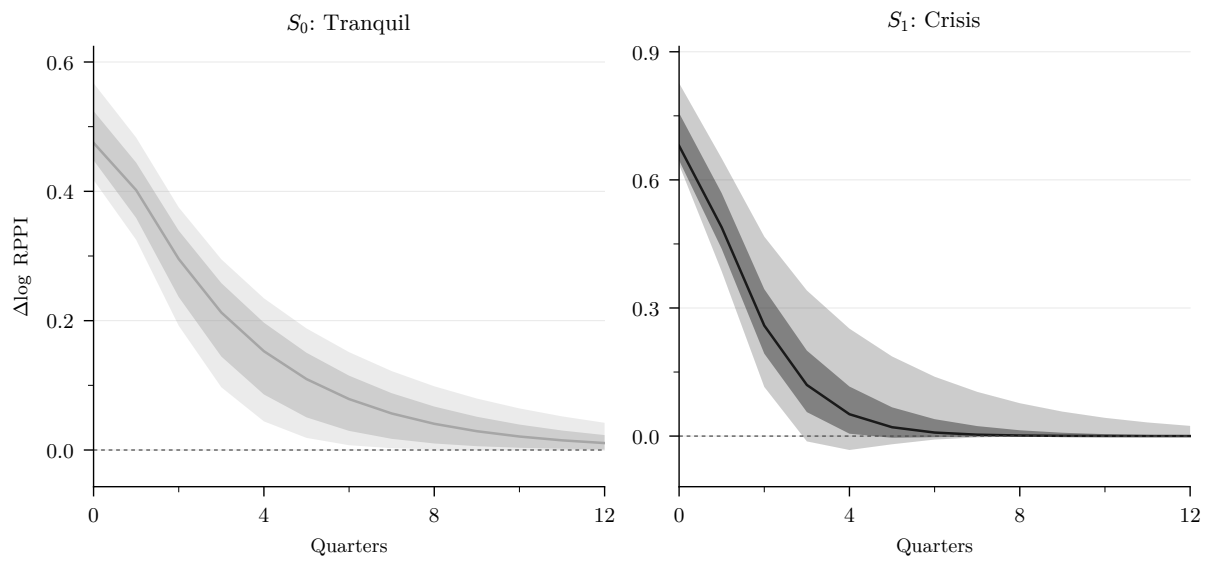


Figure 15: Stage 3 assets block GIRFs conditional on demand regime path. Cross-regime differences sharpen relative to Stage 2. The SovCISS-to-spread cross-response reports an approximately 0.5 standard deviation by quarter two in S_1 and is near zero in S_0 , with non-overlapping bands for the first four quarters. Sharpening under Stage 3 conditioning is consistent with the high coupling weight ($\hat{\alpha}_A = 0.885$): combining demand and asset-block likelihood contributions improves regime classification compared to the assets block conditioned in isolation.



RPPI univariate IRF. Bands: 68\% (dark) and 90\% (light) CI.

Figure 16: Stage 3 RPPI block univariate GIRF by regime. Bands: 68% (dark) and 90% (light) credible intervals. S_1 impact response: approximately 0.65 standard deviations. S_0 impact response: approximately 0.40 standard deviations. Posterior bands overlap substantially from quarter two onward. The overlap confirms that residential property prices do not exhibit sharp regime separation: the model cannot confidently attribute the larger S_1 response to a discrete state change rather than continuous volatility variation. These prices respond more strongly to demand shocks during high-volatility periods, but the regime differential is not as cleanly identified as in the assets or credit blocks.

A.3 Regime-Conditional Elasticities

The empirical results in Tables 16–19 report the contemporaneous demand transmission coefficients $\mathbf{B}_k^b$ for every downstream block and regime, with the headline finding that direct demand transmission into downstream blocks is weak and is statistically diffuse across both regimes.

No single demand variable reported a posterior probability of positive sign exceeding 0.90 for any downstream block variable in either regime, with marginal exceptions for MDD-to-construction-volume in S_0 ($\mathbb{P}(B_{b,k} > 0) = 0.69$) and CPI-to-construction-cost in S_0 ($\mathbb{P}(B_{b,k} > 0) = 0.84$).

This result is not a failure of the model. It reflects the block-recursive structure’s separation of demand-level effects (captured in $\mathbf{B}_k^b$) from regime-pathway effects (captured through the Hamilton filter mixture weights $\pi_{D,t}^*$). Weak direct transmission coefficients indicate that within-regime contemporaneous demand-to-supply and demand-to-credit linkages are modest, consistent with the Irish housing literature, which attributes price dynamics primarily to lagged credit and supply conditions rather than to contemporaneous demand levels (Egan and McQuinn, 2024b).

However, the regime pathway through the coupling mechanism is the primary channel through which demand propagates into downstream blocks, not the level of demand variables themselves.

A.4 Variable Definitions

Table 7: Posterior KL regime hardness. $J_b = \frac{1}{2}[\text{KL}(P_0\|P_1) + \text{KL}(P_1\|P_0)]$ between the two regime-conditional innovation distributions $\mathcal{N}(\mathbf{0}, \mathbf{\Sigma}^{(b,k)})$. Posterior medians with 95% credible intervals; directional components $\text{KL}(P_0\|P_1)$ and $\text{KL}(P_1\|P_0)$ reported as posterior medians.

Block	J_b	95% CI	$\text{KL}(P_0\ P_1)$	$\text{KL}(P_1\ P_0)$
Demand	5.25	[1.94, 12.79]	1.80	8.66
Supply	9.33	[3.42, 23.12]	2.04	16.56
Credit	5.89	[1.92, 16.81]	1.91	9.72
Assets	9.20	[3.78, 21.02]	1.95	16.42
RPPI	0.12	[0.07, 0.41]	0.09	0.15

Table 8: Regime-conditional correlation matrices: Demand block

	$\Delta \log \text{ MDD}$	$\Delta \log \text{ RTI}$	$\Delta \text{ Unempl.}$	$\Delta \log \text{ CPI}$
<i>Panel A: S_0 (tranquil)</i>				
(2)	−0.060			
	[−0.296, 0.178]			
(3)	−0.349	0.174		
	[−0.540, −0.126]	[−0.062, 0.392]		
(4)	−0.055	0.030	0.098	
	[−0.290, 0.197]	[−0.203, 0.266]	[−0.154, 0.338]	
<i>Panel B: S_1 (crisis)</i>				
(2)	−0.489			
	[−0.796, −0.011]			
(3)	−0.268	−0.062		
	[−0.634, 0.219]	[−0.489, 0.381]		
(4)	0.544	−0.054	−0.427	
	[0.074, 0.810]	[−0.468, 0.379]	[−0.735, 0.050]	
<i>Panel C: $\Delta \mathbf{R}$</i>				
(2)	−0.426			
	[−0.837, 0.128]			
(3)	+0.079	−0.234		
	[−0.363, 0.623]	[−0.735, 0.280]		
(4)	+0.593	−0.084	−0.523	
	[0.035, 0.975]	[−0.574, 0.419]	[−0.924, 0.036]	

Notes. Posterior median with 95 % equal-tailed credible intervals (lower triangle; diagonal and upper triangle omitted). **: 95 % CI excludes zero; *: 90 % CI excludes zero. Panel C: $\Delta \mathbf{R} = \mathbf{R}_{S_1} - \mathbf{R}_{S_0}$; credible intervals from posterior draws of the difference.

Table 9: Regime-conditional correlation matrices: Supply block

	$\Delta \log$ Constr.	Planning	$\Delta \log$ ULC	Δ Cost
<i>Panel A: S_0 (tranquil)</i>				
(2)	0.244			
	[−0.074, 0.492]			
(3)	0.179	0.184		
	[−0.094, 0.429]	[−0.080, 0.410]		
(4)	−0.054	0.157	0.001	
	[−0.379, 0.214]	[−0.111, 0.449]	[−0.315, 0.278]	
<i>Panel B: S_1 (crisis)</i>				
(2)	0.553			
	[0.005, 0.809]			
(3)	0.269	0.145		
	[−0.240, 0.656]	[−0.315, 0.540]		
(4)	0.041	−0.022	0.126	
	[−0.392, 0.422]	[−0.440, 0.388]	[−0.317, 0.527]	
<i>Panel C: $\Delta \mathbf{R}$</i>				
(2)	+0.311			
	[−0.366, 0.748]			
(3)	+0.088	−0.040		
	[−0.504, 0.595]	[−0.567, 0.467]		
(4)	+0.101	−0.185	+0.129	
	[−0.444, 0.603]	[−0.687, 0.333]	[−0.422, 0.651]	

Notes. Posterior median with 95 % equal-tailed credible intervals (lower triangle; diagonal and upper triangle omitted). **: 95 % CI excludes zero; *: 90 % CI excludes zero. Panel C: $\Delta \mathbf{R} = \mathbf{R}_{S_1} - \mathbf{R}_{S_0}$; credible intervals from posterior draws of the difference.

Table 10: Regime-conditional correlation matrices: Credit block

	Cost Borr.	$\Delta \log \text{LDR}$	$\Delta \text{Net Lend.}$	ΔShadow
<i>Panel A: S_0 (tranquil)</i>				
(2)	0.137			
	[−0.106, 0.365]			
(3)	0.158	−0.103		
	[−0.075, 0.379]		[−0.335, 0.143]	
(4)	0.218	0.252	−0.195	
	[−0.042, 0.465]		[0.020, 0.462] [−0.407, 0.045]	
<i>Panel B: S_1 (crisis)</i>				
(2)	0.325			
	[−0.295, 0.724]			
(3)	0.113	0.282		
	[−0.417, 0.652]		[−0.490, 0.754]	
(4)	0.065	0.162	−0.410	
	[−0.692, 0.535]		[−0.332, 0.591] [−0.770, 0.118]	
<i>Panel C: $\Delta \mathbf{R}$</i>				
(2)	+0.189			
	[−0.511, 0.676]			
(3)	−0.044	+0.385		
	[−0.641, 0.559]		[−0.496, 0.960]	
(4)	−0.144	−0.089	−0.214	
	[−1.028, 0.425]		[−0.639, 0.408] [−0.666, 0.371]	

Notes. Posterior median with 95 % equal-tailed credible intervals (lower triangle; diagonal and upper triangle omitted). **: 95 % CI excludes zero; *: 90 % CI excludes zero. Panel C: $\Delta \mathbf{R} = \mathbf{R}_{S_1} - \mathbf{R}_{S_0}$; credible intervals from posterior draws of the difference.

Table 11: Regime-conditional correlation matrices: Assets block

	Spread	$\Delta \log \text{Eq.}$	$\Delta \log \text{CB Assets}$	sovCISS
<i>Panel A: S_0 (tranquil)</i>				
(2)	−0.285			
	[−0.525, 0.005]			
(3)	−0.070	0.071		
	[−0.359, 0.229]	[−0.233, 0.374]		
(4)	−0.172	0.018	0.341	
	[−0.478, 0.159]	[−0.297, 0.348]	[0.018, 0.614]	
<i>Panel B: S_1 (crisis)</i>				
(2)	−0.408			
	[−0.674, −0.056]			
(3)	0.461	−0.506		
	[0.129, 0.711]	[−0.746, −0.124]		
(4)	0.464	−0.459	0.359	
	[0.141, 0.705]	[−0.725, −0.117]	[0.012, 0.646]	
<i>Panel C: $\Delta \mathbf{R}$</i>				
(2)	−0.123			
	[−0.525, 0.315]			
(3)	+0.526	−0.572		
	[0.079, 0.915]	[−0.981, −0.059]		
(4)	+0.632	−0.477	+0.013	
	[0.165, 1.026]	[−0.888, −0.008]	[−0.439, 0.488]	

Notes. Posterior median with 95 % equal-tailed credible intervals (lower triangle; diagonal and upper triangle omitted). **: 95 % CI excludes zero; *: 90 % CI excludes zero. Panel C: $\Delta \mathbf{R} = \mathbf{R}_{S_1} - \mathbf{R}_{S_0}$; credible intervals from posterior draws of the difference.

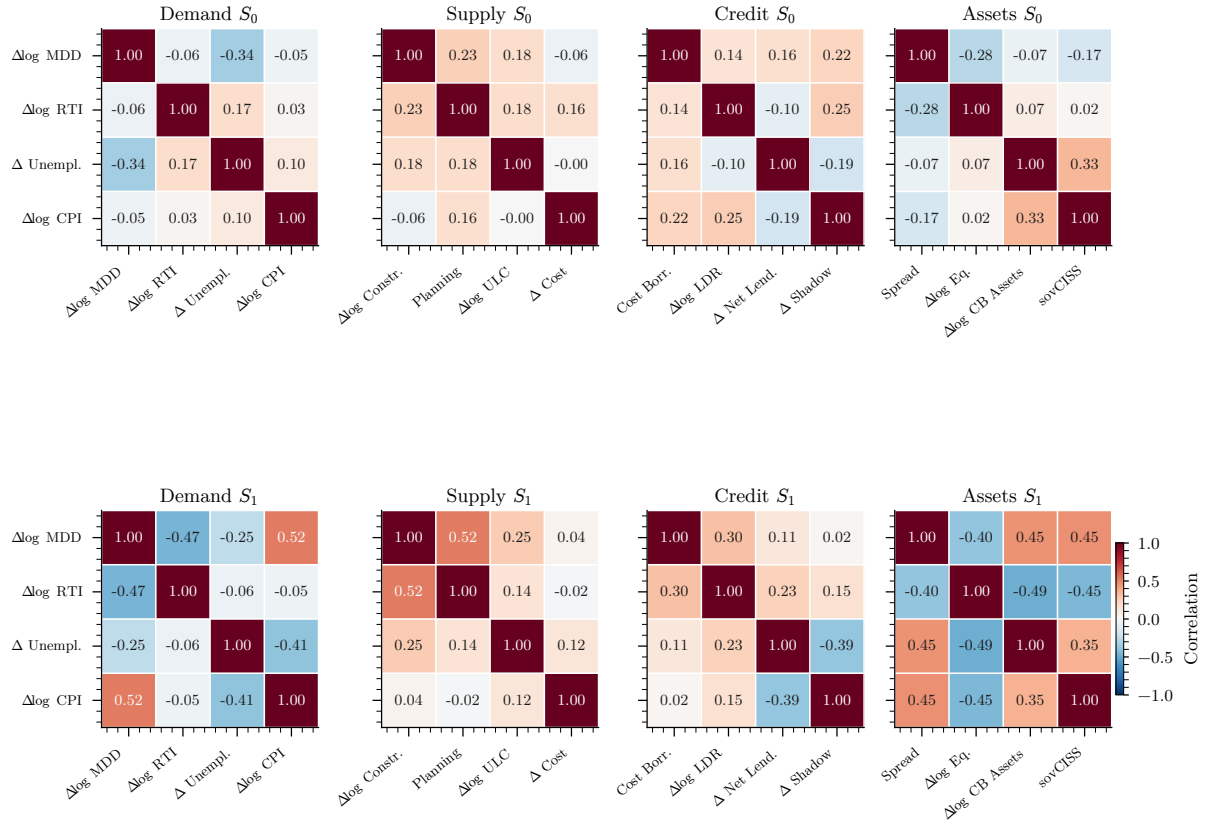


Figure 17: Stage 2 correlation heatmap.

Table 12: Posterior transition matrix and regime statistics: Supply block.

Panel A: Transition matrix $\hat{\mathbf{P}}$

Median		95 % CI	
$\begin{bmatrix} 0.903 & 0.097 \\ 0.315 & 0.685 \end{bmatrix}$		$\begin{bmatrix} [0.759, 0.973] & [0.027, 0.241] \\ [0.087, 0.633] & [0.367, 0.913] \end{bmatrix}$	

Panel B: Ergodic distribution and mixing

	$\bar{\pi}_0$		$\bar{\pi}_1$		Spectral gap		κ	
	Med.	95 % CI	Med.	95 % CI	Med.	95 % CI	Med.	95 % CI
Supply	0.766	[0.409, 0.922]	0.234	[0.078, 0.591]	0.422	[0.164, 0.768]	1.451	[0.291, 3.271]

Notes. Posterior medians with 95 % equal-tailed credible intervals in brackets. Panel A: left matrix = posterior medians; right matrix = 95 % CIs; rows index s_{t-1} , columns s_t . Panel B: $\bar{\pi}_k$ = ergodic probability; spectral gap = $1 - |\lambda_2(\hat{\mathbf{P}})|$; κ = softmax persistence concentration.

A.5 Volatility and Correlation

Table 13: Posterior transition matrix and regime statistics: Credit block.

Panel A: Transition matrix $\hat{\mathbf{P}}$

Median		95 % CI	
$\begin{bmatrix} 0.942 & 0.058 \\ 0.357 & 0.643 \end{bmatrix}$		$\begin{bmatrix} [0.828, 0.993] & [0.007, 0.172] \\ [0.039, 0.712] & [0.288, 0.961] \end{bmatrix}$	

Panel B: Ergodic distribution and mixing

	$\bar{\pi}_0$		$\bar{\pi}_1$		Spectral gap		κ	
	Med.	95 % CI	Med.	95 % CI	Med.	95 % CI	Med.	95 % CI
Credit	0.851	[0.587, 0.961]	0.149	[0.039, 0.413]	0.427	[0.056, 0.814]	1.593	[0.313, 3.871]

Notes. Posterior medians with 95 % equal-tailed credible intervals in brackets. *Panel A:* left matrix = posterior medians; right matrix = 95 % CIs; rows index s_{t-1} , columns s_t . *Panel B:* $\bar{\pi}_k$ = ergodic probability; spectral gap = $1 - |\lambda_2(\hat{\mathbf{P}})|$; κ = softmax persistence concentration.

Table 14: Posterior transition matrix and regime statistics: RPPI block.

Panel A: Transition matrix $\hat{\mathbf{P}}$

Median		95 % CI	
$\begin{bmatrix} 0.660 & 0.340 \\ 0.273 & 0.727 \end{bmatrix}$	$\begin{bmatrix} [0.239, 0.985] & [0.015, 0.761] \\ [0.026, 0.858] & [0.142, 0.974] \end{bmatrix}$		

Panel B: Ergodic distribution and mixing

	$\bar{\pi}_0$		$\bar{\pi}_1$		Spectral gap		κ	
	Med.	95 % CI	Med.	95 % CI	Med.	95 % CI	Med.	95 % CI
RPPI	0.476	[0.063, 0.951]	0.524	[0.049, 0.937]	0.648	[0.130, 0.981]	—	—

Notes. Posterior medians with 95 % equal-tailed credible intervals in brackets. *Panel A:* left matrix = posterior medians; right matrix = 95 % CIs; rows index s_{t-1} , columns s_t . *Panel B:* $\bar{\pi}_k$ = ergodic probability; spectral gap = $1 - |\lambda_2(\hat{\mathbf{P}})|$; κ = softmax persistence concentration.

Table 15: Regime-conditional volatility

Block	Variable	$\hat{\sigma}_{S_0}$	$\hat{\sigma}_{S_1}$	$\hat{\sigma}_{S_1}/\hat{\sigma}_{S_0}$	$\Pr(>1)$
Demand	$\Delta \log$ MDD	0.701 (0.063)	1.506 (0.343)	2.15 (0.53)	1.00**
	$\Delta \log$ RTI	0.655 (0.060)	1.697 (0.431)	2.60 (0.70)	1.00**
	Δ Unempl.	0.579 (0.056)	1.575 (0.407)	2.72 (0.76)	1.00**
	$\Delta \log$ CPI	0.648 (0.059)	1.259 (0.291)	1.95 (0.49)	1.00**
Supply	$\Delta \log$ Constr. Vol.	0.586 (0.085)	1.283 (0.523)	2.33 (0.93)	1.00**
	Planning Perms.	0.654 (0.074)	1.188 (0.411)	1.79 (0.69)	1.00**
	$\Delta \log$ ULC	0.682 (0.070)	1.151 (0.426)	1.68 (0.64)	1.00**
	Δ Cons. Cost	0.329 (0.075)	1.642 (0.585)	5.09 (1.41)	1.00**
Credit	Cost Borr.	0.442 (0.056)	1.350 (0.524)	3.14 (1.31)	1.00**
	$\Delta \log$ LDR	0.887 (0.083)	1.158 (0.287)	1.29 (0.37)	1.00**
	Δ Net Lend.	0.717 (0.069)	0.929 (0.212)	1.28 (0.31)	1.00**
	Δ Shadow	0.511 (0.058)	1.679 (0.518)	3.31 (1.04)	1.00**
Assets	US–IE Spread	0.371 (0.047)	0.975 (0.188)	2.64 (0.59)	1.00**
	$\Delta \log$ –Share Price.	0.653 (0.072)	0.839 (0.143)	1.27 (0.24)	1.00**
	$\Delta \log$ CB Assets	0.591 (0.069)	0.870 (0.153)	1.47 (0.31)	1.00**
	sovCISS	0.221 (0.055)	1.058 (0.185)	4.76 (1.17)	1.00**
RPPI	$\Delta \log$ RPPI	0.475 (0.045)	0.680 (0.066)	1.41 (0.14)	1.00**

Notes. Posterior medians; posterior standard deviations in parentheses. Ratio = $\hat{\sigma}_{S_1}/\hat{\sigma}_{S_0}$; hard ordering $\sigma_{S_0} \leq \sigma_{S_1}$ enforced via a sorted-exponential reparameterisation. $\Pr(>1)$: posterior probability that the ratio exceeds unity. **: $\Pr(>1) > 0.90$; *: $\Pr(>1) > 0.75$.

A.6 Demand Transmission Coefficients

Table 16: Demand transmission coefficients: Supply block.

<i>Panel A: S_0 (Tranquil)</i>				
	$\Delta \log \text{ Con.}$	Planning	$\Delta \log \text{ ULC}$	$\Delta \text{ Cost}$
$\Delta \log \text{ MDD}$	0.021 [−0.073, 0.132] (0.69)	−0.006 [−0.123, 0.095] (0.45)	−0.066 [−0.223, 0.028] (0.11)	−0.005 [−0.093, 0.074] (0.45)
$\Delta \log \text{ RTI}$	−0.007 [−0.119, 0.089] (0.43)	−0.028 [−0.161, 0.065] (0.27)	−0.060 [−0.215, 0.034] (0.13)	−0.015 [−0.105, 0.063] (0.34)
$\Delta \text{ Unempl.}$	−0.043 [−0.168, 0.049] (0.18)	−0.021 [−0.143, 0.077] (0.32)	0.003 [−0.110, 0.111] (0.53)	−0.015 [−0.108, 0.062] (0.34)
$\Delta \log \text{ CPI}$	−0.010 [−0.119, 0.089] (0.41)	−0.000 [−0.110, 0.106] (0.50)	0.005 [−0.102, 0.117] (0.55)	0.039 [−0.036, 0.139] (0.84)
<i>Panel B: S_1 (Crisis)</i>				
	$\Delta \log \text{ Con.}$	Planning	$\Delta \log \text{ ULC}$	$\Delta \text{ Cost}$
$\Delta \log \text{ MDD}$	0.033 [−0.069, 0.188] (0.74)	0.014 [−0.092, 0.153] (0.62)	−0.006 [−0.134, 0.108] (0.44)	0.001 [−0.121, 0.127] (0.51)
$\Delta \log \text{ RTI}$	−0.033 [−0.199, 0.069] (0.26)	−0.011 [−0.146, 0.098] (0.41)	−0.025 [−0.174, 0.081] (0.31)	0.001 [−0.124, 0.125] (0.51)
$\Delta \text{ Unempl.}$	−0.008 [−0.134, 0.107] (0.43)	−0.001 [−0.122, 0.114] (0.49)	−0.022 [−0.162, 0.082] (0.32)	0.005 [−0.111, 0.136] (0.55)
$\Delta \log \text{ CPI}$	0.010 [−0.102, 0.136] (0.59)	0.012 [−0.095, 0.142] (0.60)	0.003 [−0.114, 0.123] (0.53)	0.010 [−0.103, 0.143] (0.59)

Notes. Contemporaneous demand transmission coefficients $\mathbf{B}_k^b$, Stage 3 MS-VARX. Each cell: posterior median (top), 95 % CI (middle), $\Pr(> 0)$ (bottom). *: $\Pr(> 0) > 0.90$ or < 0.10 .

Table 17: Demand transmission coefficients: Credit block.

<i>Panel A: S_0 (Tranquil)</i>				
	Cost Borr.	$\Delta \log \text{LDR}$	$\Delta \text{NetLend.}$	ΔShadow
$\Delta \log \text{MDD}$	0.004 [−0.032, 0.077] (0.64)	0.000 [−0.059, 0.056] (0.50)	−0.003 [−0.076, 0.039] (0.40)	−0.000 [−0.054, 0.053] (0.48)
$\Delta \log \text{RTI}$	0.000 [−0.047, 0.057] (0.52)	−0.000 [−0.063, 0.052] (0.48)	0.004 [−0.035, 0.083] (0.63)	0.002 [−0.041, 0.068] (0.57)
$\Delta \text{Unempl.}$	−0.001 [−0.062, 0.046] (0.46)	0.001 [−0.052, 0.064] (0.53)	−0.000 [−0.062, 0.052] (0.48)	0.001 [−0.049, 0.061] (0.55)
$\Delta \log \text{CPI}$	0.005 [−0.030, 0.085] (0.66)	0.002 [−0.044, 0.074] (0.58)	0.001 [−0.048, 0.066] (0.55)	0.002 [−0.041, 0.070] (0.58)
<i>Panel B: S_1 (Crisis)</i>				
	Cost Borr.	$\Delta \log \text{LDR}$	$\Delta \text{NetLend.}$	ΔShadow
$\Delta \log \text{MDD}$	0.001 [−0.051, 0.067] (0.54)	0.000 [−0.055, 0.063] (0.52)	−0.001 [−0.069, 0.047] (0.44)	0.000 [−0.057, 0.061] (0.51)
$\Delta \log \text{RTI}$	0.001 [−0.052, 0.067] (0.54)	−0.002 [−0.075, 0.046] (0.43)	0.004 [−0.037, 0.088] (0.63)	0.001 [−0.050, 0.070] (0.55)
$\Delta \text{Unempl.}$	−0.002 [−0.077, 0.044] (0.42)	0.001 [−0.047, 0.071] (0.56)	0.001 [−0.052, 0.063] (0.53)	−0.000 [−0.063, 0.056] (0.49)
$\Delta \log \text{CPI}$	0.001 [−0.049, 0.070] (0.55)	0.000 [−0.056, 0.062] (0.51)	−0.000 [−0.063, 0.054] (0.47)	0.002 [−0.045, 0.076] (0.57)

Notes. Contemporaneous demand transmission coefficients $\mathbf{B}_k^b$, Stage 3 MS-VARX. Each cell: posterior median (top), 95 % CI (middle), $\Pr(> 0)$ (bottom). *: $\Pr(> 0) > 0.90$ or < 0.10 .

Table 18: Demand transmission coefficients: Assets block.*Panel A: S_0 (Tranquil)*

	Δ Spread	$\Delta \log$ Eq.	$\Delta \log$ CB Assets	sovCISS
$\Delta \log$ MDD	0.001 [−0.029, 0.046] (0.57)	−0.000 [−0.041, 0.038] (0.49)	0.000 [−0.037, 0.041] (0.51)	−0.001 [−0.043, 0.026] (0.43)
$\Delta \log$ RTI	0.000 [−0.033, 0.042] (0.53)	−0.002 [−0.060, 0.024] (0.37)	0.000 [−0.037, 0.041] (0.52)	0.001 [−0.027, 0.040] (0.57)
Δ Unempl.	−0.001 [−0.043, 0.034] (0.46)	−0.001 [−0.048, 0.033] (0.45)	0.002 [−0.028, 0.055] (0.60)	−0.002 [−0.048, 0.026] (0.40)
$\Delta \log$ CPI	0.000 [−0.035, 0.039] (0.52)	−0.000 [−0.042, 0.037] (0.48)	0.001 [−0.033, 0.046] (0.54)	0.001 [−0.026, 0.042] (0.59)

Panel B: S_1 (Crisis)

	Δ Spread	$\Delta \log$ Eq.	$\Delta \log$ CB Assets	sovCISS
$\Delta \log$ MDD	−0.000 [−0.045, 0.035] (0.46)	0.002 [−0.024, 0.061] (0.63)	0.001 [−0.031, 0.049] (0.57)	0.000 [−0.036, 0.043] (0.53)
$\Delta \log$ RTI	0.001 [−0.034, 0.046] (0.54)	−0.002 [−0.058, 0.025] (0.38)	−0.000 [−0.044, 0.035] (0.46)	−0.001 [−0.050, 0.032] (0.44)
Δ Unempl.	−0.000 [−0.044, 0.036] (0.47)	−0.000 [−0.040, 0.039] (0.50)	0.001 [−0.028, 0.053] (0.59)	0.000 [−0.037, 0.043] (0.52)
$\Delta \log$ CPI	−0.000 [−0.041, 0.039] (0.49)	−0.000 [−0.044, 0.035] (0.47)	−0.000 [−0.044, 0.035] (0.47)	0.000 [−0.035, 0.046] (0.54)

Notes. Contemporaneous demand transmission coefficients $\mathbf{B}_k^b$, Stage 3 MS-VARX. Each cell: posterior median (top), 95 % CI (middle), $\Pr(> 0)$ (bottom). *: $\Pr(> 0) > 0.90$ or < 0.10 .

Table 19: Demand transmission coefficients: RPPI block.

	S_0 : Tranquil	S_1 : Crisis
	$\Delta \log \text{RPPI}$	$\Delta \log \text{RPPI}$
$\Delta \log \text{MDD}$	0.043 [−0.054, 0.200] (0.79)	0.024 [−0.094, 0.173] (0.69)
$\Delta \log \text{RTI}$	0.003 [−0.130, 0.130] (0.53)	0.003 [−0.117, 0.136] (0.53)
$\Delta \text{Unempl.}$	−0.010 [−0.147, 0.122] (0.41)	−0.063 [−0.250, 0.044] (0.14)
$\Delta \log \text{CPI}$	−0.006 [−0.142, 0.111] (0.44)	0.046 [−0.068, 0.219] (0.79)

Notes. Contemporaneous demand transmission coefficients $\mathbf{B}_k^b$, Stage 3 MS-VARX. Each cell: posterior median (top), 95 % CI (middle), $\Pr(> 0)$ (bottom). *: $\Pr(> 0) > 0.90$ or < 0.10 .

B. The JA Prior

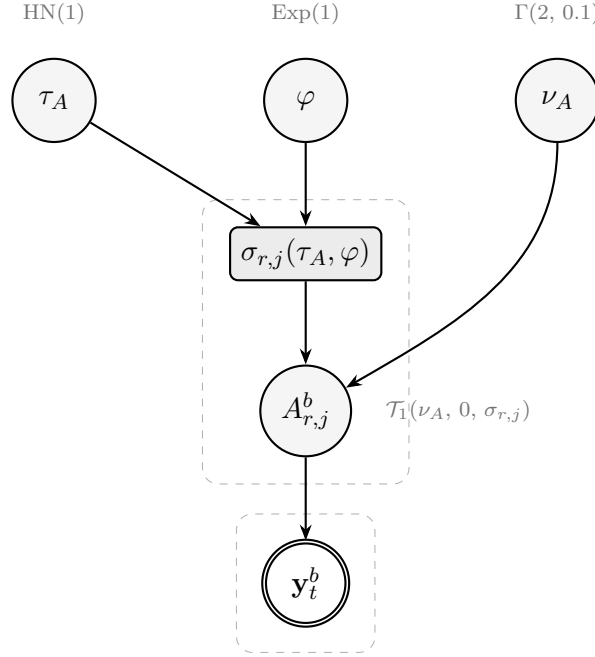


Figure 18: Plate diagram for the $\mathfrak{JA}$ prior. Circles denote stochastic nodes; the double circle denotes observed data; the rectangle denotes a deterministic transformation. Hyperparameters τ_A , φ , and ν_A are estimated jointly with the VAR coefficients via NUTS. The scale $\sigma_{r,j}(\tau_A, \varphi) = \tau_A \ell(r)^{-\varphi}$ encodes lag-decay, and ν_A governs tail thickness. As $\nu_A \rightarrow \infty$ and $\varphi \rightarrow 1$ the prior collapses to the Minnesota prior (Lemma B.3, Proposition B.4).

Remark B.1 (Spurious Regime Suppression). The heavy-tailed Student- t coefficient prior also suppresses weakly identified or spurious regimes. When the data provide little evidence for a genuinely distinct dynamic structure, the prior tends to shrink the associated regime-specific coefficient deviations toward zero, making that regime increasingly similar to the baseline dynamics. In practice, this reduces the model’s tendency to retain unnecessary regimes merely because additional flexibility is available. Unlike hard restrictions on the transition matrix, this mechanism is posterior-driven and does not require the researcher to pre-specify which regimes should be active.

Remark B.2 (Generated-Regressor Bias). Using the posterior-mean smoothed demand-regime probabilities from Stage 2 as fixed inputs in Stage 3 would induce generated-regressor bias (Pagan, 1984). This would suppress uncertainty in $\mathbf{P}$ and κ , understate posterior variance in all Stage 3 parameters, and invalidate inference on impulse responses. The reparameterisation in (66) –(73) avoids this by introducing $(\gamma_Z, \gamma_\kappa)$ as auxiliary

parameters sampled within the NUTS posterior, so that the Hamilton filter is re-evaluated at each leapfrog iteration of $(\mathbf{P}^*, \kappa^*)$.

The integral representation and Jacobian are provided in Appendix B.5

The explicit multivariate Student- t log-density is given by

$$e_{t,k}^b = \log \Gamma \left(\frac{\nu^b + n_b}{2} \right) - \log \Gamma \left(\frac{\nu^b}{2} \right) - \frac{n_b}{2} \log (\nu^b \pi) - \sum_{i=1}^{n_b} \log [\mathbf{L}_{\Omega,k}^b]_{ii} \\ - \frac{\nu^b + n_b}{2} \log \left(1 + \frac{(\mathbf{y}_t^b - \mathbf{A}_k^b \mathbf{x}_t^b)' (\mathbf{L}_{\Omega,k}^b \mathbf{L}_{\Omega,k}^{b'})^{-1} (\mathbf{y}_t^b - \mathbf{A}_k^b \mathbf{x}_t^b)}{\nu^b} \right). \quad (45)$$

$$\sigma_{r,j}(\tau_A, \varphi) = \begin{cases} 10 & r = 0, \\ \tau_A \left\lceil \frac{r}{n_b} \right\rceil & \end{cases} \quad \tau_A \sim \text{HN}(1), \quad \varphi \sim \text{Exp}(1). \quad (46)$$

$$\log p(\mathbf{A}^b) = \sum_{r=0}^{n_q-1} \sum_{j=1}^{n_b} \log \Gamma \left(\frac{\nu_A + 1}{2} \right) - \log \Gamma \left(\frac{\nu_A}{2} \right) - \frac{1}{2} \log (\nu_A \pi) - \log \sigma_{r,j}(\tau_A, \varphi) \\ - \frac{\nu_A + 1}{2} \log \left(1 + \frac{(A_{r,j}^b)^2}{\nu_A [\sigma_{r,j}(\tau_A, \varphi)]^2} \right). \quad (47)$$

B.1 Multivariate Student- t Emission Density

For a random vector $\mathbf{y} \in \mathbb{R}^n$, the multivariate Student- t log-density with degrees of freedom $\nu > 0$, location $\boldsymbol{\mu} \in \mathbb{R}^n$, and lower-triangular Cholesky factor $\mathbf{L}_{\Omega} \in \mathbb{R}^{n \times n}$ is

$$\log f_{\mathcal{T}_n}(\mathbf{y} \mid \nu, \boldsymbol{\mu}, \mathbf{L}_{\Omega}) = \log \Gamma \left(\frac{\nu + n}{2} \right) - \log \Gamma \left(\frac{\nu}{2} \right) - \frac{n}{2} \log (\nu \pi) - \sum_{i=1}^n \log [\mathbf{L}_{\Omega}]_{ii} \\ - \frac{\nu + n}{2} \log \left(1 + \frac{(\mathbf{y} - \boldsymbol{\mu})' (\mathbf{L}_{\Omega} \mathbf{L}_{\Omega}')^{-1} (\mathbf{y} - \boldsymbol{\mu})}{\nu} \right). \quad (48)$$

where $[\mathbf{L}_{\Omega}]_{ii}$ denotes the i -th diagonal entry. The Mahalanobis-type quadratic form in (48) is evaluated by forward substitution:

$$(\mathbf{y} - \boldsymbol{\mu})'(\mathbf{L}_\Omega \mathbf{L}_\Omega')^{-1}(\mathbf{y} - \boldsymbol{\mu}) = \left\| \mathbf{L}_\Omega^{-1}(\mathbf{y} - \boldsymbol{\mu}) \right\|_2^2 = \|\mathbf{v}\|_2^2, \quad \mathbf{L}_\Omega \mathbf{v} = \mathbf{y} - \boldsymbol{\mu}. \quad (49)$$

As $\nu \rightarrow \infty$, (48) converges to the multivariate Gaussian log-density. The univariate specialisation ($n = 1$, $L_\Omega = \sigma$) recovers the scalar Student- t used in the RPPI MS-ARX and the JA coefficient prior.

B.2 Explicit Form of the kappa-Inflated Transition Matrix

For $K = 2$, substituting (27)–(29) yields the explicit κ -inflated row-normalised exponential representation:

$$P_{ij} = \frac{\exp([\mathbf{Z}]_{ij} + \kappa \delta_{ij})}{\sum_{m=1}^K \exp([\mathbf{Z}]_{im} + \kappa \delta_{im})}, \quad (50)$$

where δ_{ij} denotes the Kronecker delta ($\delta_{ij} = 1$ if $i = j$, zero otherwise). For $K = 2$, this yields the explicit form:

$$\mathbf{P} = \begin{pmatrix} \frac{e^{\zeta_{11} + \kappa}}{e^{\zeta_{11} + \kappa} + e^{\zeta_{12}}} & \frac{e^{\zeta_{12}}}{e^{\zeta_{11} + \kappa} + e^{\zeta_{12}}} \\ \frac{e^{\zeta_{21}}}{e^{\zeta_{21}} + e^{\zeta_{22} + \kappa}} & \frac{e^{\zeta_{22} + \kappa}}{e^{\zeta_{21}} + e^{\zeta_{22} + \kappa}} \end{pmatrix}, \quad (51)$$

such that as $\kappa \rightarrow \infty$, $p_{ii} \rightarrow 1$ for both regimes and the chain becomes absorbing. As $\kappa \rightarrow 0$, (51) reduces to an unconstrained row-normalised exponential with no persistence prior. The posterior mean $\bar{\kappa} \approx 2$ across all blocks implies moderate persistence boosting, consistent with the observed regime durations reported in Table 2.

B.3 Implied Volatility Scale Under the JA Prior

The regime-specific volatility vector $\boldsymbol{\sigma}_k^b$ is computed using the method described in Section 2.6.2. For variable j in block b , the implied marginal prior on $\sigma_{j,k}^b$ is log-normal. In particular,

$$\log \sigma_{j,k}^b = \log \hat{\sigma}_j^b + \tau_\sigma u_{j,(k+1)}, \quad k \in \{0, 1\}, \quad (52)$$

which implies

$$\sigma_{j,k}^b \mid \tau_\sigma \sim \text{LogNormal}(\log \hat{\sigma}_j^b, \tau_\sigma^2), \quad (53)$$

where $\hat{\sigma}_j^b$ is the Stage 1 posterior mean residual standard deviation serving as a centring point. The prior mean and variance of $\sigma_{j,k}^b$ are then given by

$$\mathbb{E}[\sigma_{j,k}^b \mid \tau_\sigma] = \hat{\sigma}_j^b e^{\tau_\sigma^2/2}, \quad (54)$$

$$\text{Var}[\sigma_{j,k}^b \mid \tau_\sigma] = (\hat{\sigma}_j^b)^2 e^{\tau_\sigma^2} (e^{\tau_\sigma^2} - 1). \quad (55)$$

B.4 Minnesota Nesting and Tikhonov Correspondence

Lemma B.3 (Student- t to Gaussian Convergence). *As $\nu_A \rightarrow \infty$, the univariate Student- t density $\mathcal{T}_1(A_{r,j}^b \mid \nu_A, 0, \sigma_{r,j})$ converges pointwise to $\mathcal{N}(A_{r,j}^b \mid 0, \sigma_{r,j}^2)$.*

Proof. The Student- t density is

$$p(A_{r,j}^b \mid \nu_A, \sigma_{r,j}) = \frac{\Gamma(\frac{\nu_A+1}{2})}{\Gamma(\frac{\nu_A}{2}) \sqrt{\pi} \nu_A \sigma_{r,j}} \left(1 + \frac{1}{\nu_A} \frac{(A_{r,j}^b)^2}{\sigma_{r,j}^2} \right)^{-(\nu_A+1)/2}. \quad (56)$$

Using the standard limit $(1 + u/\nu_A)^{-(\nu_A+1)/2} \rightarrow \exp(-u/2)$ as $\nu_A \rightarrow \infty$ with $u = (A_{r,j}^b)^2/\sigma_{r,j}^2$, together with Stirling's approximation for the normalising constant, the density converges pointwise to

$$\lim_{\nu_A \rightarrow \infty} p(A_{r,j}^b \mid \nu_A, \sigma_{r,j}) = \frac{1}{\sqrt{2\pi} \sigma_{r,j}} \exp\left(-\frac{(A_{r,j}^b)^2}{2\sigma_{r,j}^2}\right), \quad (57)$$

which is the $\mathcal{N}(0, \sigma_{r,j}^2)$ density. □

Proposition B.4 (Minnesota Nesting). *As $\nu_A \rightarrow \infty$ and $\varphi \rightarrow 1$, the $\mathfrak{JA}$ prior on the lag coefficients converges to the Minnesota prior of [Litterman \(1986\)](#) with harmonic lag decay and global scale τ_A .*

Proof. By Lemma B.3, as $\nu_A \rightarrow \infty$ the Student- t marginal converges pointwise to a Gaussian. Taking additionally $\varphi \rightarrow 1$ in the scale specification (8d) yields $\sigma_{r,j} = \tau_A/\ell(r)$,

so that

$$\lim_{\substack{\nu_A \rightarrow \infty \\ \varphi \rightarrow 1}} \underbrace{\mathcal{T}_1 \left(A_{r,j}^b \mid \nu_A, 0, \tau_A \ell(r)^{-\varphi} \right)}_{\mathfrak{J}\mathfrak{A} \text{ prior}} = \underbrace{\mathcal{N} \left(A_{r,j}^b \mid 0, \frac{\tau_A^2}{\ell(r)^2} \right)}_{\text{Minnesota prior (Litterman, 1986)}}, \quad (58)$$

with intercept scale $\sigma_{0,j} = 10$ retained as the diffuse element, matching the standard Minnesota treatment of the constant term. This establishes the distributional limit; the corresponding modal (Tikhonov-ridge) correspondence is derived below.

The posterior mode under the $\mathfrak{J}\mathfrak{A}$ prior satisfies the element-wise first-order condition:

$$\frac{\partial \mathcal{L}}{\partial A_{r,j}} - \frac{\partial}{\partial A_{r,j}} \left[\frac{A_{r,j}^2}{\sigma_{r,j}^2} \left(1 + \frac{A_{r,j}^2}{\nu_A \sigma_{r,j}^2} \right)^{-1} \right] = 0. \quad (59)$$

Evaluating the penalty derivative gives

$$\frac{\partial}{\partial A_{r,j}} \left[\frac{A_{r,j}^2}{\sigma_{r,j}^2} \left(1 + \frac{A_{r,j}^2}{\nu_A \sigma_{r,j}^2} \right)^{-1} \right] = \frac{2A_{r,j}/\sigma_{r,j}^2}{\left(1 + A_{r,j}^2/(\nu_A \sigma_{r,j}^2) \right)^2}. \quad (60)$$

As $\nu_A \rightarrow \infty$, the denominator in (60) converges to unity and the first-order condition reduces to

$$\frac{\partial \mathcal{L}}{\partial A_{r,j}} - \frac{2A_{r,j}}{\sigma_{r,j}^2} = 0, \quad (61)$$

which corresponds to a $\mathcal{N}(0, \sigma_{r,j}^2)$ prior. Setting $\lambda = 1$ then gives $\sigma_{r,j} = \tau_A/\ell(r)$, recovering the harmonic lag-decay structure of the Minnesota prior with tightness τ_A . The intercept receives a diffuse scale $\sigma_{0,j} = 10$, consistent with standard Minnesota prior practices. $\square$

Proposition B.5 (Tikhonov Modal Correspondence). *The posterior mode of $\mathbf{A}^b$ under the $\mathfrak{J}\mathfrak{A}$ prior satisfies a generalised Tikhonov regularisation problem. For fixed (τ_A, φ, ν_A) , the mode solves:*

$$\hat{\mathbf{A}}^b = \arg \min_{\mathbf{A}} \left\{ \text{tr} \left[(\mathbf{Y} - \mathbf{X} \mathbf{A}) \boldsymbol{\Omega}^{(b)-1} (\mathbf{Y} - \mathbf{X} \mathbf{A})' \right] + \sum_{r,j} \frac{A_{r,j}^2}{\sigma_{r,j}^2} \left(1 + \frac{A_{r,j}^2}{\nu_A \sigma_{r,j}^2} \right)^{-1} \right\}. \quad (62)$$

The Student- t prior induces a non-quadratic shrinkage penalty that converges to

the standard ℓ_2 (ridge) penalty as $\nu_A \rightarrow \infty$, recovering the Gaussian prior case and a conventional Tikhonov regularisation problem.

For finite ν_A , the penalty becomes progressively more tolerant of large coefficient deviations as ν_A decreases, accommodating the heavy-tailed dynamics of crisis episodes. The degree of tail heaviness is itself estimated from the data through the posterior of ν_A .

Proposition B.6 (Huber-type Robustness). *The shrinkage penalty induced by the $\mathfrak{JA}$ prior has a bounded influence function. For any coefficient $A_{r,j}$, the marginal contribution to the penalty in (62) satisfies*

$$\lim_{A_{r,j} \rightarrow \infty} \psi(A_{r,j}) \equiv \frac{\partial}{\partial A_{r,j}} \left[\frac{A_{r,j}^2}{\sigma_{r,j}^2} \left(1 + \frac{A_{r,j}^2}{\nu_A \sigma_{r,j}^2} \right)^{-1} \right] \equiv 0 \quad (63)$$

The effective-shrinkage force on large coefficients thus diminishes rather than grows without bound. This mirrors Huber's M -estimator: the penalty transitions from a quadratic regime near the origin to a plateau for large deviations, preventing the prior from excessively penalising coefficients that represent genuine structural breaks or crisis-period dynamics. The threshold between the two regimes is determined by ν_A , which is estimated from the data.

B.5 Generated-Regressor Bias and Reparameterisation

The demand block is summarised by a moment-matched handoff from Stage 2 to Stage 3. The posterior distribution over $(\mathbf{Z}, \log \kappa)$ is approximated by its first two moments, computed from the S MCMC draws $\{(\mathbf{Z}^{(s)}, \kappa^{(s)})\}_{s=1}^S$:

$$\boldsymbol{\mu}_Z = \frac{1}{S} \sum_{s=1}^S \mathbf{Z}^{(s)}, \quad \boldsymbol{\sigma}_Z = \left(\frac{1}{S-1} \sum_{s=1}^S (\mathbf{Z}^{(s)} - \boldsymbol{\mu}_Z)^{\odot 2} \right)^{\odot 1/2}, \quad (64)$$

$$\mu_\kappa = \frac{1}{S} \sum_{s=1}^S \log \kappa^{(s)}, \quad \sigma_\kappa = \left(\frac{1}{S-1} \sum_{s=1}^S (\log \kappa^{(s)} - \mu_\kappa)^2 \right)^{1/2}. \quad (65)$$

Here $\odot$ denotes element-wise (Hadamard) operations; $(\cdot)^{\odot 2}$ and $(\cdot)^{\odot 1/2}$ denote element-wise squaring and element-wise square-root respectively. The posterior of $\mathbf{Z}$ and $\log \kappa$ is

well-approximated as Gaussian, so moment-matching captures the key features of the distribution, including posterior uncertainty in the transition process.

Stage 2.1 then precomputes regime-conditional emission log-probabilities for both the demand block and each downstream block using posterior parameter values, and moment-matched regime-specific log-volatilities and log-degrees-of-freedom. These objects enter Stage 3 through the combined Hamilton filter, which computes the smoothed probabilities $\pi_{D,t}^*$ of the latent demand regime process without conditioning on a fixed smoothed regime path.

Treating the posterior-mean smoothed probabilities $\bar{\rho}_D$ as fixed inputs in Stage 3 would induce generated-regressor bias (Pagan, 1984), suppressing uncertainty in $\mathbf{P}$ and κ and understating the posterior variance in all Stage 3 parameters, which would invalidate inference on impulse responses.

The reparameterisation introduces $(\gamma_Z, \gamma_\kappa)$ as auxiliary parameters within the NUTS posterior; at each leapfrog step a draw of $(\mathbf{P}^*, \kappa^*)$ is constructed by

$$\mathbf{Z}^* = \boldsymbol{\mu}_Z + \boldsymbol{\sigma}_Z \odot \boldsymbol{\gamma}_Z, \quad (66)$$

$$\kappa^* = \exp(\mu_\kappa + \sigma_\kappa \gamma_\kappa), \quad (67)$$

$$P_{ij}^* = \frac{\exp(\zeta_{ij}^* + \kappa^* \delta_{ij})}{\sum_{m=1}^K \exp(\zeta_{im}^* + \kappa^* \delta_{im})}, \quad i, j \in \{0, 1\}, \quad (68)$$

$$\mathbf{P}^* = [P_{ij}^*]_{i,j=1}^K,$$

$$\pi^*(i) = \left\{ \boldsymbol{\pi} \in \Delta^{K-1} : \mathbf{P}^{*'} \boldsymbol{\pi} = \boldsymbol{\pi}, \mathbf{1}' \boldsymbol{\pi} = 1 \right\}, \quad (69)$$

and $\mathcal{H}$ is recomputed on the combined log-emission (32).

Denoting the filter state as a ket $|\pi_t\rangle = \sum_{k \in \mathcal{S}} \langle k | \pi_t \rangle |k\rangle$, the transition step is

$$|\pi_t^-\rangle = \mathbf{P}^{*'} |\pi_{t-1}\rangle, \quad (70)$$

and the emission-reweighted update gives

$$\langle k | \pi_0 \rangle = \frac{\pi^*(k) \exp(e_{1,k,b}^\circ)}{\sum_{k' \in \mathcal{S}} \pi^*(k') \exp(e_{1,k',b}^\circ)}, \quad (71)$$

$$\langle k | \pi_t \rangle = \frac{\langle k | \pi_t^- \rangle \exp(e_{t,k,b}^\circ)}{\sum_{k' \in \mathcal{S}} \langle k' | \pi_t^- \rangle \exp(e_{t,k',b}^\circ)}, \quad (72)$$

$$|\pi_t\rangle = \mathcal{H}(\mathbf{P}^*, |e_{t,b}^\circ\rangle) |\pi_{t-1}\rangle \in \mathbb{R}^K, \quad (73)$$

where $|e_{t,b}^\circ\rangle = \sum_{k \in \mathcal{S}} e_{t,k,b}^\circ |k\rangle$ collects the combined log-emissions across regime states.

The filtered probabilities $|\pi_t\rangle$ are then passed to the block likelihood. Marginalising over $(\gamma_Z, \gamma_\kappa)$ is equivalent to integrating out the Stage 2 posterior:

$$p(\mathbf{y}^b | \mathcal{D}) = \int p(\mathbf{y}^b | \boldsymbol{\pi}^*(\mathbf{P}, \kappa), \boldsymbol{\theta}^b) p(\mathbf{P}, \kappa | \mathbf{y}^D) d\mathbf{P} d\kappa. \quad (74)$$

In general, this integral does not admit a closed-form expression and is instead evaluated implicitly through posterior simulation over the auxiliary latent variables $(\gamma_Z, \gamma_\kappa)$.

The Jacobian of the reparameterisation is denoted in vectorised form as

$$\frac{\partial \text{ved}(\mathbf{P}^*, \kappa^*)}{\partial \text{ved}(\gamma_Z, \gamma_\kappa)} = \begin{pmatrix} \text{diag}(\text{ved}(\boldsymbol{\sigma}_Z)) & \mathbf{0} \\ \mathbf{0} & \sigma_\kappa \kappa^* \end{pmatrix}. \quad (75)$$

This reparameterisation ensures that gradients propagate through the Hamilton filter, with no Metropolis–Hastings step and no discrete integration required. At each leapfrog step, the NUTS update is

$$(\gamma_Z^{(\ell+1)}, \gamma_\kappa^{(\ell+1)}) = (\gamma_Z^{(\ell)}, \gamma_\kappa^{(\ell)}) + \epsilon \mathbf{r}^{(\ell)}, \quad \mathbf{r}^{(\ell)} \leftarrow \mathbf{r}^{(\ell)} - \frac{\epsilon}{2} \nabla_{(\gamma_Z, \gamma_\kappa)} U(\gamma_Z^{(\ell)}, \gamma_\kappa^{(\ell)}), \quad (76)$$

where ϵ is the step size, $\mathbf{r}$ is the auxiliary momentum, and $U = -\log p(\gamma_Z, \gamma_\kappa | \mathbf{y})$ is the potential energy.

B.6 Explicit Recursive Form of Propagated Regime Weights

Substituting (66)–(69) into (72)–(73) and normalising yields a recursion for the filtered regime probabilities, evaluated at each leapfrog step:

$$\langle k | \pi_t \rangle = \frac{f_{\mathcal{T}_{n_D}}(\mathbf{y}_t^D \mid \bar{\mathbf{A}}_k^D, \bar{\mathbf{L}}_{\Omega, k}^D, \bar{\nu}^D) \sum_{j \in \mathcal{S}} \frac{e^{[\mu_Z + \sigma_Z \odot \gamma_Z]_{jk} + e^{\mu_\kappa + \sigma_\kappa \gamma_\kappa} \delta_{jk}}}{\sum_{m \in \mathcal{S}} e^{[\mu_Z + \sigma_Z \odot \gamma_Z]_{jm} + e^{\mu_\kappa + \sigma_\kappa \gamma_\kappa} \delta_{jm}}} \langle j | \pi_{t-1} \rangle}{\sum_{k' \in \mathcal{S}} f_{\mathcal{T}_{n_D}}(\mathbf{y}_t^D \mid \bar{\mathbf{A}}^{(D, k')}, \bar{\mathbf{L}}_{\Omega}^{(D, k')}, \bar{\nu}^D) \sum_{j \in \mathcal{S}} \frac{e^{[\mu_Z + \sigma_Z \odot \gamma_Z]_{jk'} + e^{\mu_\kappa + \sigma_\kappa \gamma_\kappa} \delta_{jk'}}}{\sum_{m \in \mathcal{S}} e^{[\mu_Z + \sigma_Z \odot \gamma_Z]_{jm} + e^{\mu_\kappa + \sigma_\kappa \gamma_\kappa} \delta_{jm}}} \langle j | \pi_{t-1} \rangle}, \quad (77)$$

Initialised at $\langle k | \pi_0 \rangle = \pi^*(k)$ the expression is a deterministic, differentiable function of $(\gamma_Z, \gamma_\kappa, \tilde{\alpha}_b)$ alone.

B.7 IRF Stabilisation

Posterior draws are not constrained to lie inside the stationarity region during estimation; a diffuse-tailed prior on the VAR coefficients admits a small fraction of draws with spectral radius $\rho(\mathbf{F}_k^b) \geq 1$. Such draws would generate divergent impulse-response paths under the MA recursion $\Phi_h = (\mathbf{F}_k^b)^h$. To prevent these draws from contaminating the reported IRFs while preserving the rest of the posterior, a per-draw Schur-based stabilisation is applied post-hoc at IRF computation time, leaving the estimated posterior itself unchanged.

For each posterior draw s and each regime k , let $\mathbf{F}_k^{b, (s)}$ denote the companion matrix from the stacked coefficient matrix $\mathbf{A}_k^{b, (s)}$. If $\rho(\mathbf{F}_k^{b, (s)}) < 1$, the matrix passes through unchanged. If $\rho(\mathbf{F}_k^{b, (s)}) \geq 1$, the complex Schur decomposition:

$$\mathbf{F}_k^{b, (s)} = \mathbf{Q} \mathbf{T} \mathbf{Q}^H \quad (78)$$

is formed, where $\mathbf{T}$ is upper triangular with eigenvalues on its diagonal and $\mathbf{Q}$ is unitary. Each diagonal element of $\mathbf{T}$ with $|T_{ii}| \geq 1$ is individually rescaled by $\rho^*/|T_{ii}|$ with $\rho^* = 0.9999$, after which the stabilised companion matrix is reconstructed as

$$\tilde{\mathbf{F}}_k^{b, (s)} = \text{Re}(\mathbf{Q} \tilde{\mathbf{T}} \mathbf{Q}^H). \quad (79)$$

The procedure clips only the explosive eigenvalues and leaves non-explosive eigenvalues untouched, preserving the relative structure of the non-explosive dynamics. The resulting $\tilde{\mathbf{F}}_k^{b,(s)}$ is used in place of $\mathbf{F}_k^{b,(s)}$ in the IRF recursion. The fraction of stabilised draws is reported alongside each IRF and provides a diagnostic for the share of posterior mass in the explosive region.

C. MCMC Implementation

C.1 Hamiltonian Monte Carlo

Hamiltonian Monte Carlo (HMC) augments the target distribution $p(\boldsymbol{\theta} \mid \mathcal{Y})$ with an auxiliary momentum variable $\mathbf{r} \sim \mathcal{N}(\mathbf{0}, \mathbf{M})$, where $\mathbf{M}$ is a mass matrix, and defines a joint Hamiltonian:

$$\mathcal{H}_{\text{HMC}}(\boldsymbol{\theta}, \mathbf{r}) = \underbrace{-\log p(\boldsymbol{\theta} \mid \mathcal{Y})}_{\text{potential energy}} + \underbrace{\frac{1}{2}\mathbf{r}'\mathbf{M}^{-1}\mathbf{r}}_{\text{kinetic energy}}. \quad (80)$$

Starting from a current state $(\boldsymbol{\theta}^{(\ell)}, \mathbf{r}^{(\ell)})$, HMC proposes a new state by simulating Hamiltonian dynamics for L leapfrog steps of size ϵ :

$$\mathbf{r}^{(\ell+1/2)} = \mathbf{r}^{(\ell)} - \frac{\epsilon}{2} \nabla_{\boldsymbol{\theta}} U(\boldsymbol{\theta}^{(\ell)}), \quad (81)$$

$$\boldsymbol{\theta}^{(\ell+1)} = \boldsymbol{\theta}^{(\ell)} + \epsilon \mathbf{M}^{-1} \mathbf{r}^{(\ell+1/2)}, \quad (82)$$

$$\mathbf{r}^{(\ell+1)} = \mathbf{r}^{(\ell+1/2)} - \frac{\epsilon}{2} \nabla_{\boldsymbol{\theta}} U(\boldsymbol{\theta}^{(\ell+1)}), \quad (83)$$

where $U(\boldsymbol{\theta}) = -\log p(\boldsymbol{\theta} \mid \mathcal{Y})$ is the potential energy and $\nabla_{\boldsymbol{\theta}} U$ is its gradient. Since the leapfrog integrator is symplectic and reversible, the proposal is accepted with probability: $\min\{1, \exp(-\mathcal{H}_{\text{HMC}}(\boldsymbol{\theta}^{(\ell+1)}, \mathbf{r}^{(\ell+1)}) + \mathcal{H}_{\text{HMC}}(\boldsymbol{\theta}^{(\ell)}, \mathbf{r}^{(\ell)}))\}$, which is close to 1 for small ϵ .

- If ϵ is too large, the leapfrog integrator will produce inaccurate proposals with low acceptance rates.
- If ϵ is too small, the chain will mix slowly due to small moves.
- If L is too small, the chain will make short, random-walk-like moves.

- If L is too large, the chain will retrace its path and waste computation.

The No-U-Turn Sampler addresses these tuning challenges by adaptively determining L at each iteration based on the trajectory’s geometry.

C.2 The No-U-Turn Criterion

The No-U-Turn Sampler (Hoffman and Gelman, 2014) eliminates the need to specify L by terminating the leapfrog trajectory automatically when it begins to turn back on itself. The criterion is:

$$(\boldsymbol{\theta}^+ - \boldsymbol{\theta}^-) \cdot \mathbf{r}^- < 0 \quad \text{or} \quad (\boldsymbol{\theta}^+ - \boldsymbol{\theta}^-) \cdot \mathbf{r}^+ < 0, \quad (84)$$

Here, $\boldsymbol{\theta}^-$ and $\boldsymbol{\theta}^+$ denote the leftmost and rightmost points of the current trajectory, and $\mathbf{r}^-$, $\mathbf{r}^+$ are the corresponding momenta. Once (84) is satisfied and the trajectory has turned around, additional steps would retrace previously explored space.

NUTS therefore builds a balanced binary tree of leapfrog proposals, doubling the trajectory length at each level and stopping once the no-U-turn condition is met; proposals are drawn uniformly from the valid leaves of the tree, ensuring detailed balance without the need for a Metropolis correction at each leapfrog step.

C.3 Dual Averaging for Step-Size Adaptation

The step size ϵ is adapted during warmup using the dual-averaging scheme of Hoffman and Gelman (2014) to target a user-specified acceptance probability δ (set to 0.90 in all estimations reported here). Dual averaging accumulates a running mean of the log step size, weighted to down-weight early iterates, and converges to the ϵ that achieves the target acceptance rate under the current mass matrix. After warmup, ϵ is fixed, and the mass matrix $\mathbf{M}$ is set to a dense estimate of the posterior covariance computed from the warmup draws.

C.4 Application in This Paper

In this paper, NUTS is applied to the continuous parameter vector $\boldsymbol{\theta}^b$ with the discrete regime sequence $\{s_t\}$ analytically marginalised at each leapfrog step via the Hamilton forward filter.

The potential energy gradient $\nabla_{\theta}U$ therefore propagates through the filter recursion, requiring the filter to be implemented as a differentiable computation graph. This is achieved in JAX Bradbury et al. (2018) using automatic differentiation through the log-sum-exp forward pass.

Then during Stage 3, the same leapfrog update is applied to the auxiliary reparameterisation variables $(\gamma_Z, \gamma_{\kappa})$, with gradients propagating through the κ -inflated softmax and the Hamilton filter jointly (see Appendix B.5 and equation (76)).

The Jacobian of the full reparameterisation is block-diagonal (equation (75)), to ensure well-conditioned gradient flow.

D. Convergence Diagnostics

Split- $\hat{R}$ with Rank Normalisation

Let ψ denote a scalar parameter with M chains each of length N .

Step 1: Chain splitting. Split each chain in half to produce $2M$ chains of length $\lfloor N/2 \rfloor$. Denote the resulting draws ψ_{mj} for $m = 1, \dots, 2M$ and $j = 1, \dots, \lfloor N/2 \rfloor$, with $S = 2M\lfloor N/2 \rfloor$ total draws.

Step 2: Rank normalisation. Replace each draw ψ_{mj} with its rank r_{mj} in the pooled sample of all S draws, then apply the Blom (1958) transformation to approximate normality:

$$z_{mj} = \Phi^{-1}\left(\frac{r_{mj} - 3/8}{S + 1/4}\right), \quad (85)$$

where Φ^{-1} denotes the standard normal quantile function.

Step 3: Between- and within-chain variance. Compute the between-chain variance B and within-chain variance W of the rank-normalised draws $\{z_{mj}\}$:

$$B = \frac{S}{2M-1} \sum_{m=1}^{2M} (\bar{z}_{m\cdot} - \bar{z}_{\cdot\cdot})^2, \quad (86)$$

$$W = \frac{1}{2M} \sum_{m=1}^{2M} s_m^2, \quad (87)$$

where $\bar{z}_{m\cdot}$ is the within-chain mean, $\bar{z}_{\cdot\cdot}$ is the grand mean, and s_m^2 is the within-chain sample variance.

Step 4: Marginal variance and $\hat{R}$. The marginal posterior variance is estimated as a weighted average:

$$\widehat{\text{var}}^+(\psi \mid y) = \frac{S-1}{S} W + \frac{1}{S} B. \quad (88)$$

The rank-normalised split- $\hat{R}$ is then

$$\hat{R} = \sqrt{\frac{\widehat{\text{var}}^+(\psi \mid y)}{W}}. \quad (89)$$

D.1 Kullback-Leibler (KL) Divergence

For discrete distributions P and Q over the same support $\mathcal{X}$, the KL divergence from Q to P is defined as

$$D_{\text{KL}}(P \parallel Q) = \sum_{x \in \mathcal{X}} P(x) \log \left(\frac{P(x)}{Q(x)} \right). \quad (90)$$

For continuous distributions with densities p and q over $\mathcal{X}$, it is defined as

$$D_{\text{KL}}(P \parallel Q) = \int_{\mathcal{X}} p(x) \log \left(\frac{p(x)}{q(x)} \right) dx. \quad (91)$$

The KL divergence diagnostic is used to compare the smoothed regime probabilities across chains, with P and Q representing the distributions of smoothed probabilities from different chains.

D.2 Hurst Exponents

Tables 20 and 21 report Hurst exponent estimates for all variables using two methods: detrended fluctuation analysis of order 2 (DFA₂) and rescaled-range (R/S) analysis as a robustness check. Values above 0.5 indicate long memory (persistent dynamics); values below 0.5 indicate mean reversion; values near 0.5 are consistent with a random walk. The estimates corroborate the regime structure identified by the MS-VAR and provide independent evidence for several of the economic mechanisms discussed in Section 4.

Financial stress variables carry the strongest long memory in the dataset.

The sovCISS series has DFA₂ $H = 0.545$ (full sample) and R/S $H = 0.933$ (95% CI: [0.865, 1.001]), the highest of any variable. The sovereign spread has R/S $H = 0.777$. Long memory in these series means that shocks to financial stress decay slowly and that past stress is informative about future stress well beyond what a standard AR process would predict. This is the long-memory signature of the financial accelerator: the 28-quarter assets S_1 episode is not merely high persistence within a Markov regime but reflects a genuine memory structure in the underlying variables. The assets block's spectral gap of 0.195 and the Hurst evidence are therefore complementary: the regime model identifies the state, and the Hurst analysis identifies why that state is difficult to exit.

RPPI has negative or near-zero Hurst estimates, confirming mean reversion at business-cycle frequencies. The wavelet estimator for $\Delta \log \text{RPPI}$ gives $H = -0.173$ for the full sample (95% CI: [-1.413, 1.066]), with the wide interval reflecting the short and diffuse S_1 episodes that limit the precision of the estimate. The low-volatility sample yields $H = 0.254$, also indicating mean reversion. Mean reversion at these frequencies implies that house price growth tends to self-correct, but the timing of the correction is not observable from the price series itself because the RPPI transition matrix is nearly uninformative. A buyer who knows prices are mean-reverting but cannot identify where in the cycle they are faces the worst-case information problem: they know a correction will eventually come, but cannot condition on when. This is the information environment referred to in Section 4, and it is consistent with the candidate behavioural interpretation discussed there, though the Hurst estimates themselves do not identify any particular

demand-side mechanism.

Construction costs show moderate positive memory. Δ Construction Cost has DFA₂ $H = 0.247$ (full sample) and R/S $H = 0.663$, with the S_0 R/S estimate of 0.613 suggesting that the persistence partially reflects the high-volatility regime rather than structural long memory. This is consistent with costs behaving more like a financial asset under S_1 , as argued in Section 4: elevated memory in stress periods means that a cost spike today is informative about costs several quarters ahead, raising the real option value of waiting for developers.

Rent-to-income shows strong persistence. RTI has R/S $H = 0.788$ (full sample), rising to 0.881 in the S_0 sample. High persistence in RTI implies that rental-affordability conditions are slow to revert: periods of high rental costs relative to income tend to persist, so households observing a tight rental market cannot assume it will loosen quickly.

$$H = \frac{\log \left(\frac{\max_{1 \leq k \leq n} \left(\sum_{t=1}^k (X_t - \bar{X}_n) \right) - \min_{1 \leq k \leq n} \left(\sum_{t=1}^k (X_t - \bar{X}_n) \right)}{\sqrt{\frac{1}{n} \sum_{t=1}^n (X_t - \bar{X}_n)^2}} \right)}{\log n} \quad (92)$$

$$\frac{\log(\text{Var}(d_j))}{\log(2^j)} = 2H + 1 \quad (93)$$

$$H = \frac{1}{2} \left(\frac{\log(\text{Var}(d_j))}{\log(2^j)} - 1 \right) \quad (94)$$

Table 20: Primary Hurst exponent estimates (DFA₂ and Wavelet) by variable and demand regime. **Bold:** lower CI bound exceeds 0.5.

Block	Variable	Regime sample	
		<i>Full</i>	Low-vol
Demand	Δ MDD	0.074 (0.058, 0.090)	0.011 (−0.131, 0.153)
	Δ RTI	0.192 (0.120, 0.263)	0.119 (−0.055, 0.293)
	Δ Unemp.	0.156 (0.046, 0.266)	0.099 (0.066, 0.131)
	Δ CPI	0.383 (0.131, 0.636)	0.153 (0.034, 0.273)
Supply	Δ Constr.Vol	0.079 (0.039, 0.119)	0.132 (0.047, 0.217)
	Plan.Perm	0.058 (0.008, 0.108)	0.015 (−0.220, 0.250)
	Δ ULC	0.068 (0.060, 0.075)	0.092 (0.056, 0.129)
	Δ ConstrCost	0.247 (0.143, 0.351)	0.262 (0.086, 0.439)
Credit	BorrCost	0.312 (0.137, 0.488)	0.217 (0.156, 0.279)
	Δ LDR	0.123 (0.058, 0.189)	0.054 (−0.068, 0.177)
	Net Lending	0.031 (0.011, 0.052)	0.076 (−0.090, 0.243)
	Δ Shadow r	0.239 (0.182, 0.296)	0.118 (−0.462, 0.698)
Assets	Spread	0.294 (0.036, 0.552)	0.536 (0.031, 1.041)
	Share Price	0.177 (0.105, 0.249)	0.228 (0.046, 0.409)
	Δ CB Assets	0.112 (0.026, 0.198)	0.095 (−0.010, 0.200)
	sovCISS	0.545 (0.104, 0.985)	0.463 (0.304, 0.621)
RPPI	Δ RPPI	−0.173 (−1.413, 1.066)	0.254 (−2.592, 3.101)

Notes. DFA₂: detrended fluctuation analysis, order 2 (Peng et al., 1994). RPPI block uses Wavelet (db4 DWT, up to 8 levels; Abry and Veitch, 1998). $H > 0.5$: persistence; $H < 0.5$: mean reversion. Dashes: $T < 20$.

Table 21: Robustness Hurst exponent estimates (R/S) by variable and demand regime. **Bold:** lower CI bound exceeds 0.5.

Block	Variable	Regime sample	
		<i>Full</i>	Low-vol
Demand	Δ MDD	0.814 (0.611, 1.017)	0.761 (0.508, 1.013)
	Δ RTI	0.788 (0.494, 1.082)	0.881 (0.570, 1.191)
	Δ Unemp.	0.873 (0.695, 1.052)	1.001 (0.808, 1.194)
	Δ CPI	0.755 (0.477, 1.034)	0.751 (0.498, 1.003)
Supply	Δ Constr.Vol	0.754 (0.652, 0.856)	0.875 (0.685, 1.065)
	Plan.Perm	0.637 (0.392, 0.883)	0.648 (0.347, 0.948)
	Δ ULC	0.720 (0.500, 0.940)	0.641 (0.228, 1.054)
	Δ ConstrCost	0.663 (0.596, 0.729)	0.613 (0.423, 0.803)
Credit	BorrCost	0.579 (0.033, 1.124)	0.739 (0.667, 0.812)
	Δ LDR	0.751 (0.527, 0.976)	0.760 (0.582, 0.938)
	Net Lending	0.548 (0.461, 0.635)	0.443 (0.379, 0.507)
	Δ Shadow r	0.772 (0.507, 1.038)	0.769 (0.685, 0.852)
Assets	Spread	0.777 (0.591, 0.962)	0.809 (0.594, 1.025)
	Share Price	0.709 (0.573, 0.845)	0.610 (−0.166, 1.386)
	Δ CB Assets	0.693 (0.607, 0.780)	0.704 (0.684, 0.724)
	sovCISS	0.933 (0.865, 1.001)	0.920 (0.835, 1.005)

Notes. R/S: rescaled-range analysis (Hurst, 1951); acts as a robustness check against DFA₂. $H > 0.5$: persistence; $H < 0.5$: mean reversion. Dashes: $T < 20$.

D.3 Unit Root and Autocorrelation Tests

Table 22: Unit-root and stationarity tests for transformed series

Block	Variable	T	$\hat{p}$	ADF test		KPSS test		Verdict
				ADF stat.	ADF p	KPSS stat.	KPSS p	
Demand	$\Delta \log \text{MDD}$	99	3	−2.897*	0.046	0.253	≥ 0.100	✓
	$\Delta \log \text{RTI}$	98	2	−4.451**	0.000	0.210	≥ 0.100	✓
	$\Delta \text{Unemployment}$	99	3	−3.237*	0.018	0.268	≥ 0.100	✓
	$\Delta \log \text{CPI}$	99	0	−5.465**	0.000	0.068	≥ 0.100	✓
Supply	$\Delta \log \text{Constr. Vol.}$	99	2	−3.657**	0.005	0.225	≥ 0.100	✓
	$\Delta \text{Planning Perms}$	99	2	−4.951**	0.000	0.273	≥ 0.100	✓
	$\Delta \log \text{ULC}$	98	0	−12.032**	0.000	0.205	≥ 0.100	✓
	$\Delta \text{Constr. Cost}$	99	1	−4.590**	0.000	0.177	≥ 0.100	✓
Credit	Mortgage Rate	99	1	−5.765**	0.000	0.057	≥ 0.100	✓
	$\Delta \log \text{LDR}$	99	2	−4.377**	0.000	0.628	0.020*	(✓) ^b
	$\Delta \text{Net Lending}$	99	3	−3.838**	0.003	0.178	≥ 0.100	✓
	$\Delta \text{Shadow Rate}$	99	1	−3.799**	0.003	0.244	≥ 0.100	✓
Assets	ΔSpread	99	0	−5.200**	0.000	0.100	≥ 0.100	✓
	$\Delta \log \text{Share Prices}$	99	1	−5.648**	0.000	0.114	≥ 0.100	✓
	$\Delta \log \text{CB Assets}$	99	0	−7.063**	0.000	0.154	≥ 0.100	✓
	sovCISS	99	1	−3.102*	0.026	0.187	≥ 0.100	✓
RPPI	$\Delta \log \text{RPPI}$	99	4	−2.480	0.120	0.188	≥ 0.100	(✓) ^a

Notes. ADF: Augmented Dickey–Fuller, H_0 : unit root. A constant is included; lag order $\hat{p}$ is selected by AIC (maximum 4). KPSS: Kwiatkowski–Phillips–Schmidt–Shin, H_0 : level stationarity. Bandwidth is selected automatically using Newey–West. Significance markers: ** $p < 0.01$, * $p < 0.05$, and † $p < 0.10$ (ADF only). Verdicts: ✓ = both tests support stationarity; (✓)^a = ADF is borderline ($p > 0.05$) but KPSS does not reject stationarity, so the series is treated as stationary; (✓)^b = ADF rejects a unit root but KPSS rejects level stationarity, consistent with long memory or fractional integration (see Hurst analysis, Appendix H); × = evidence of a unit root. KPSS p -values are clipped to the interval [0.01, 0.10] by the test implementation.

Table 23: Ljung–Box residual autocorrelation tests. H_0 : no autocorrelation up to lag h .

Block	Variable	Lag $h = 4$		Lag $h = 8$	
		Q_4	p -value	Q_8	p -value
Demand	$\Delta \log$ MDD	7.11	0.130	14.22	0.076
	$\Delta \log$ RTI	17.45	0.002*	20.12	0.010*
	Δ Unempl.	6.97	0.137	27.59	0.001*
	$\Delta \log$ CPI	2.59	0.629	13.28	0.103
Supply	$\Delta \log$ Constr.	12.23	0.016*	19.90	0.011*
	Planning	10.84	0.028*	17.29	0.027*
	$\Delta \log$ ULC	5.01	0.286	7.98	0.436
	Δ Cost	5.23	0.265	9.27	0.320
Credit	Cost Borr.	4.47	0.346	6.83	0.555
	$\Delta \log$ LDR	6.94	0.139	9.07	0.336
	Δ Net Lend.	62.72	0.000*	108.14	0.000*
	Δ Shadow	2.09	0.720	6.65	0.574
Assets	Spread	5.16	0.271	8.15	0.419
	Δ Share prices.	16.21	0.003*	19.30	0.013*
	$\Delta \log$ CB Assets	2.84	0.586	6.39	0.604
	sovCISS	185.79	0.000*	231.20	0.000*
RPPI	$\Delta \log$ RPPI	39.47	0.000*	68.80	0.000*

Notes. Ljung–Box statistic $Q_h = n(n+2) \sum_{k=1}^h \hat{\rho}_k^2 / (n-k)$, distributed $\chi^2(h)$ under H_0 . *: $p < 0.05$. Residuals standardised by regime-weighted posterior mean volatility.

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